

THE UPPER LIMB

THE UL

Outline

- ◆ Shoulder: pectoral girdle
- ◆ Arm
- ◆ Forearm
- ◆ Hand
- ◆ Joints
- ◆ Clinical correlations

BONES OF UL

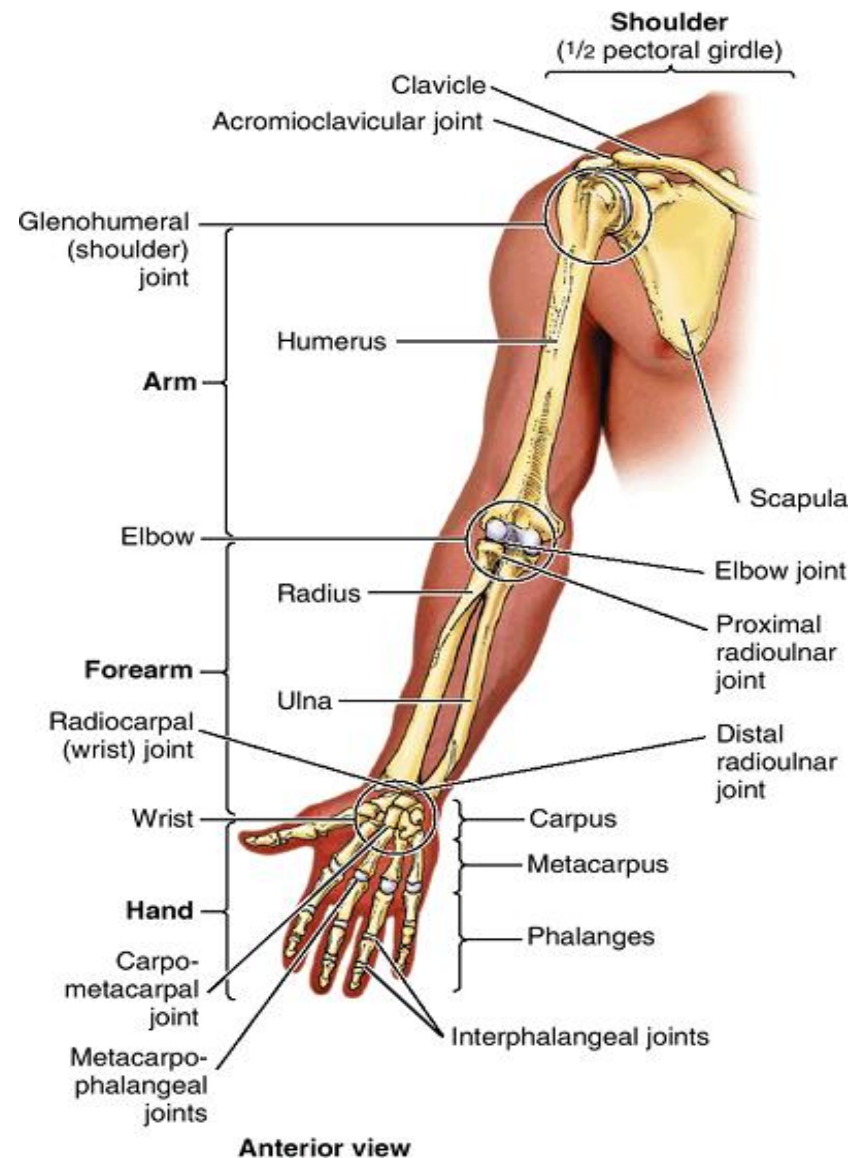
The superior appendicular skeleton

◆ Composed of :

- Pectoral girdle
- Free part

◆ Articulates with the axial skeleton only at the **sternoclavicular joint**.

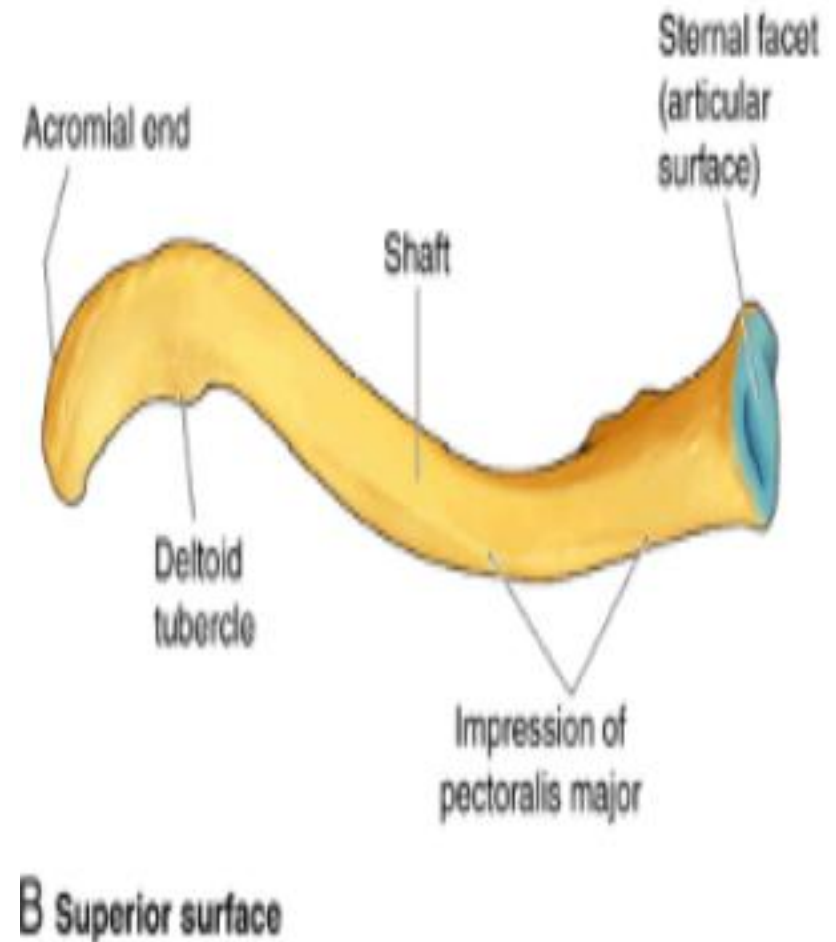
◆ Allows great mobility



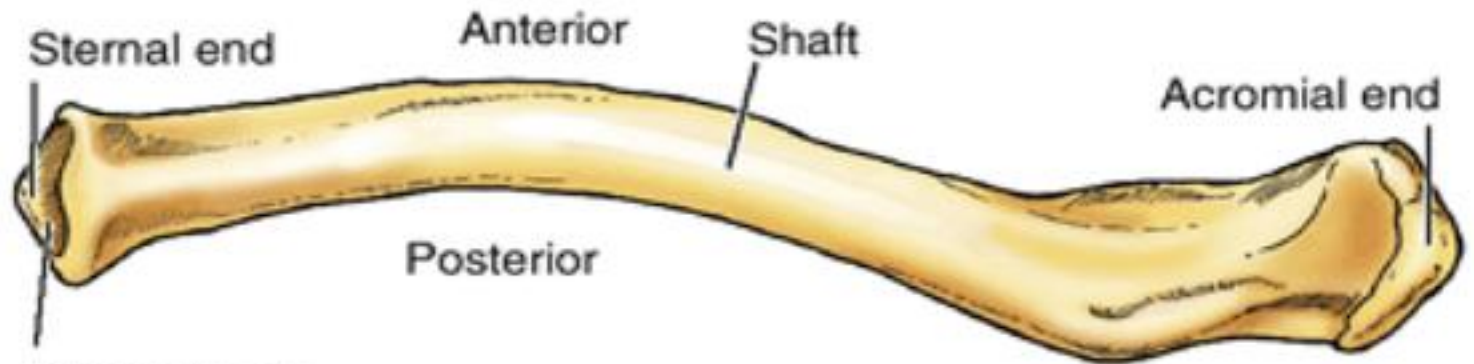
BONES OF UL

Clavicle/Collar bone

- Connects the upper limb to the trunk.
- **Sternal end:** At manubrium of the sternum at the SC joint.
- **Acromial end:** Acromion of the scapula at the AC joint.
- The medial 2/3rd of the shaft **convex** anteriorly whereas the lateral third is flattened and **concave** anteriorly.

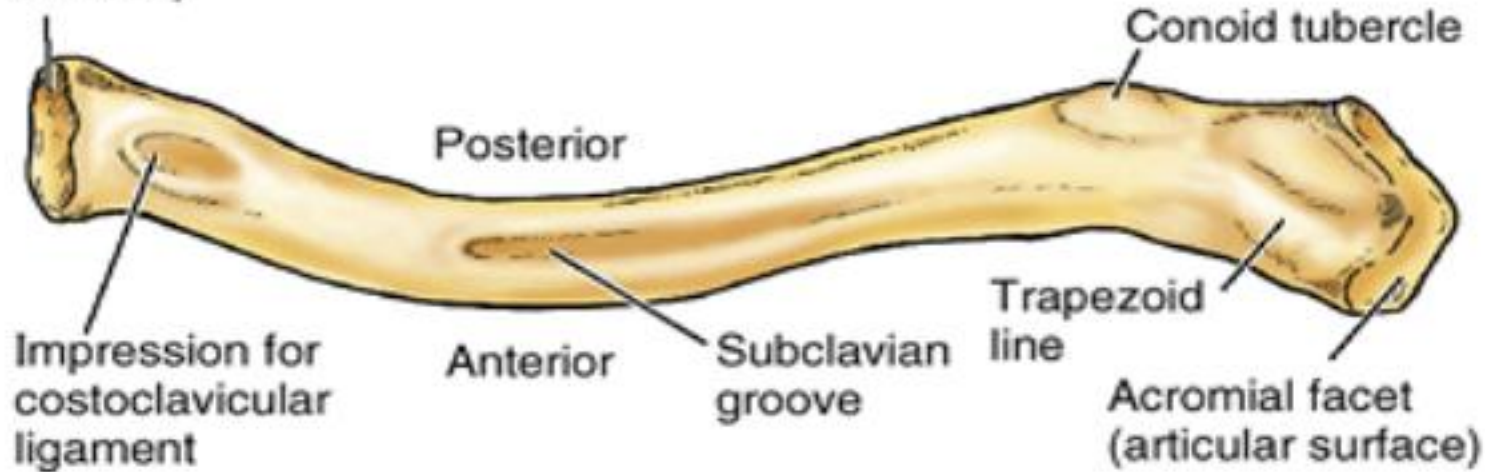


Right clavicle



Sternal facet
(articular
surface)

Superior surface

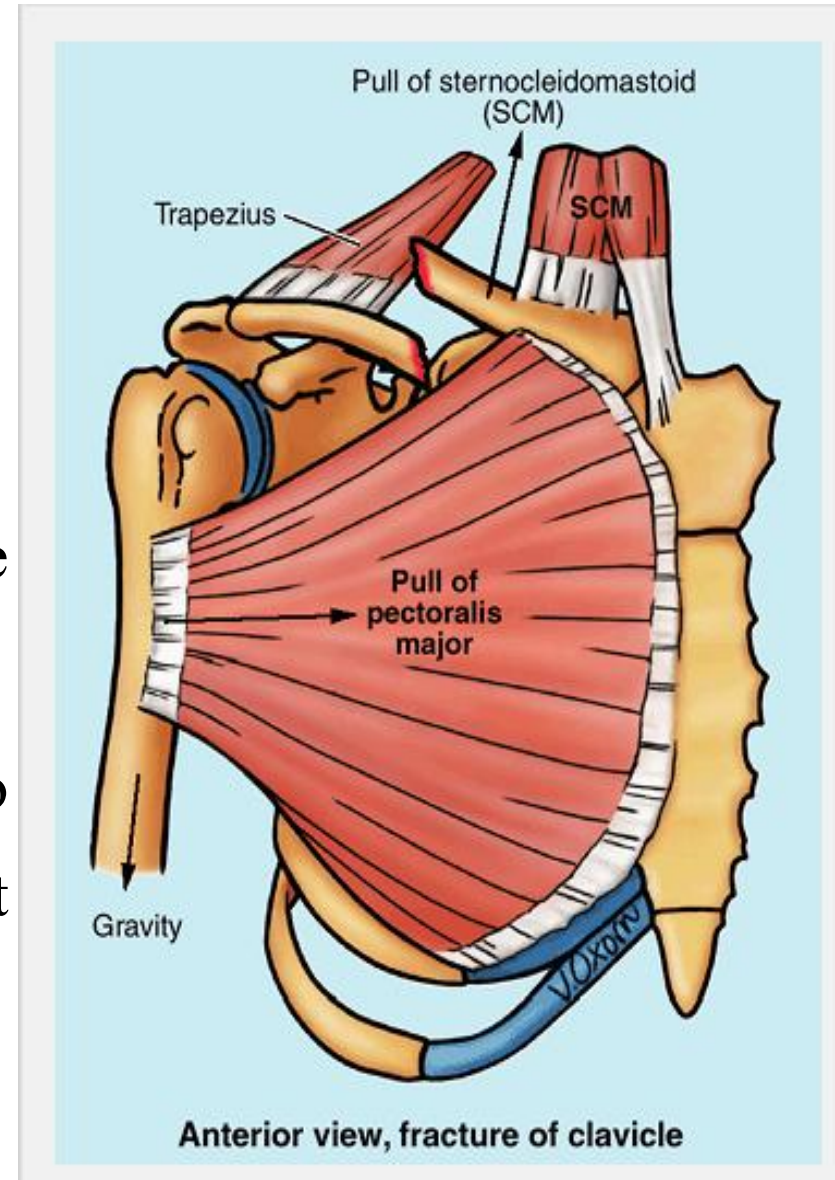


Inferior surface

The weakest part is at the
**junction of its middle and
lateral thirds**

After fracture

- ◆ SCM muscle elevates the medial fragment of bone
- ◆ Trapezius muscle is unable to hold up the lateral fragment owing to the weight of the upper limb or **shoulder drops**



BONES OF UL.....CONT'D

Scapula

- ◆ The scapula (shoulder blade) is a triangular flat bone that lies on the posterolateral aspect of the thorax, overlying the 2nd - 7th ribs.
- ◆ Posterior surface
 - ✓ Convex, divided by its spine into small **supraspinous** fossa and large **infraspinous** fossa.
- ◆ Anterior or concave costal surface
 - ✓ has a large **subscapular** fossa.
- ◆ The triangular body (blade) of the scapula is thin.

SCAPULA

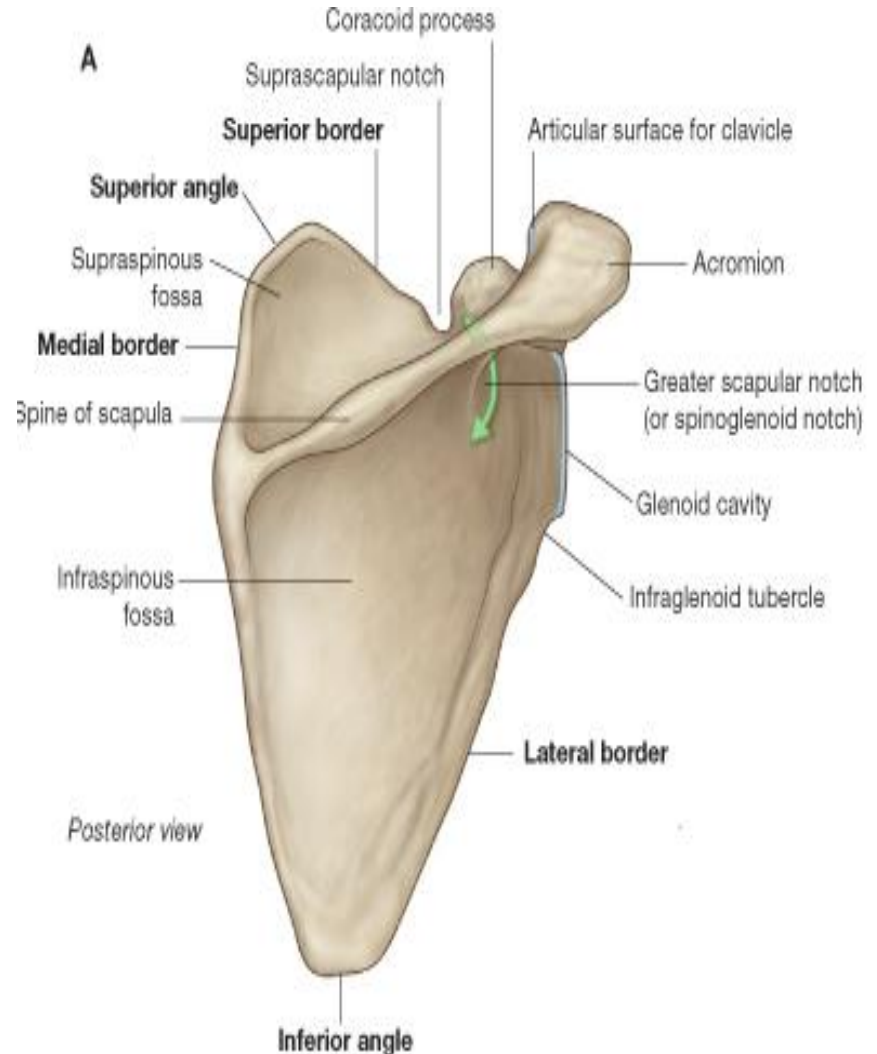
The scapula has: two **angles**

(superior, and inferior)

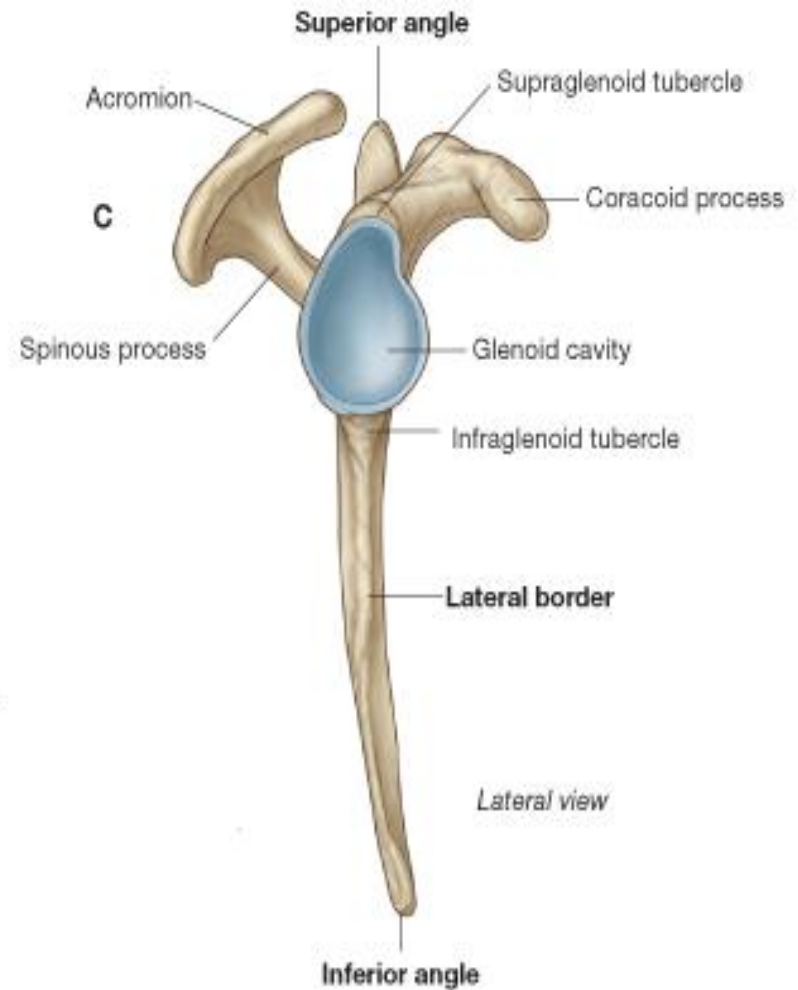
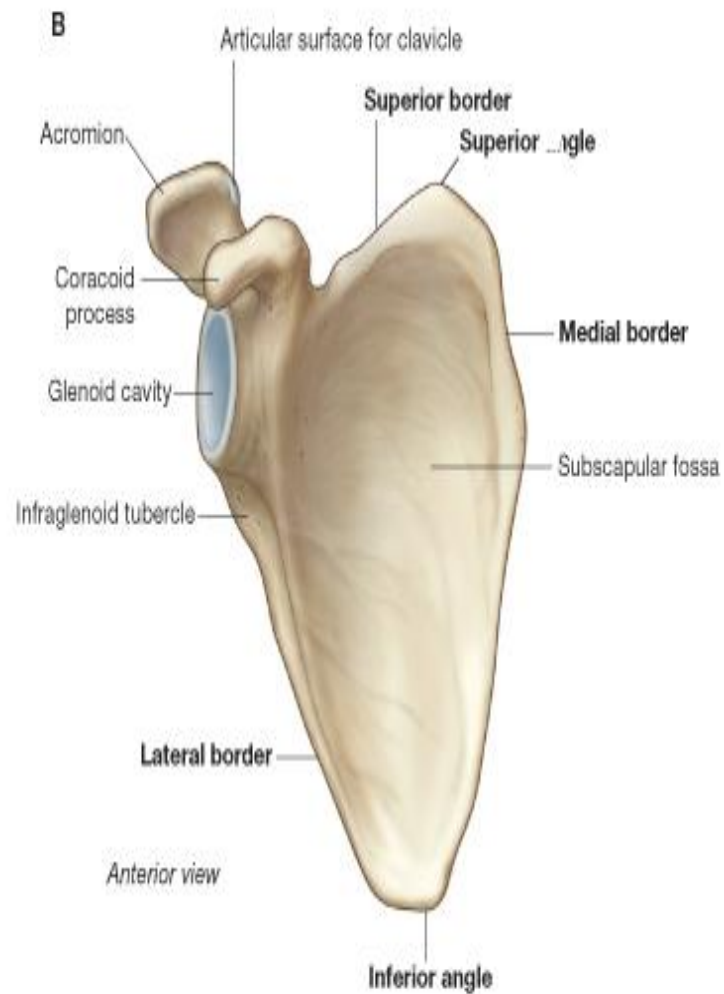
three **borders** (superior, lateral, and medial);

two **surfaces** (costal and posterior); and

Two **processes** (acromion, and coracoid process)



SCAPULA...

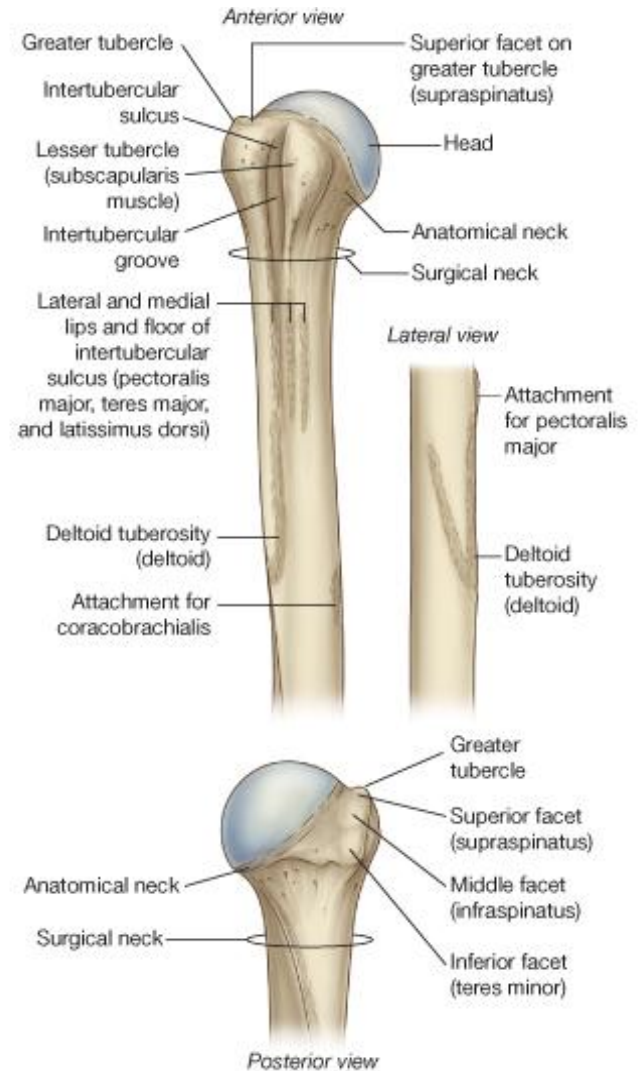


HUMERUS

◆ The proximal end of the humerus consists

- ✓ Head
- ✓ Neck(Anatomical and surgical)
- ✓ Tubercles(Greater and Lesser)
- ✓ Superior half of the shaft

◆ Surgical neck- the most frequent site of fracture .



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HUMERUS

◆ Shaft (body):- features

✓ **Deltoid tuberosity**

✓ **Radial groove**

◆ **Inferior end**

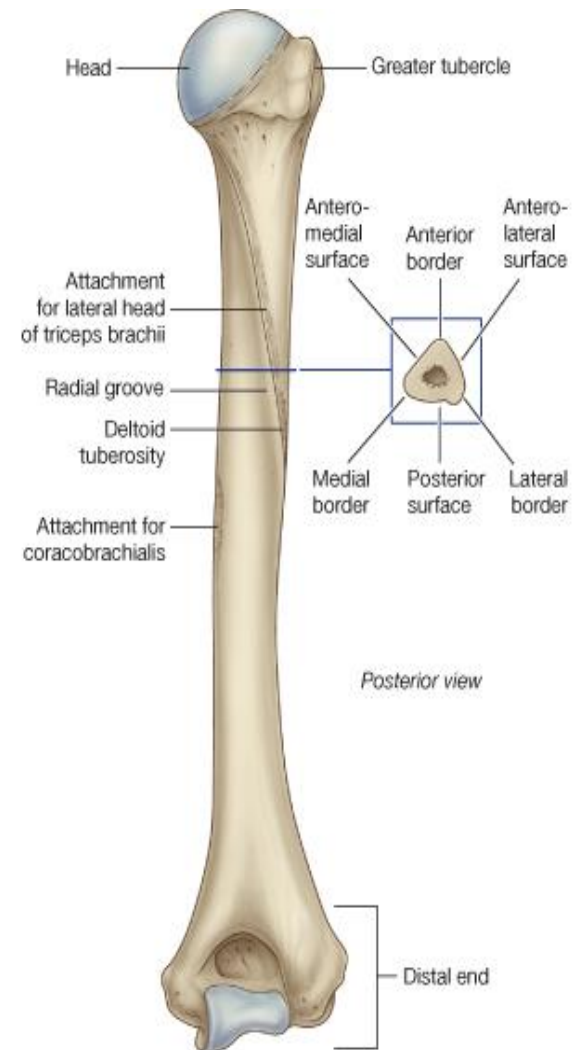
✓ Medial and lateral

supracondylar ridges

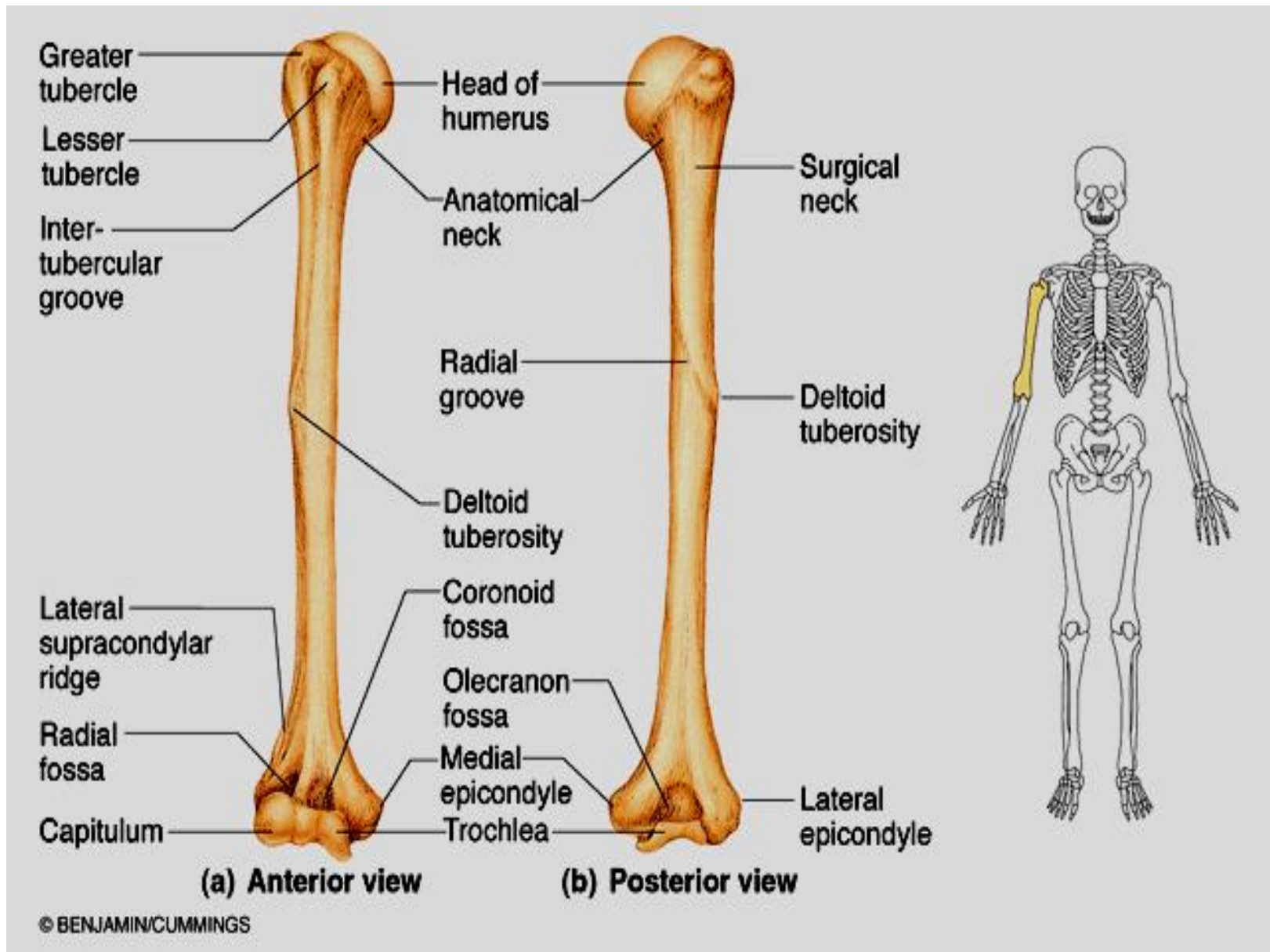
✓ medial and lateral
epicondyles

✓ **The radial, Olecranon, &
Coronoid fossa**

✓ **Trochlea and capitulum**

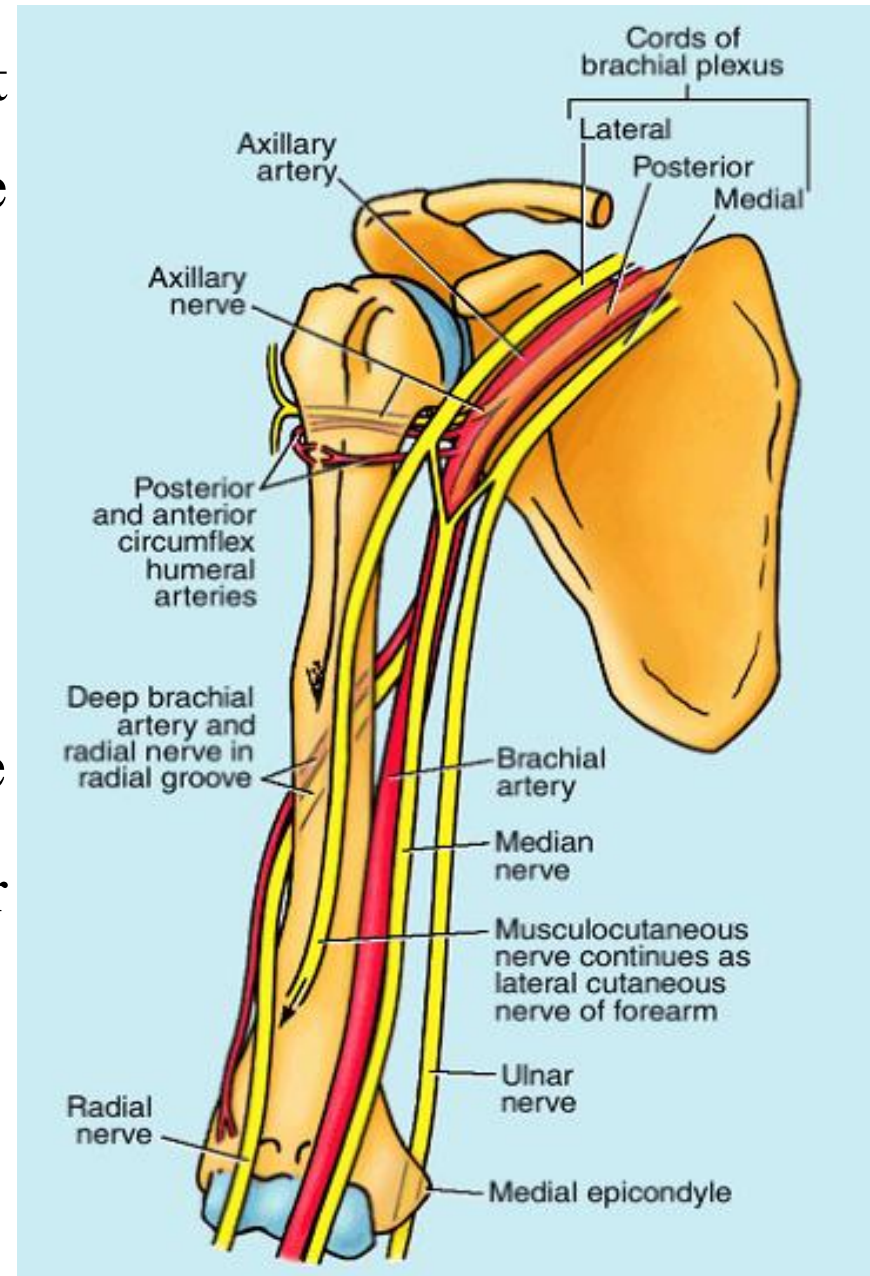


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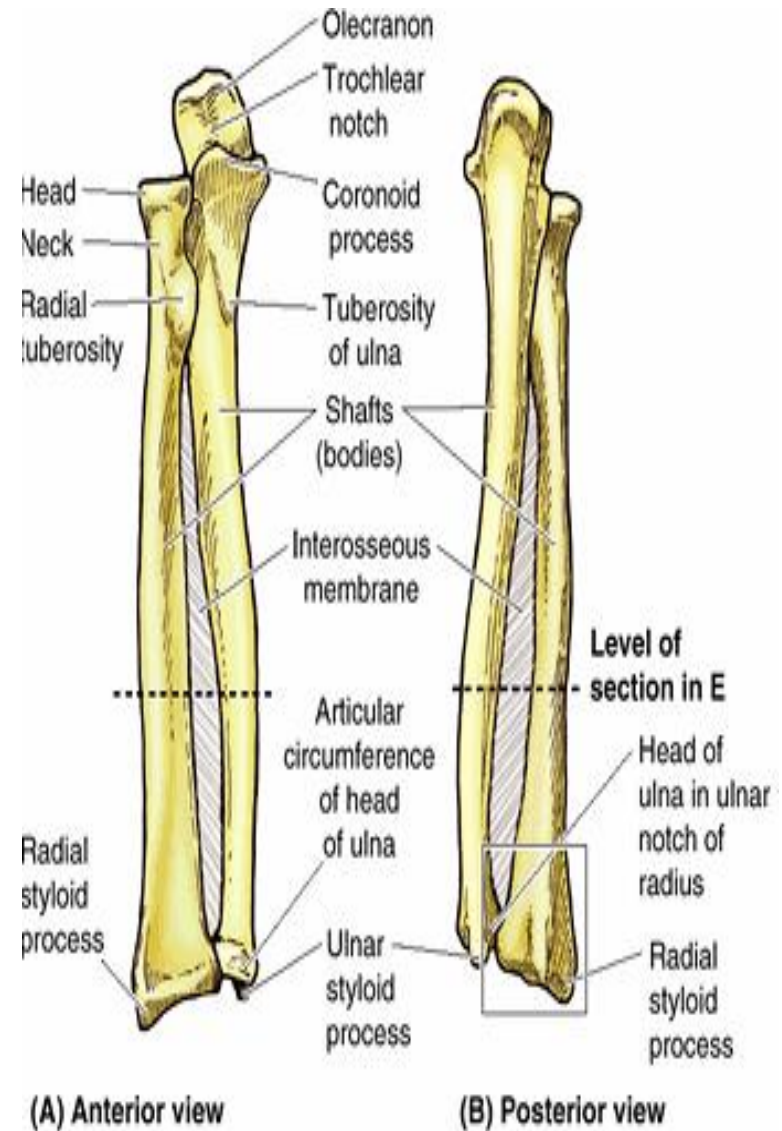
Because nerves are in contact with the humerus, they may be injured when the associated part of the humerus is fractured

- **Surgical neck** – axillary nerve
- **Radial groove** – radial nerve
- **Distal humerus** – median nerve
- **Medial epicondyle** – ulnar nerve



ULNA

- ◆ The medial and longer of the two forearm bones
- ◆ Its proximal end has **two** prominent projections
 - **Olecranon posteriorly** – resemble the upper jaw
 - **Coronoid process anteriorly** – form the walls of the trochlear notch



Ulna

Radial notch

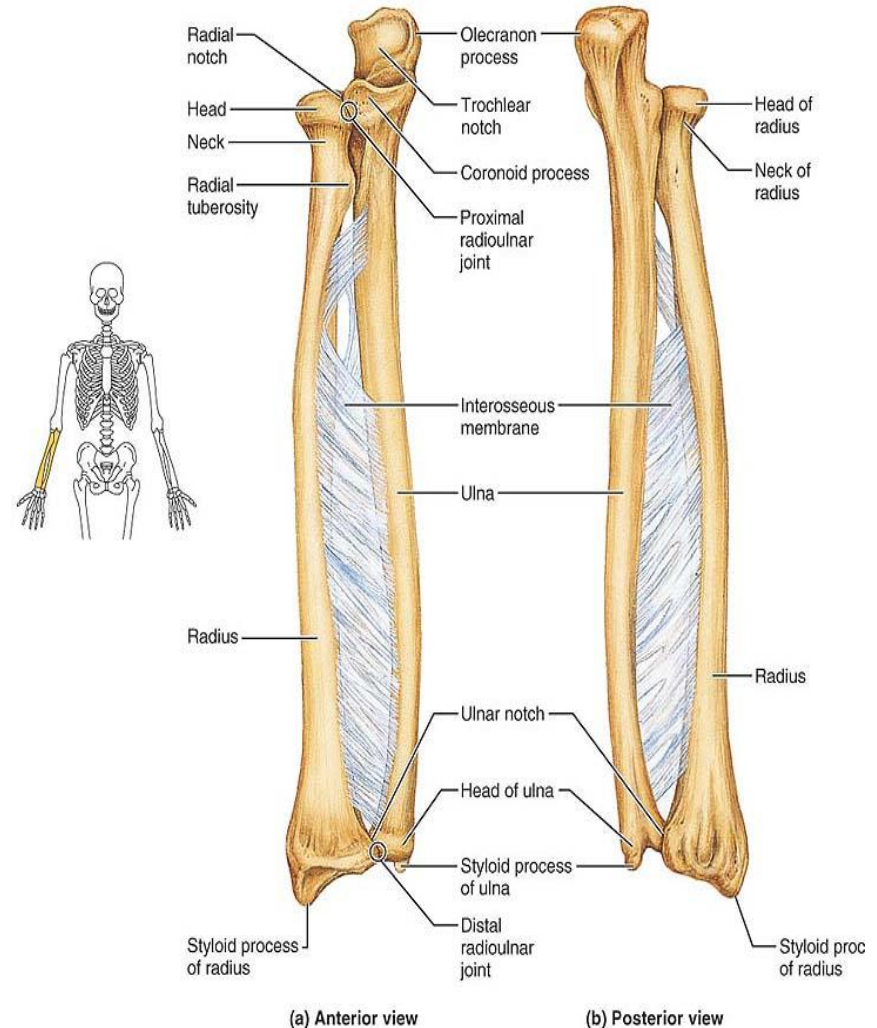
- ✓ on the lateral side of the coronoid process
- ✓ articulates with the head of radius

Shaft (body)

- proximally it is thick, but it tapers distally

Head

- Narrow distal end
- Rounded with small, conical ulnar styloid process
- Ulna does not reach, and therefore does not participate in the wrist (radiocarpal) joint

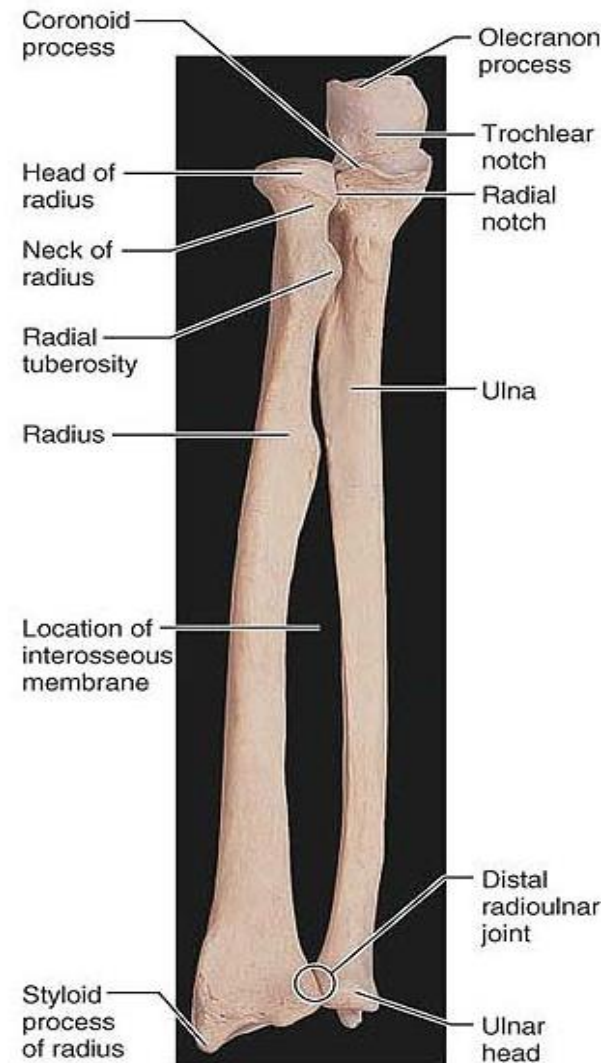


RADIUS

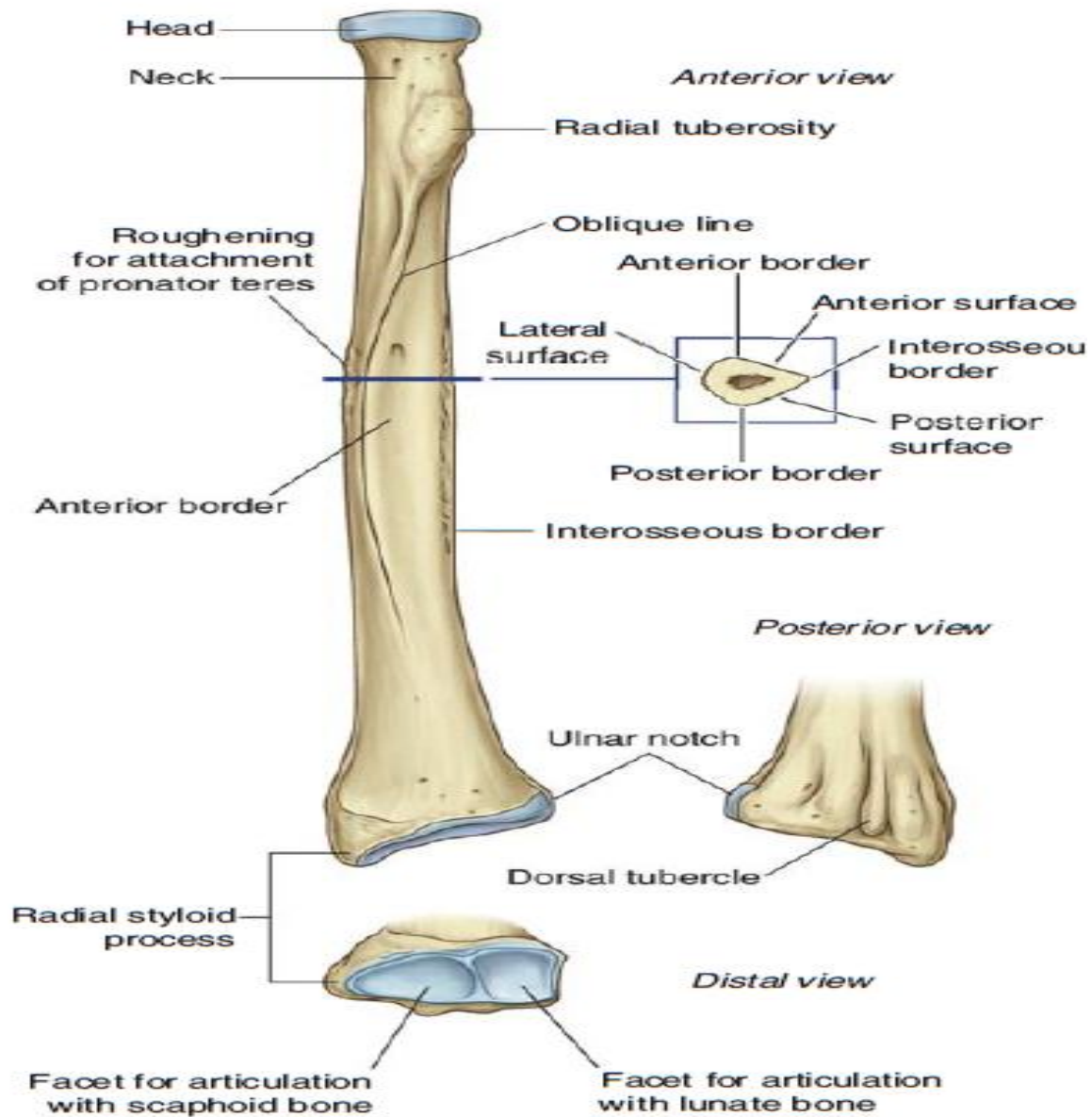
- ◆ The lateral and shorter of the two forearm bones
- ◆ Its proximal end consists of a cylindrical **head**, a **short neck**, a projection from the medial surface (**radial tuberosity**)

Head

- superior surface is concave for articulation with **capitulum of humerus**
- medially articulate with **radial notch of ulna**



Radius (left) and ulna (right), anterior view



A

Neck

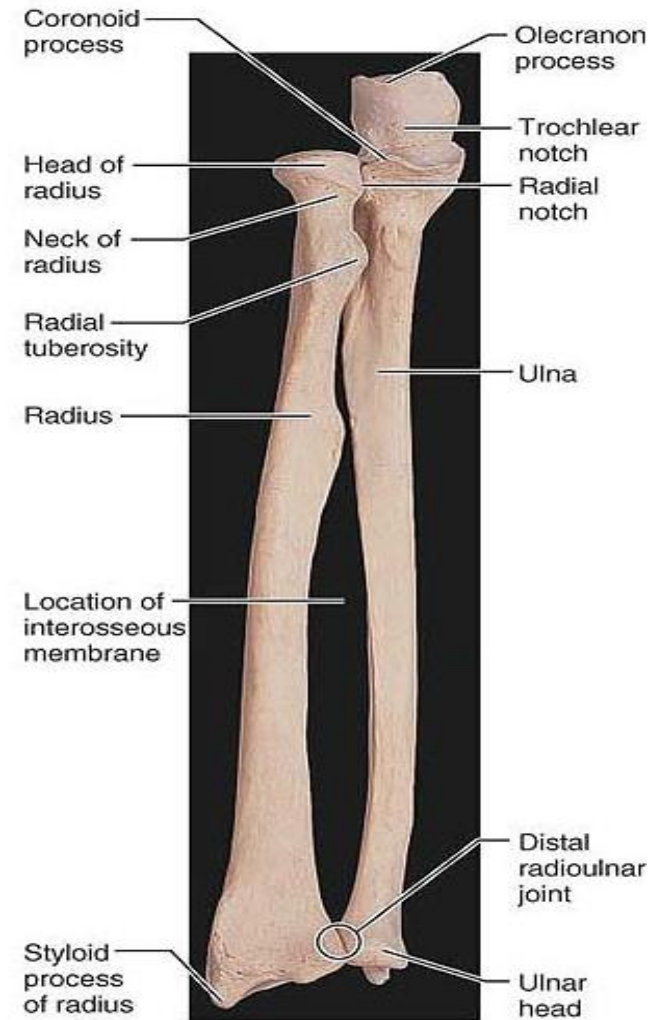
Narrow part between head and radial tuberosity.

Shaft (body)

Gradually enlarges distally

Distal end

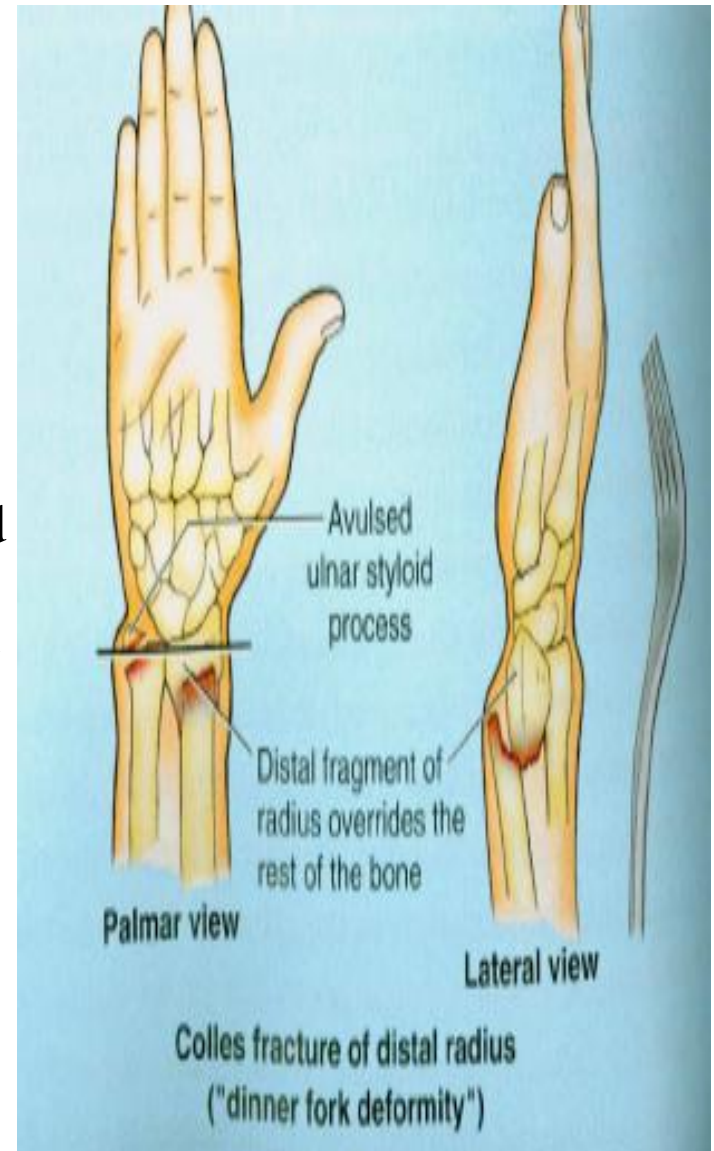
- medially - ulnar notch for head of ulna
- laterally - radial styloid process



Radius (left) and ulna (right), anterior view

Colles fracture

- A complete fracture of the distal **end of the radius**
- The most common fracture of the forearm
- Results from **forced dorsiflexion** of the hand
- The distal fragment of the radius is displaced dorsally and often broken into pieces
- Often, the ulnar styloid process is avulsed (broken off)

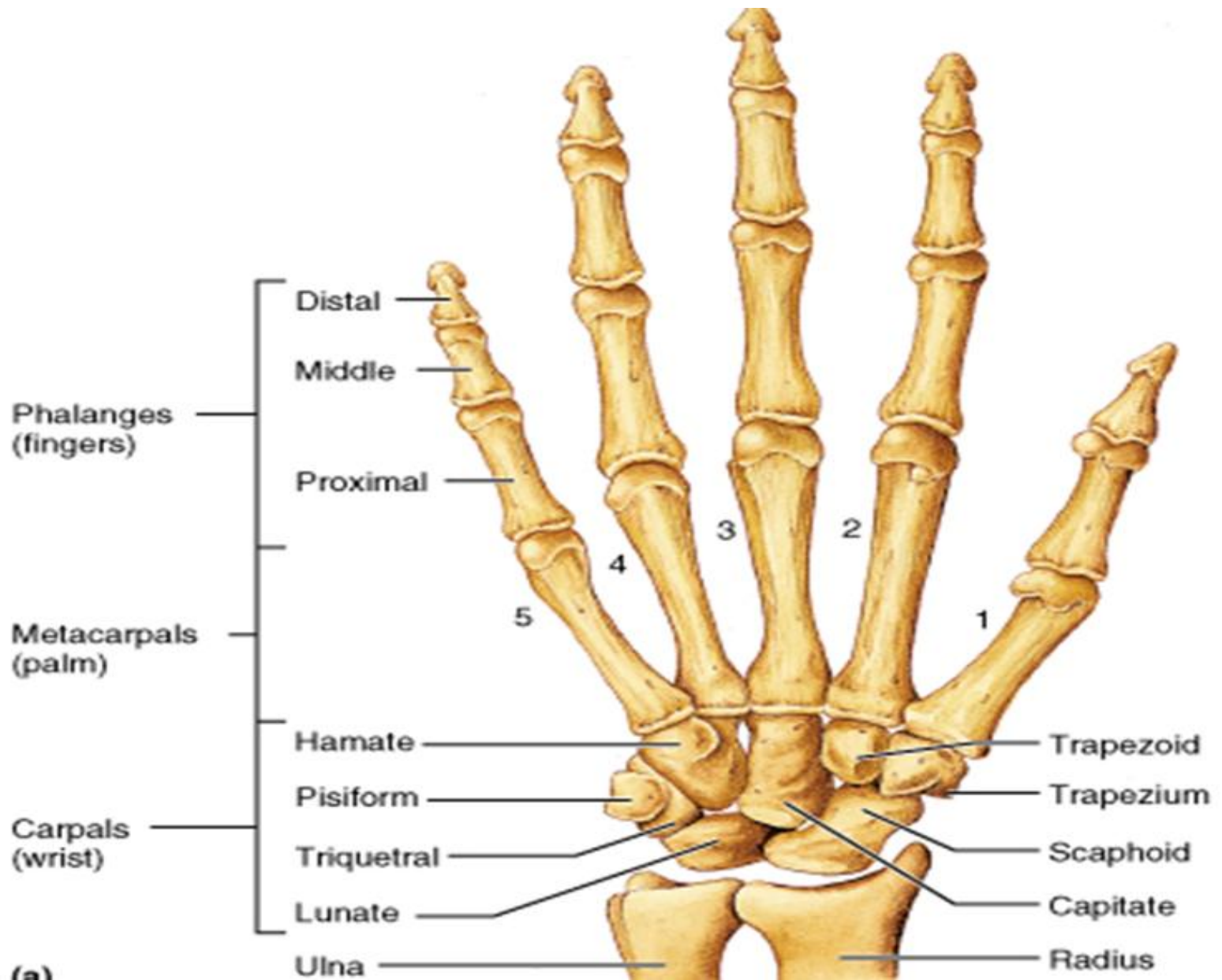


BONES OF THE HAND

- The wrist, or carpus, is composed of eight carpal bones (carpals) arranged in proximal and distal rows of four.
- The carpus is markedly convex from side to side posteriorly and concave anteriorly.

Proximal row from lateral to medial:

- Scaphoid: boat-shaped with scaphoid tubercle
- Lunate: moon-shaped
- Triquetrum: pyramidal
- Pisiform: pea-shaped; lies on palmar surface of triquetrum



DISTAL ROW

- ◆ Articulate with proximal row of carpals proximally and metacarpals distally.

Distal row from lateral to medial:

- Trapezium: four sided
- Trapezoid: wedge-shaped
- Capitate: head like; largest
- Hamate: wedge-shaped, has hooked process (hook of hamate)

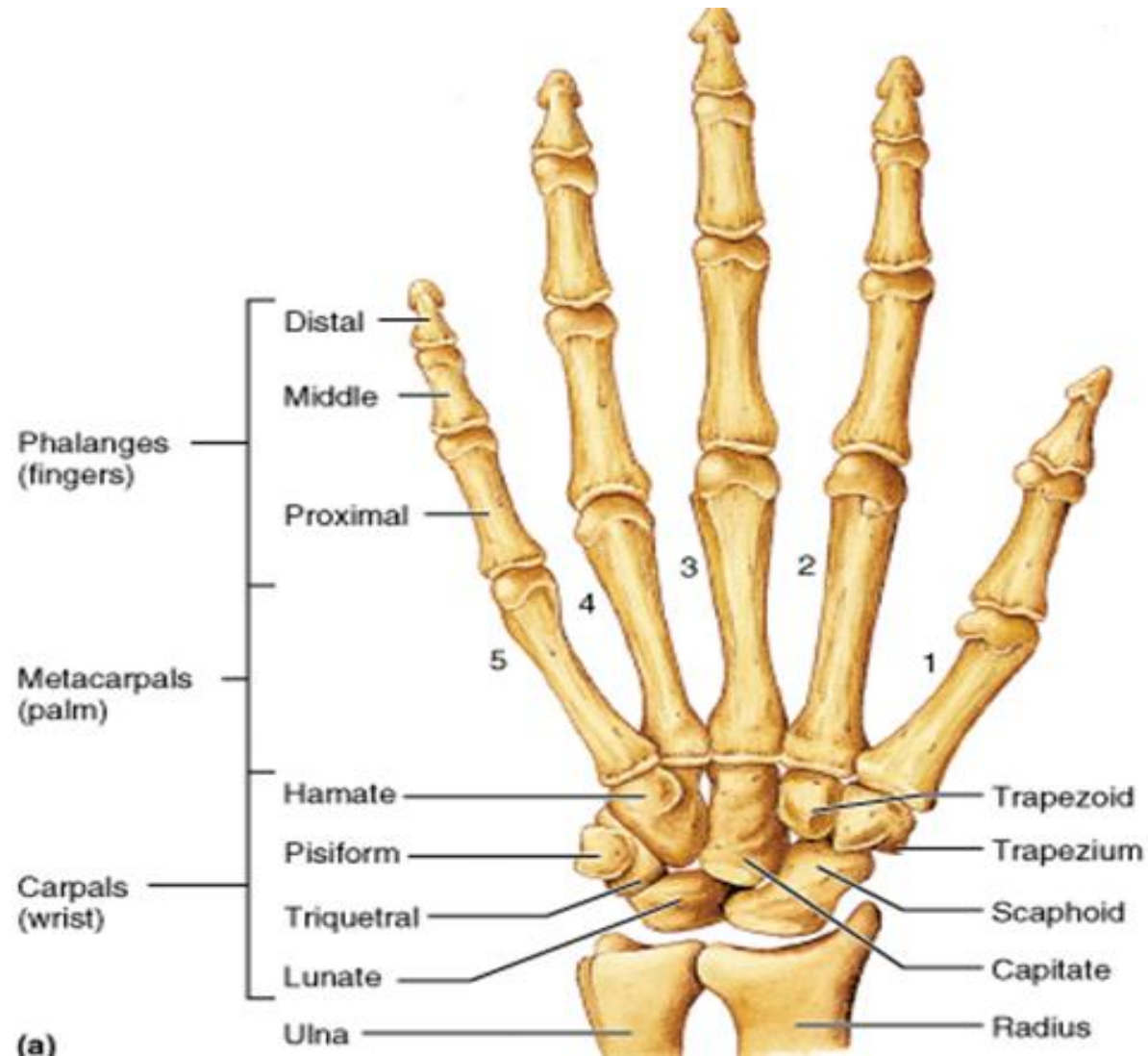
METACARPALS

- ◆ Forms the skeleton of the palm.
- ◆ Composed of five metacarpal bones
- ◆ Each metacarpal consists of a base (near carpal), shaft, and head.
- ◆ The proximal bases of the metacarpals articulate with the carpal bones.
- ◆ The distal heads of the metacarpals articulate with the proximal phalanges and form the knuckles

PHALANGES

- ◆ 14 bones; form the fingers
- ◆ Each digit has three phalanges (proximal, middle, and distal) except for the first (thumb), which has only two (proximal and distal)
- ◆ Each phalanx has a base proximally, a shaft (body), and a head distally
- ◆ The distal phalanges are flattened and expanded at their distal ends, which underlie the nail beds

THE HAND BONES

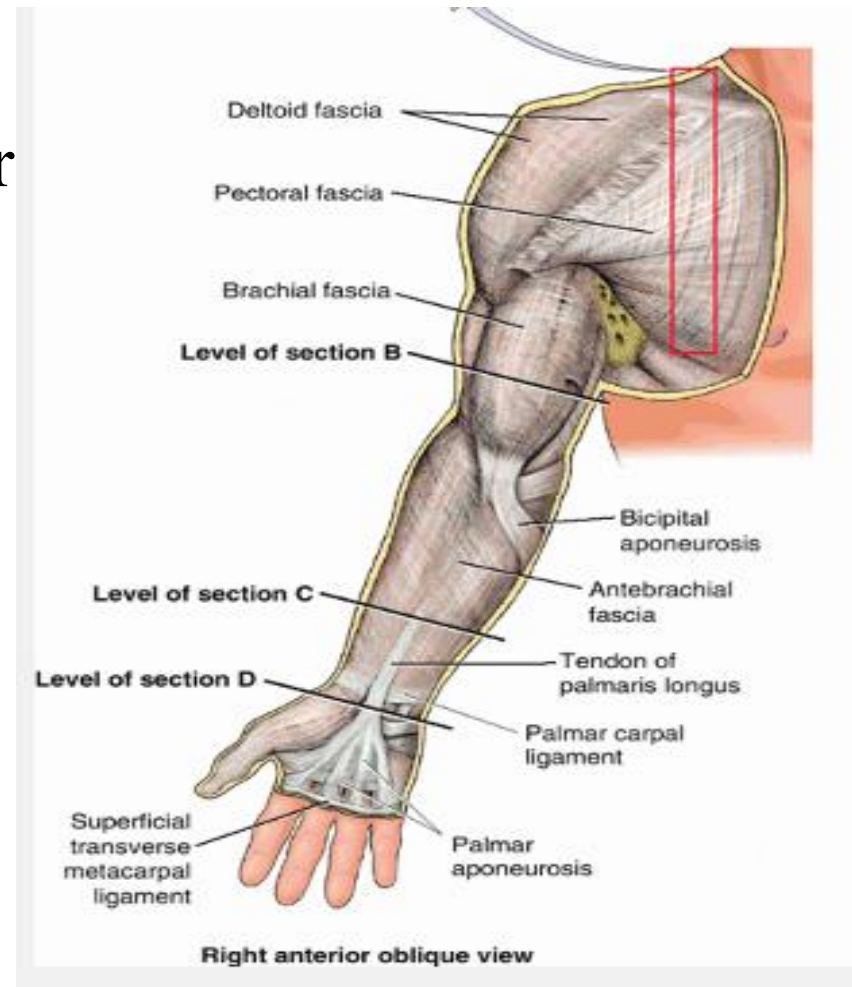


SUPERFICIAL STRUCTURES OF UPPER LIMB

FASCIA OF UPPER LIMB

Pectoral fascia

- Invests the pectoralis major
- Continuous inferiorly with fascia of anterior abdominal wall
- Leaves lateral border of pectoralis major and becomes axillary fascia



Deltoid fascia: invests deltoid muscle

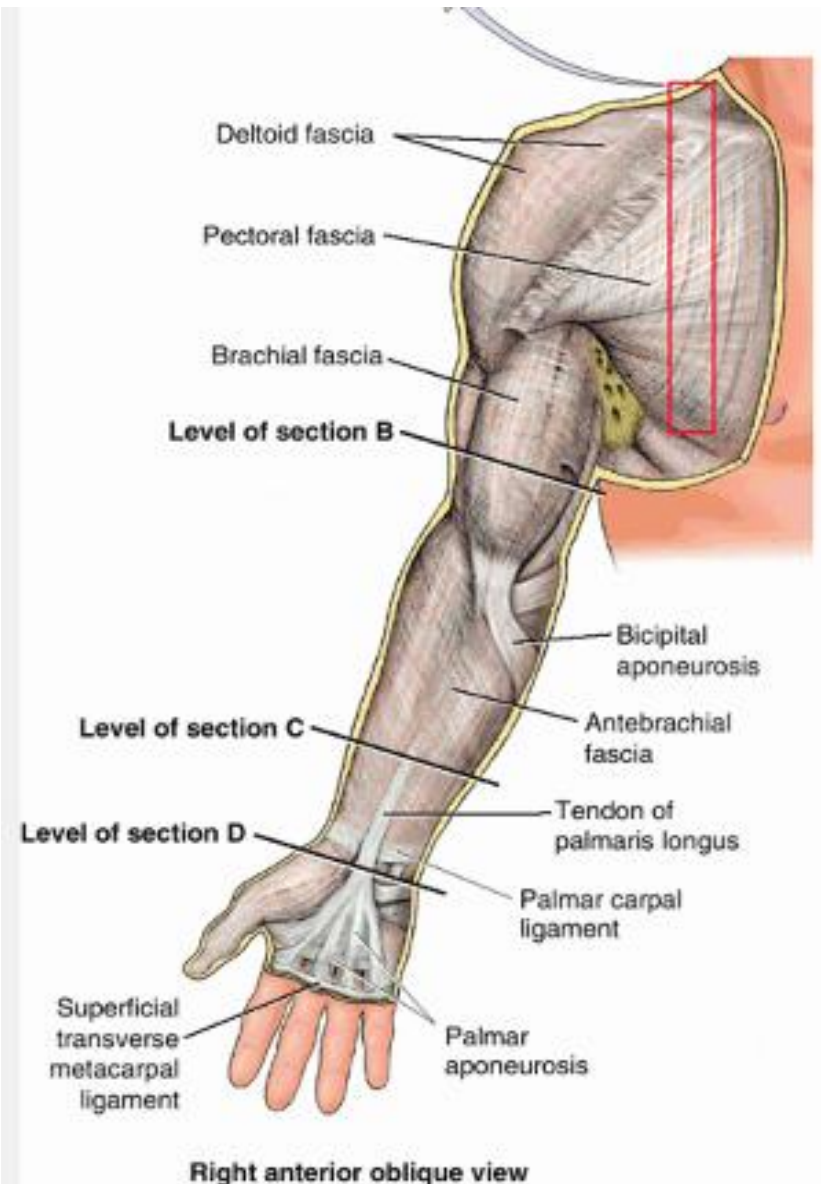
■ **Brachial fascia:** encloses the arm

■ continuous with deltoid, pectoral, axillary and antebrachial fasciae

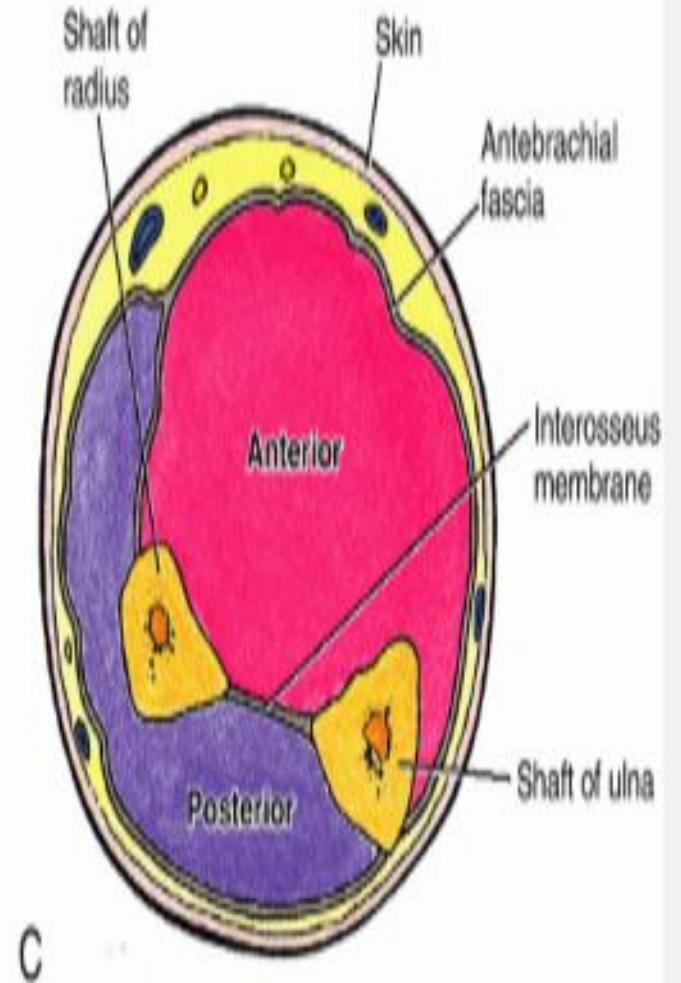
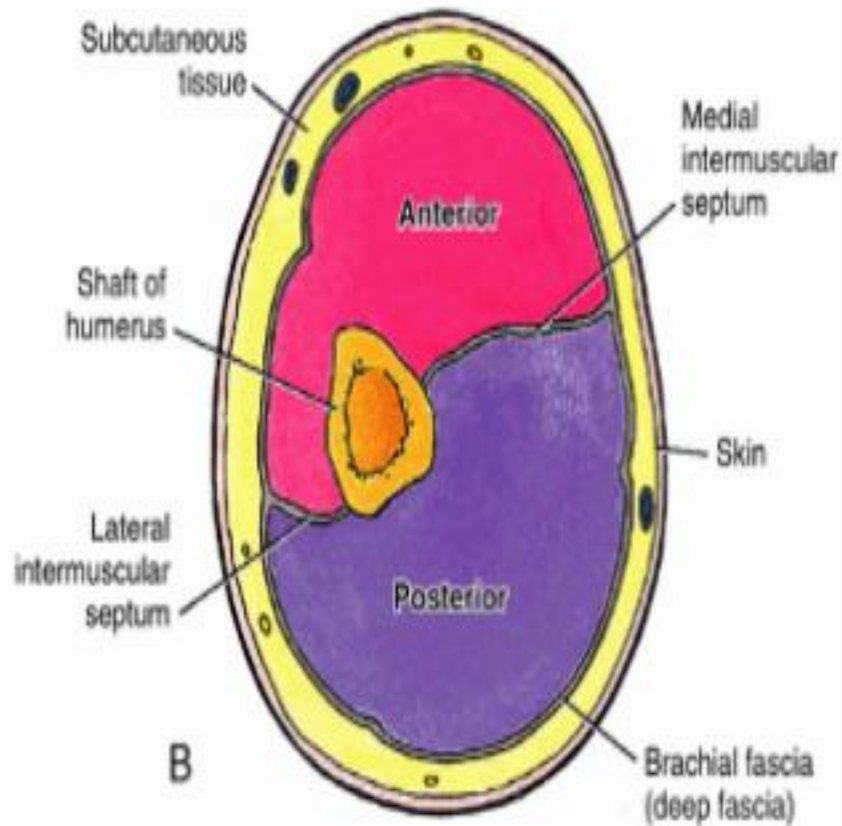
■ Form two intermuscular septa

○ Medial and lateral

○ Divide the arm into anterior (flexor) and posterior (extensor) fascial

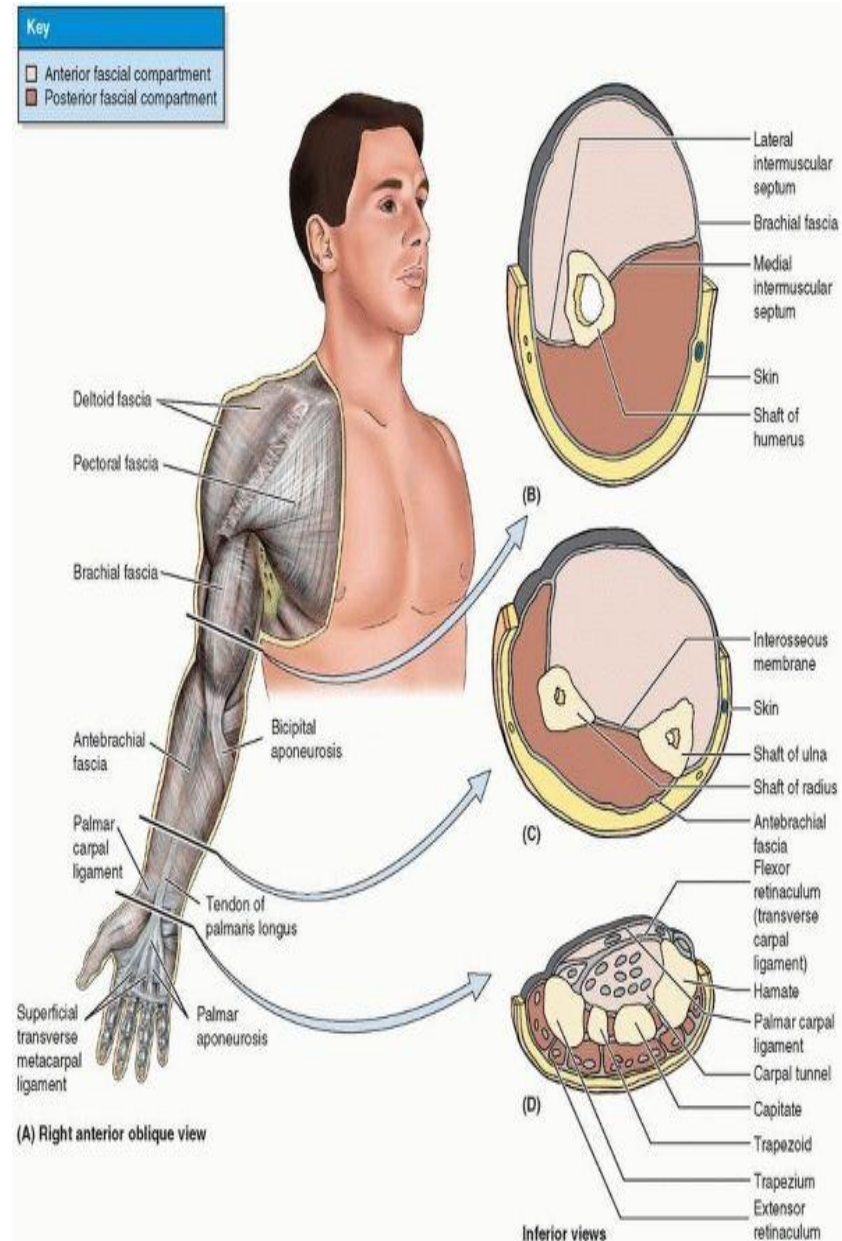


Fascia in A		Fascial compartments in B & C	
	Deep cervical		Pectoral
	Clavipectoral		Axillary
			Anterior
			Posterior



Antebrachial fascia: Forearm

- Surrounds fascial compartments
- Posterior thickening** over the distal end to form a **transverse band**,
 - Extensor retinaculum**, which holds the extensor tendons in position.
- Anterior thickening**: at distal end forms:
 - palmar carpal ligament and **FR**



Palmar fascia

- Extension of deep fascia beyond retinacula.
- **Palmar aponeurosis**
 - Central part, thick, triangular
 - Forms four thickenings that radiate to the base of fingers
 - Superficial **transverse metacarpal ligament**: traverse bands distally
 - Ligaments extend from aponeurosis to skin, hold the skin closer



CUTANEOUS NERVES OF UPPER LIMB

- ◆ Developmentally the limbs grow as lateral protrusions of trunk, with thumb located on cranial side.
- ◆ Most cutaneous nerves of upper limb are derived from brachial plexus
- ◆ Cutaneous nerves to shoulder are derived from cervical plexus

C4: base of the neck, shoulder

C5: arm laterally

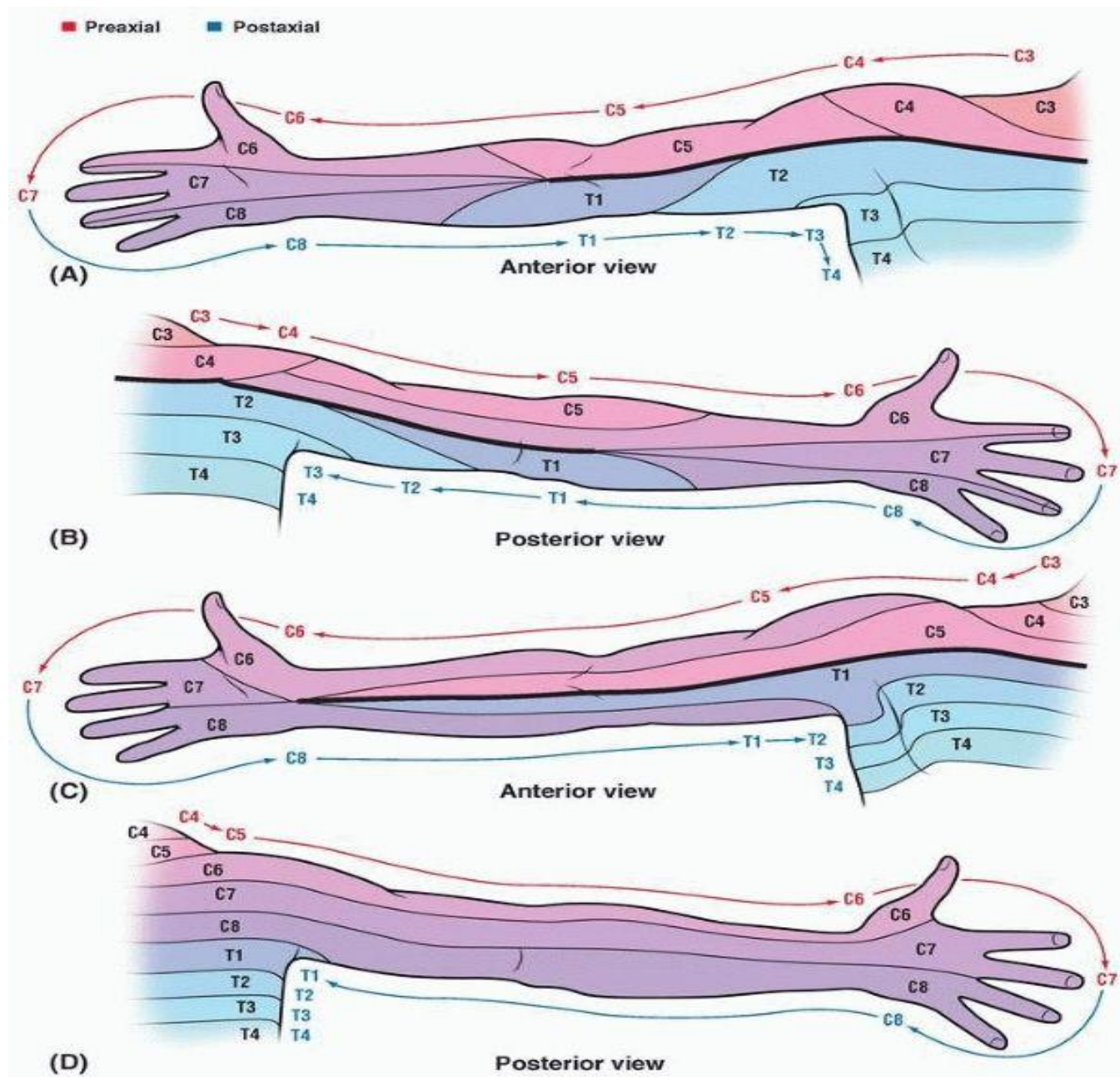
C6: forearm laterally and thumb

C7: middle three fingers and middle of posterior surface

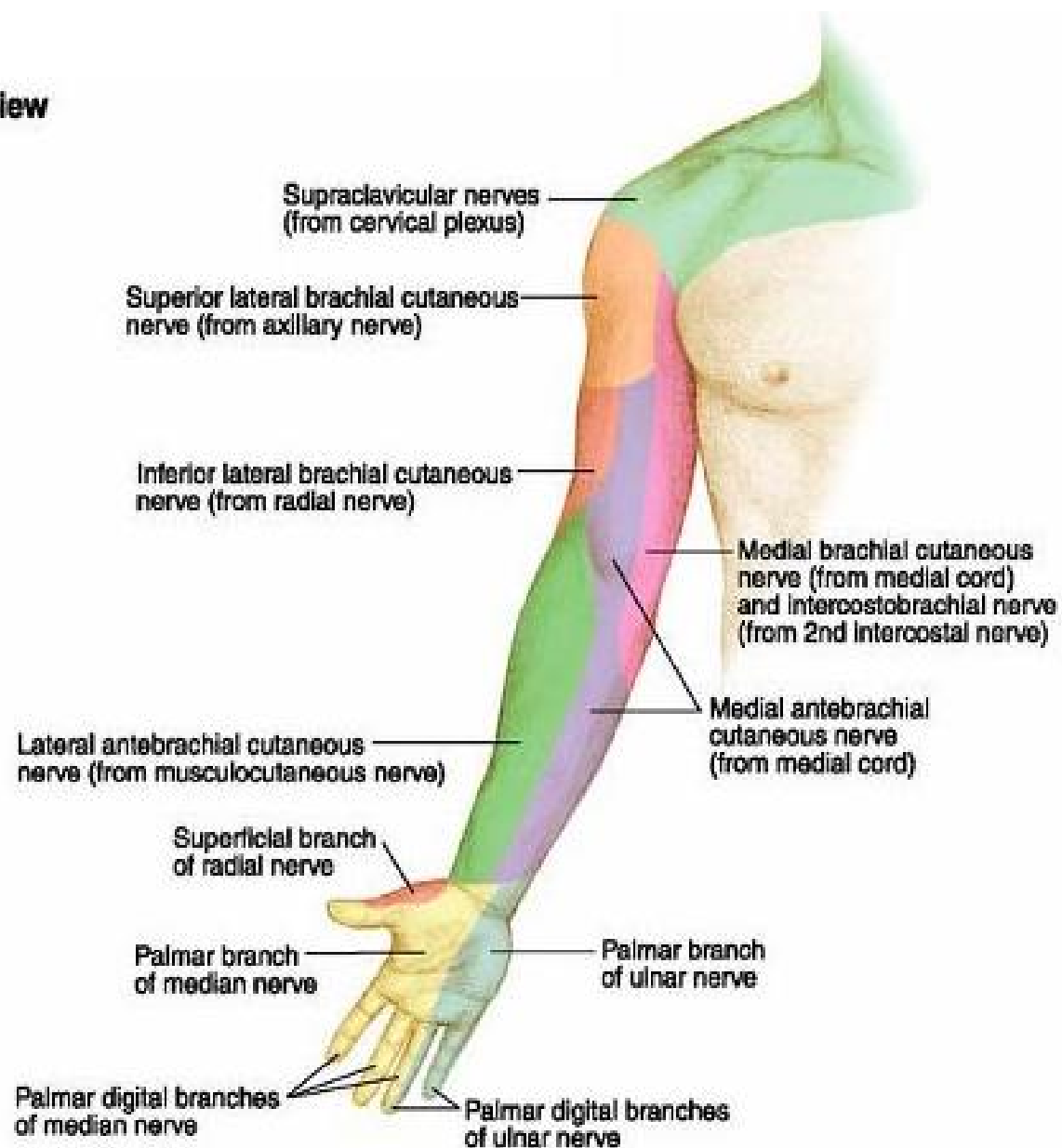
C8: little finger, the medial side

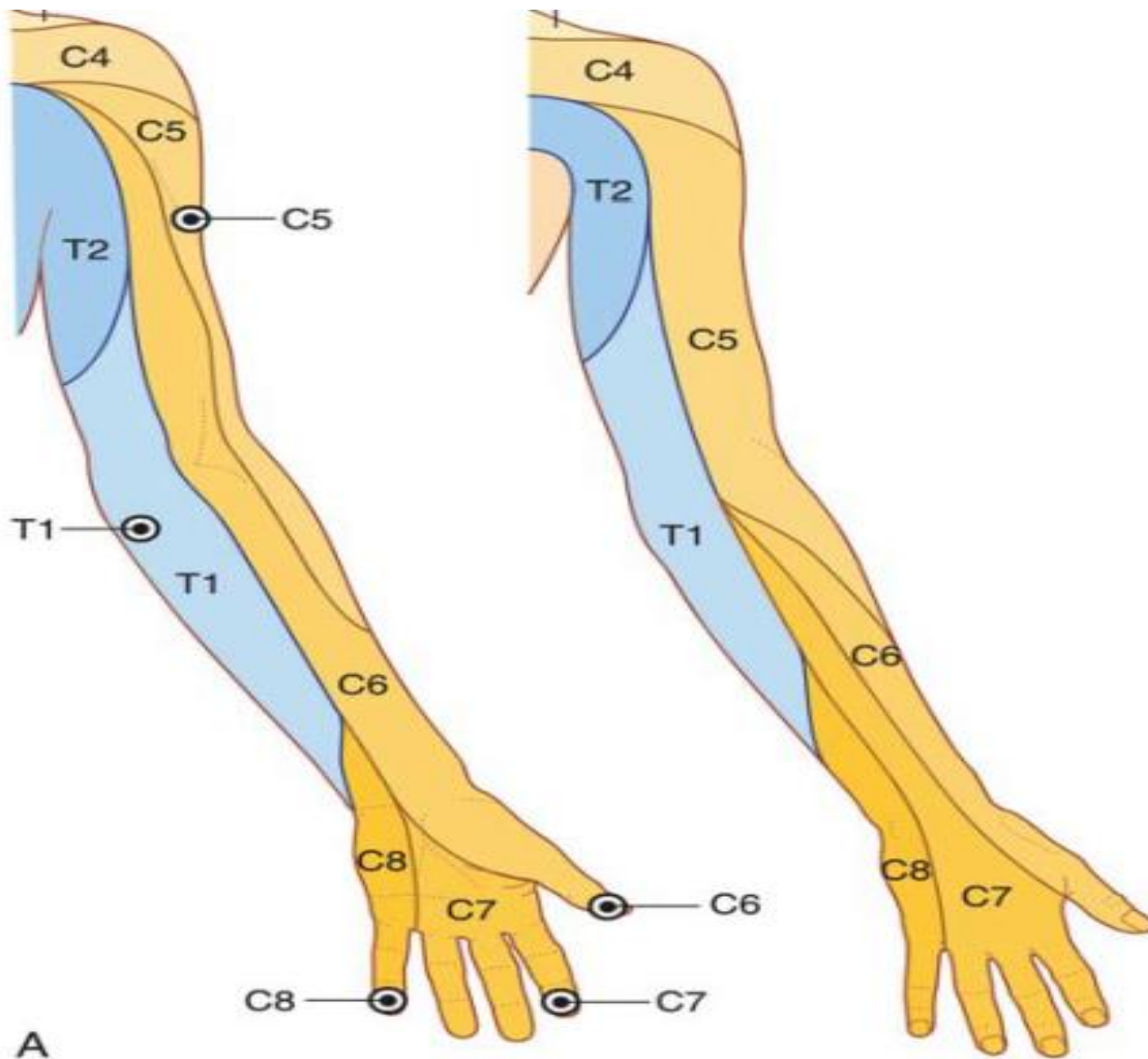
T1: middle of forearm to axilla

T2: small part of arm and axilla

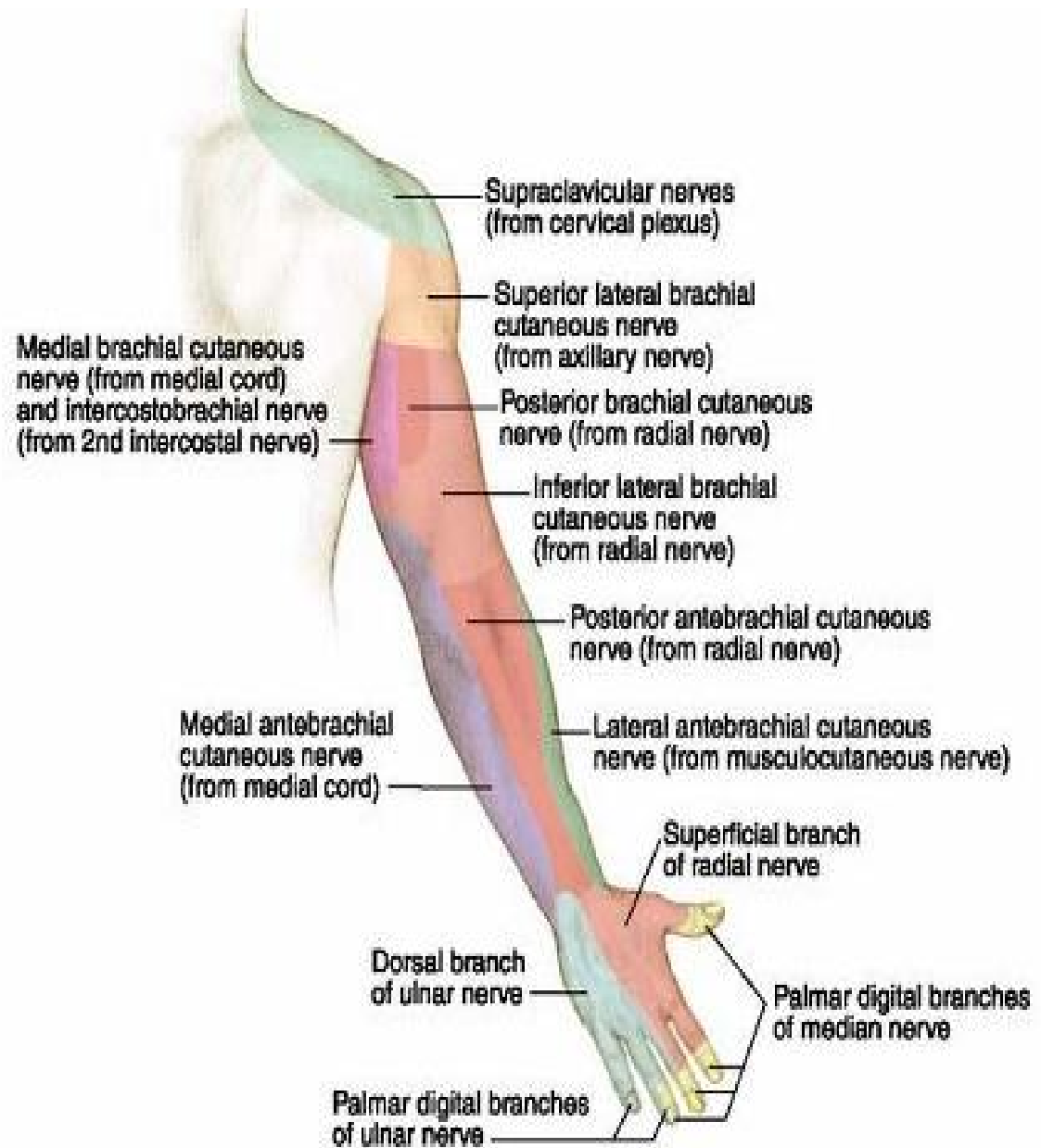


A. Anterior view





B. Posterior view



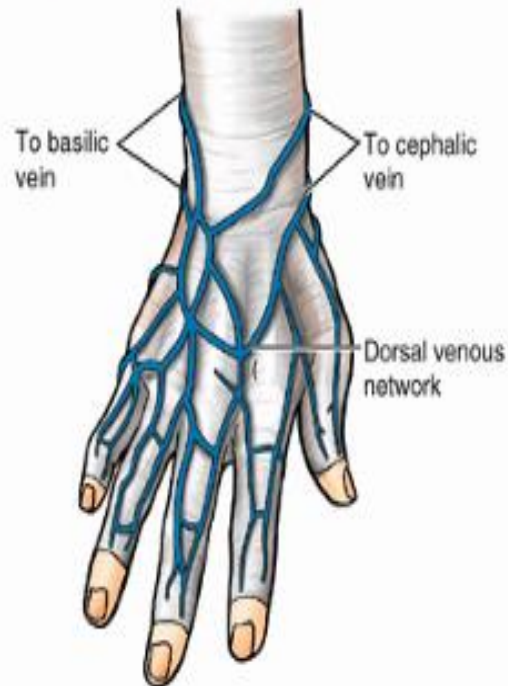
SUPERFICIAL VESSELS OF UPPER LIMB

Dorsal venous network

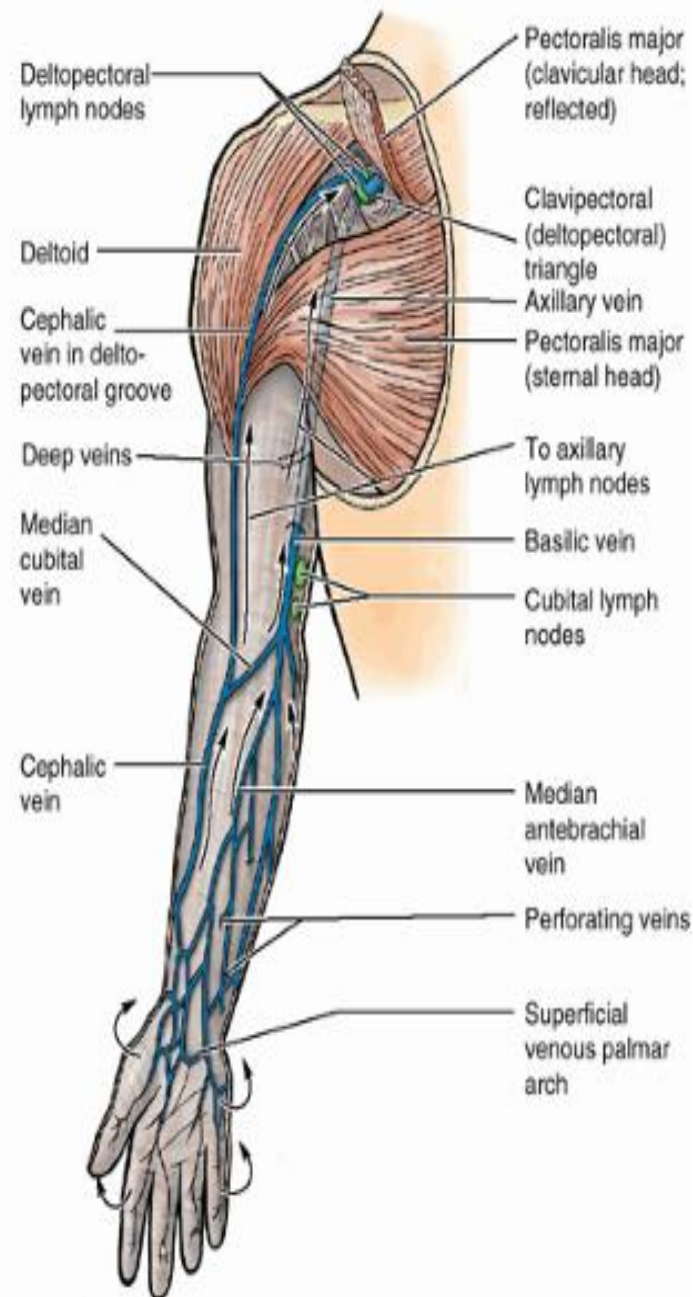
- On dorsum of the hand
- Give rise to cephalic and basilic veins

Cephalic vein

- Ascends from lateral part of dorsal venous network
- Proceed along anterolateral surface of forearm and arm
- Anterior to elbow communicate with median cubital vein
- Superiorly passes between deltoid and pectoralis major muscles and enters clavipectoral triangle
- Pierces clavipectoral fascia and joins axillary vein



(A) Posterior (dorsal) views



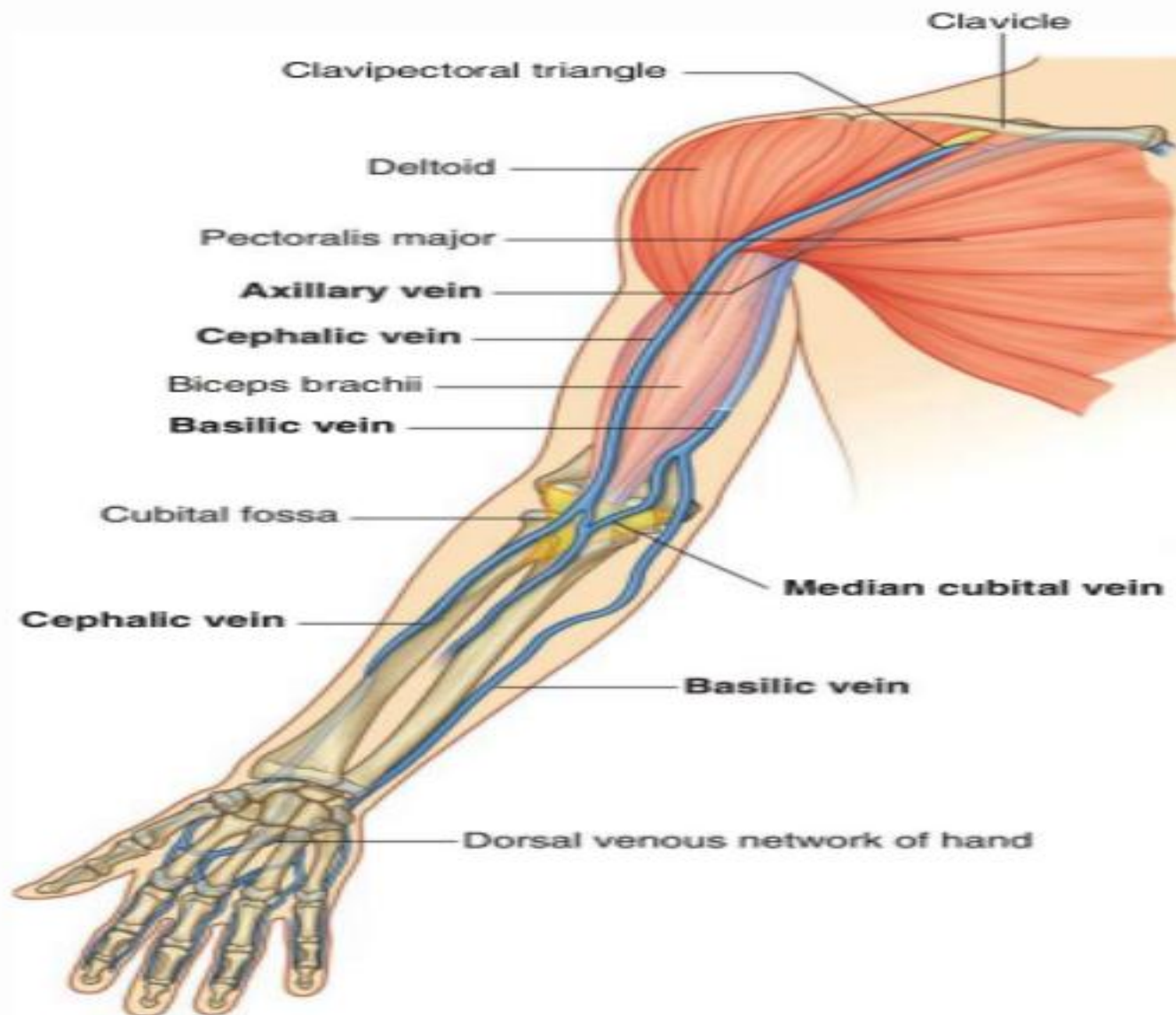
(B) Anterior (palmar) views

Basilic vein

- Ascends from medial part of dorsal venous network
- Proceed along medial side of forearm and inferior part of arm
- Passes deeply near the junction of the middle and inferior thirds of the arm, piercing the brachial fascia and running superiorly parallel to the brachial artery
- Merges with the accompanying veins (venae comitantes) of the axillary artery to form the axillary vein

Median antebrachial vein (median vein of forearm)

- Ascends in the middle of anterior aspect of forearm between cephalic and basilic veins



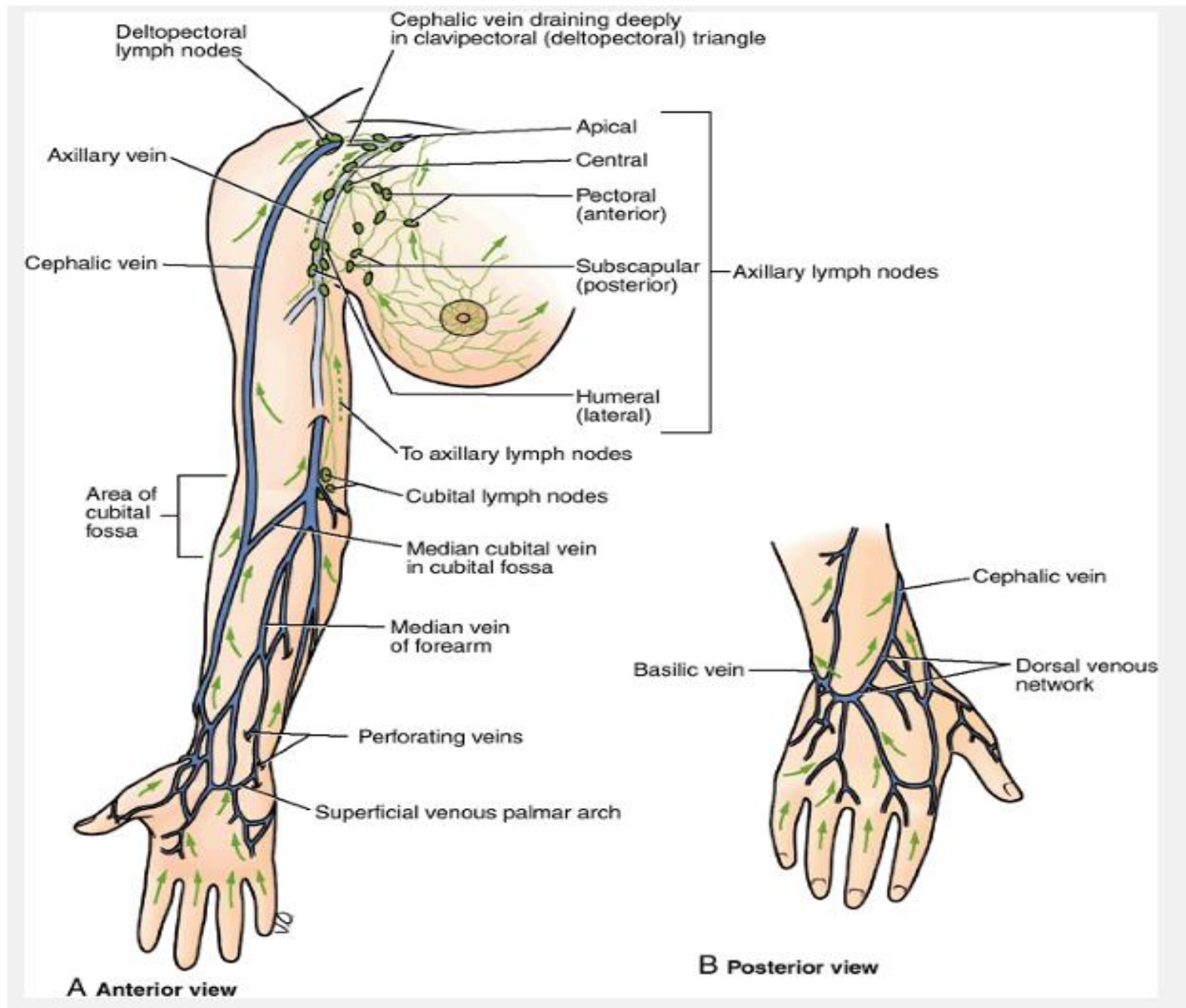
LYMPHATIC VESSELS OF UPPER LIMB

Superficial lymphatic vessels

- Arise from lymphatic plexuses in the skin of fingers, palm, and dorsum of the hand.
- Ascend mostly with superficial veins.
- Some accompanying the basilic vein enter cubital lymph nodes
- Most lymphatic vessels accompanying the cephalic vein and enter **apical axillary lymph nodes**

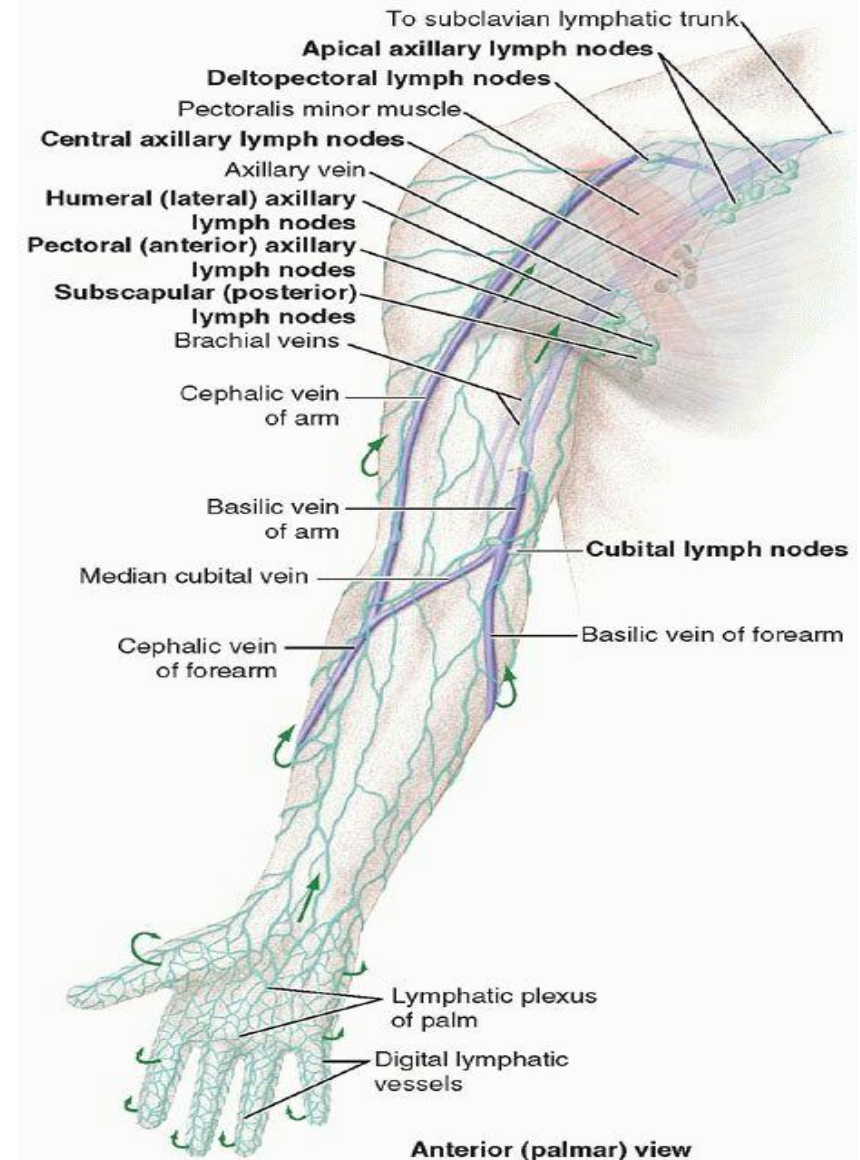
Deep lymphatic vessels

- Accompany major veins and terminate in humeral **axillary lymph nodes**



VENIPUNCTURE

- Puncture of a vein to draw blood or inject a solution
- B/C **Prominence** and **Accessibility** **SV** are commonly used for venipuncture.
- The **median cubital vein** is commonly used for venipuncture.
- The veins forming dorsal venous network and the cephalic and basilic veins arising from it are commonly used for long-term introduction of fluids



AXIO-APPENDICULAR MUSCLES

Anterior axioappendicular muscles

Four muscles that move the pectoral girdle

- Pectoralis major
- Pectoralis minor
- Subclavius
- Serratus anterior

PECTORALIS MAJOR

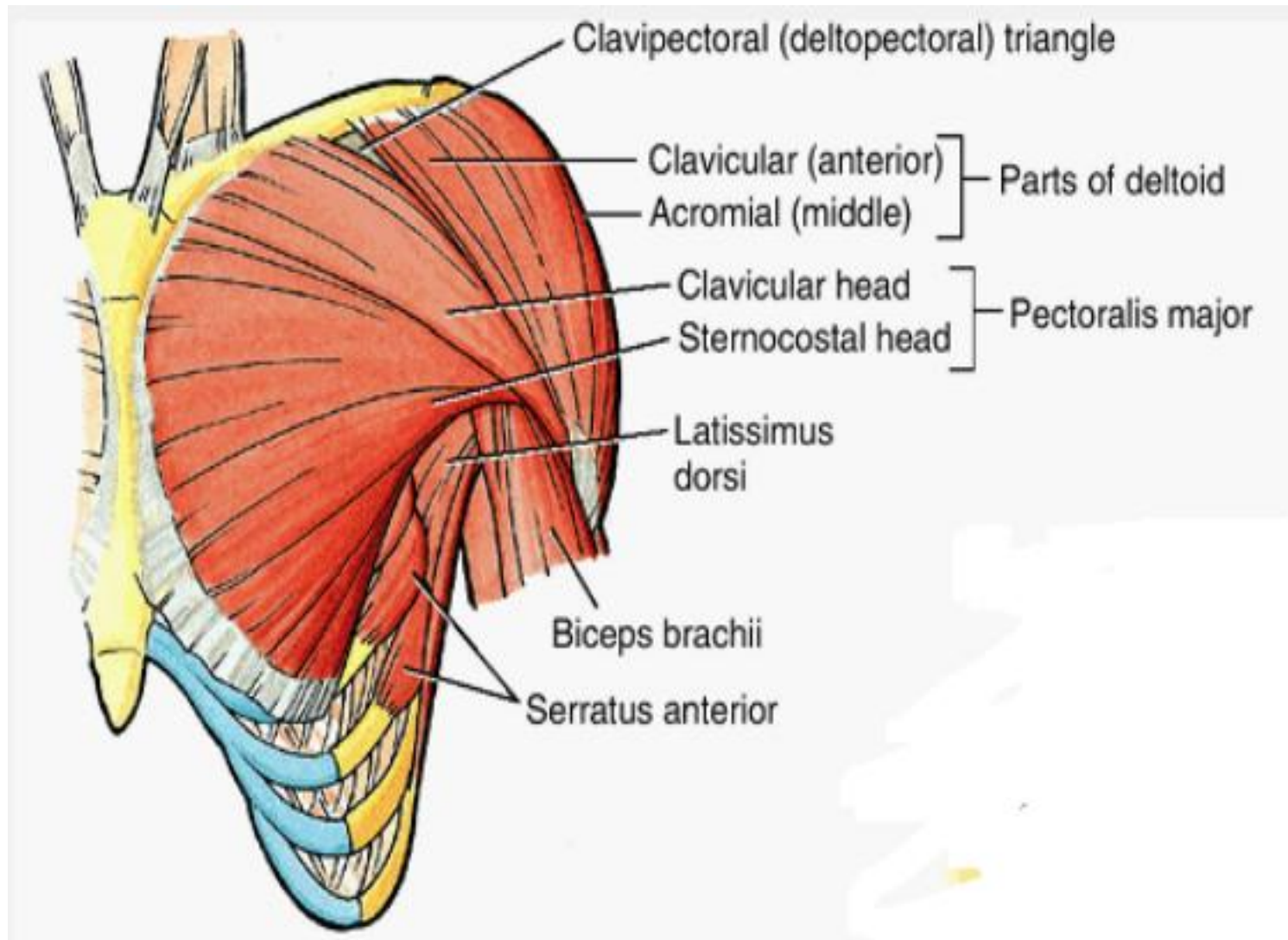
- ◆ Fan-shapes, covers superior part of thorax.
- ◆ Lateral border forms anterior wall of axilla (anterior axillary fold)
- ◆ With deltoid form **deltopectoral** groove
 - In which the cephalic vein runs

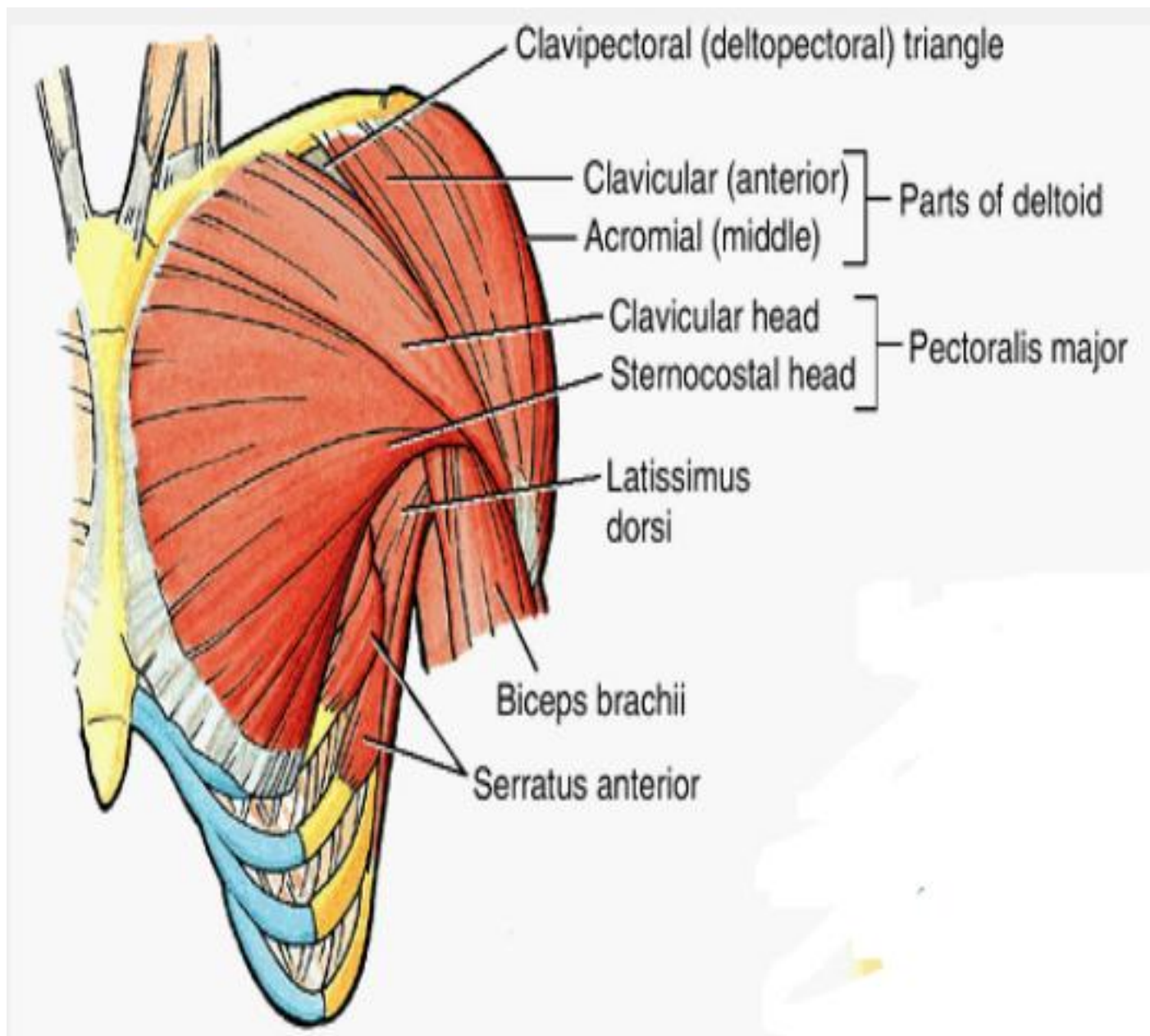
diverge slightly from deltoid superiorly and, along with the clavicle, form the **clavipectoral (delto-pectoral) triangle**.

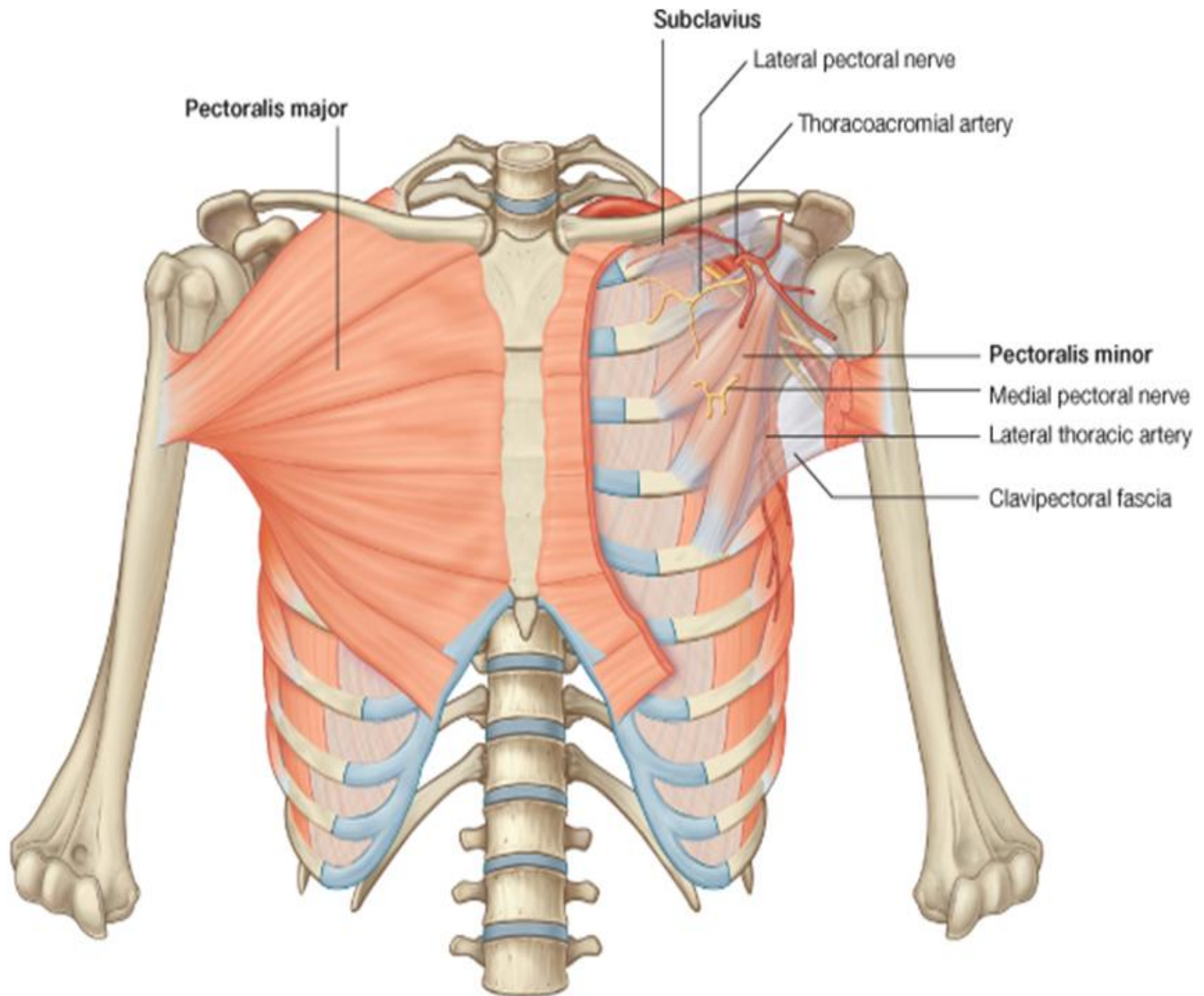
Origin

- Clavicular head: medial half of clavicle
- Sternocostal head: sternum, superior six costal cartilages, aponeurosis of external oblique muscle

PECTORALIS MAJOR







PECTORALIS MINOR

Triangular, lies in anterior wall of axilla

Covered by pectoralis major

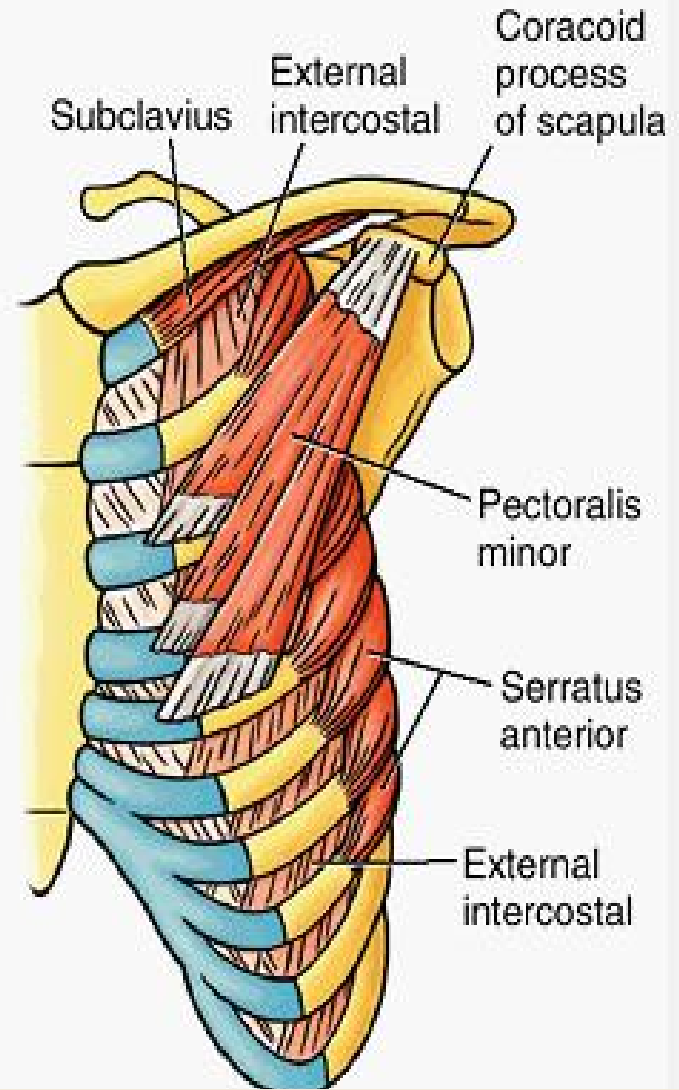
Origin: 3rd to 5th ribs near their costal cartilages

Insertion: coracoid process of scapula

Innervation: Medial pectoral nerve

Action:

- Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
- Elevates ribs in forced inspiration



It is used when stretching the upper limb forward to touch an object that is just out of reach.

SUB-CLAVIUS

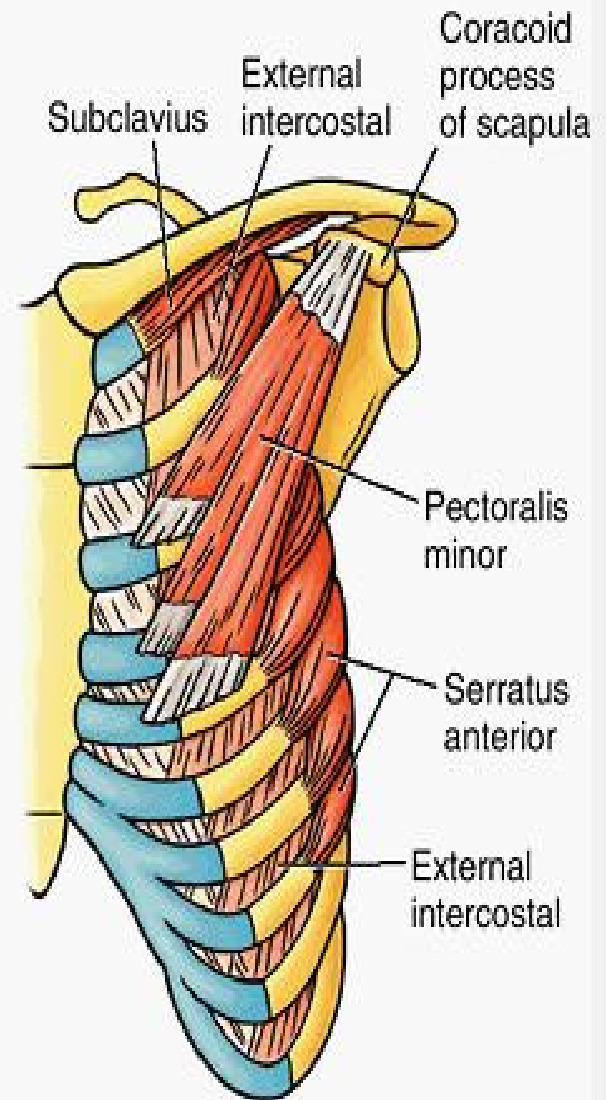
- Small, round and lies horizontally inferior to clavicle.
- Protect subclavian vessels and superior trunk of brachial plexus

Origin: Junction of 1st rib and its costal cartilage

Insertion: Inferior surface of middle third of clavicle

Innervation: Nerve to subclavius

Action: Anchors and depresses clavicle



SERRATUS ANTERIOR

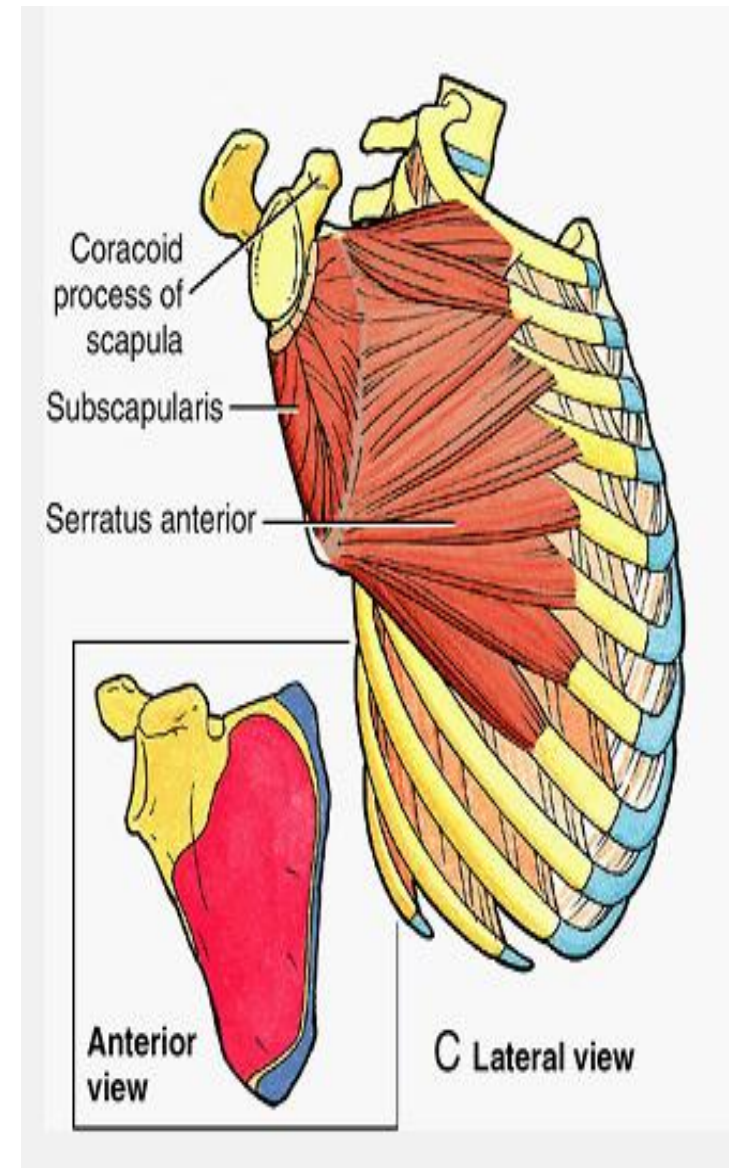
- Overlies lateral part of thorax and forms medial wall of axilla.
- Anchors scapula, enabling other muscles to use it as a fixed bone for movement of humerus

Origin: External surfaces of lateral parts of 1st to 8th ribs

Insertion: Anterior surface of medial border of scapula

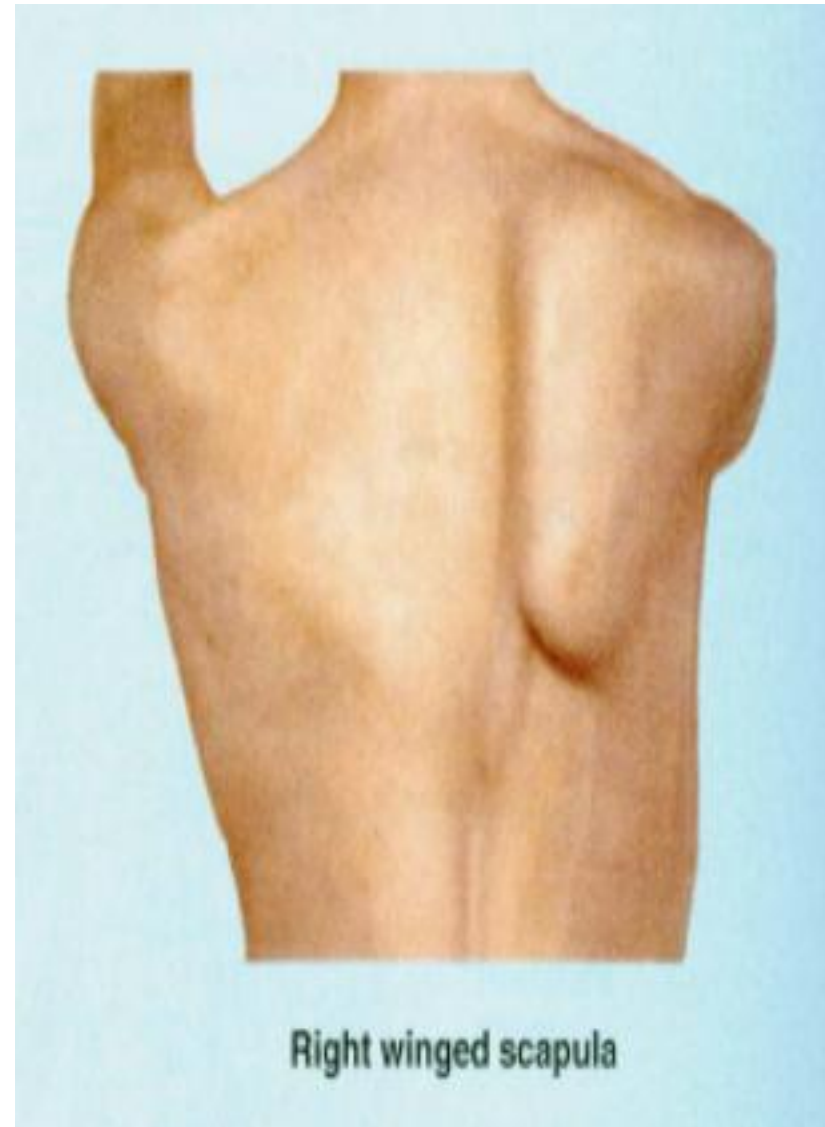
Innervation: Long thoracic nerve

Action: Protracts scapula and holds it against thoracic wall; rotates scapula



PARALYSIS OF SERRATUS ANTERIOR

- ◆ long thoracic nerve **pass on the superficial aspect of serratus anterior.**
- ◆ Due to injury of long thoracic nerve
- ◆ Medial border of the scapula moves laterally and posteriorly away from thoracic wall.
- ◆ **winged scapula**
 - Arm cannot be abducted above horizontal position



THE BREAST

Are modified sweat glands, have no special capsule, produce milk

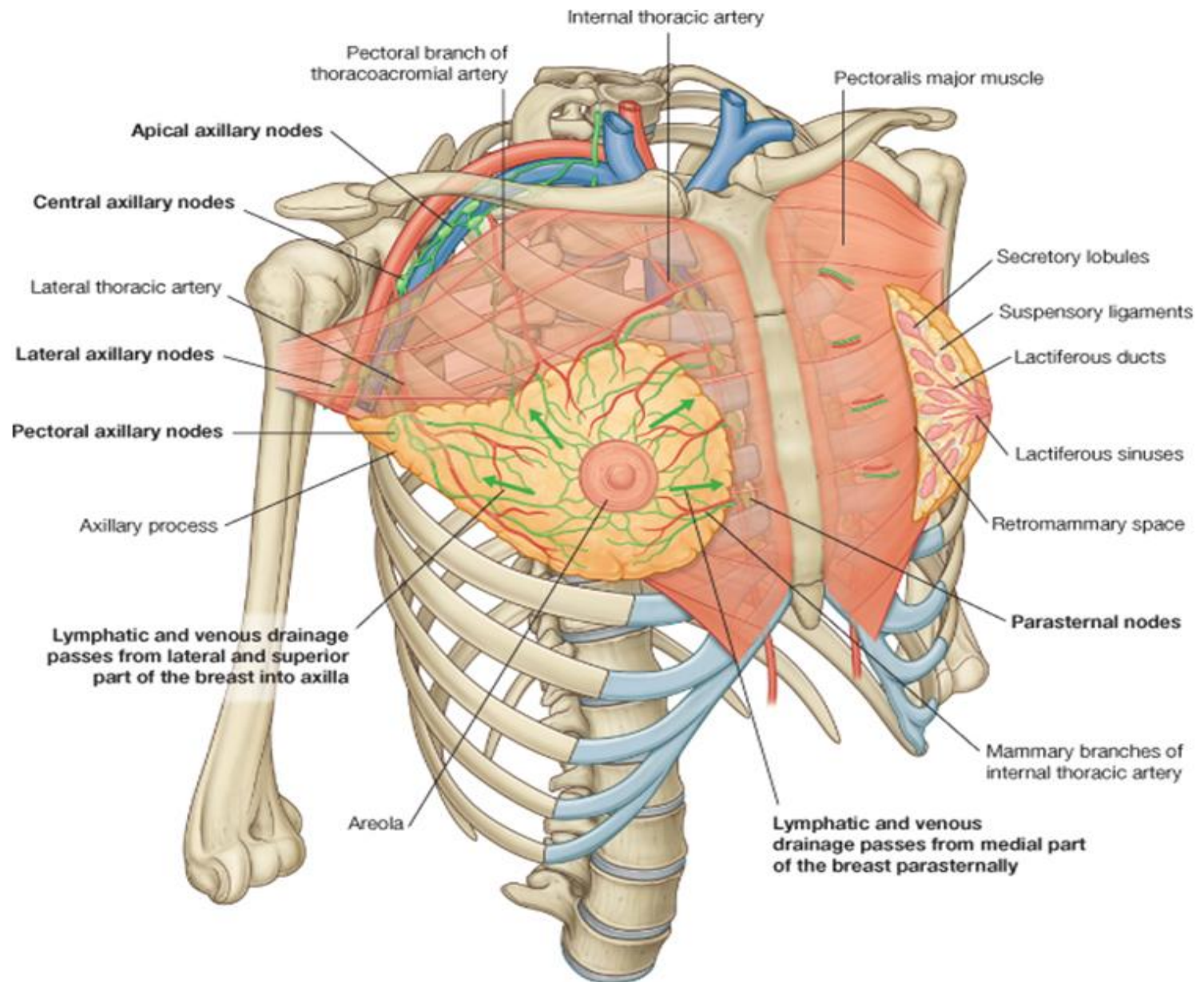
Situated on the anterior surface of the thorax, overlying **pectoral major (2/3)** and **serratus anterior (1/3)** mm

They often extend towards the axilla forming axillary tails

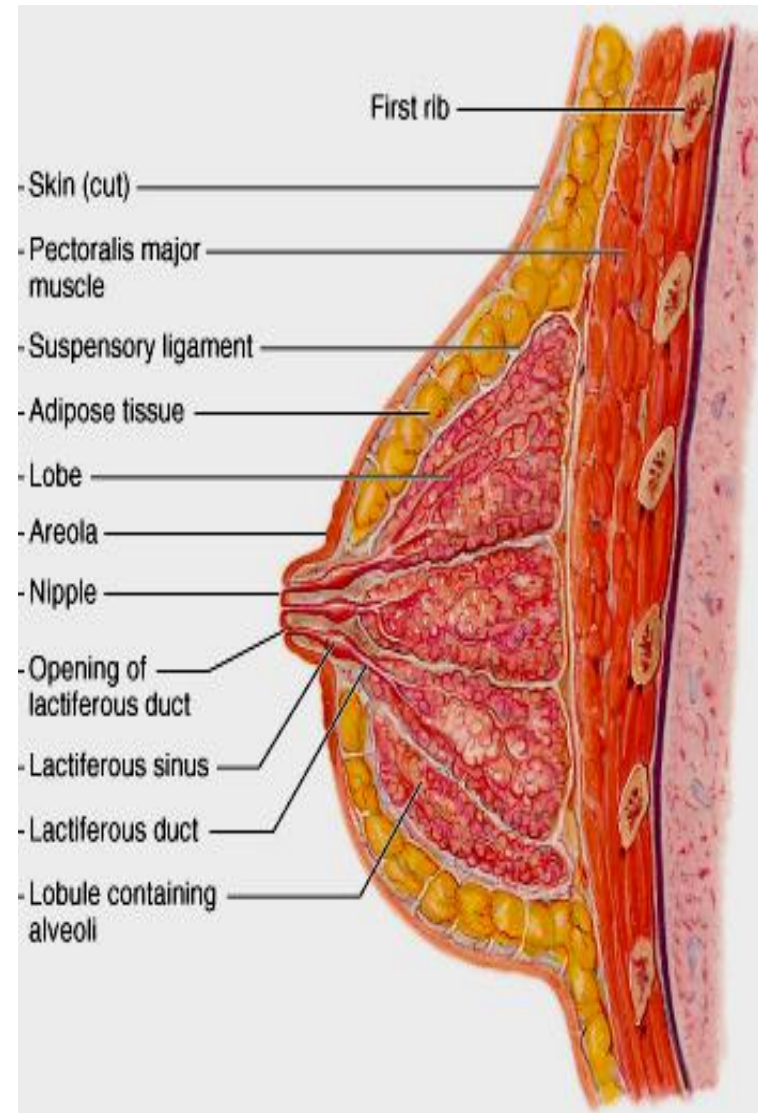
Lies in superficial fascia

Between breast and pectoral fascia is **retromammary space**; allows breast to move freely

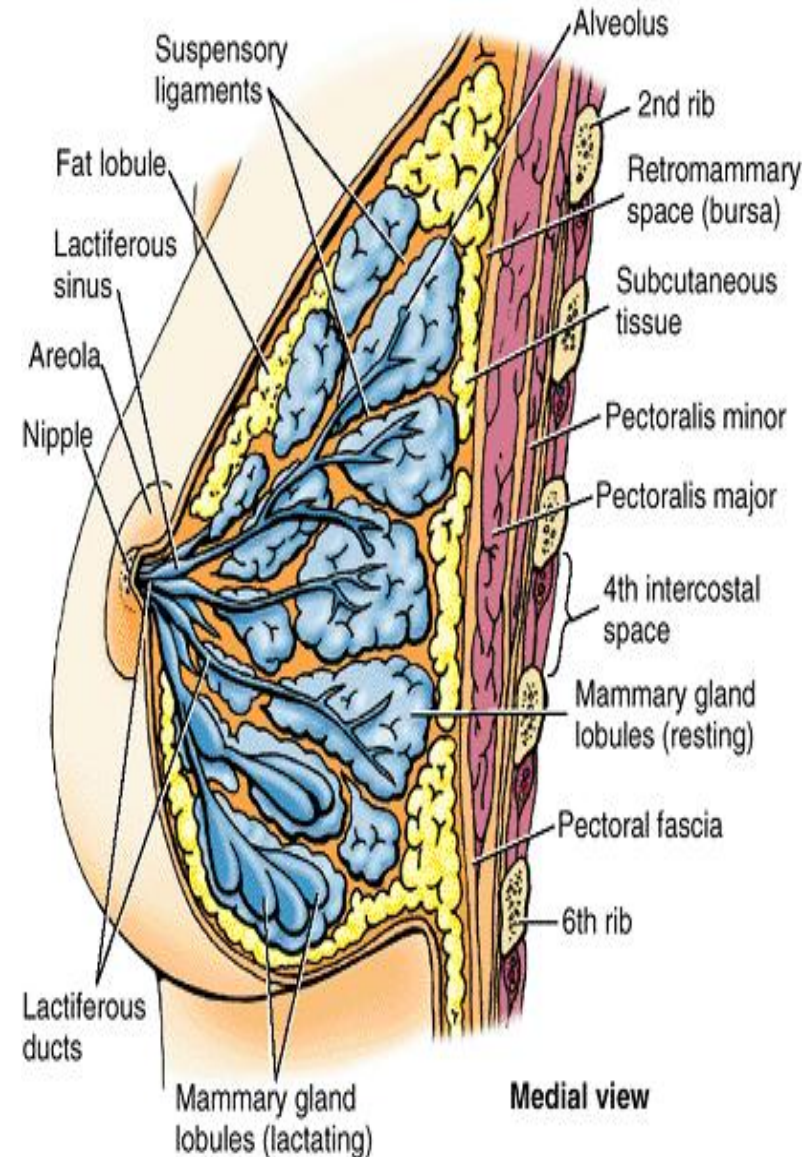
The amount of fat surrounding the glandular tissue determine the size of the breast



- ◆ Each mammary gland consists of **15 to 25 lobes**
- ◆ The lobes are separated by **fat** and **fibrous connective tissue**.
- ◆ The interlobar connective tissue forms **suspensory ligaments (Cooper's ligament)** that attach the breast to the underlying muscle fascia and to the overlying skin.
- ◆ Within the lobes (**lobules**) which contain glandular **alveoli** that produce **milk**.

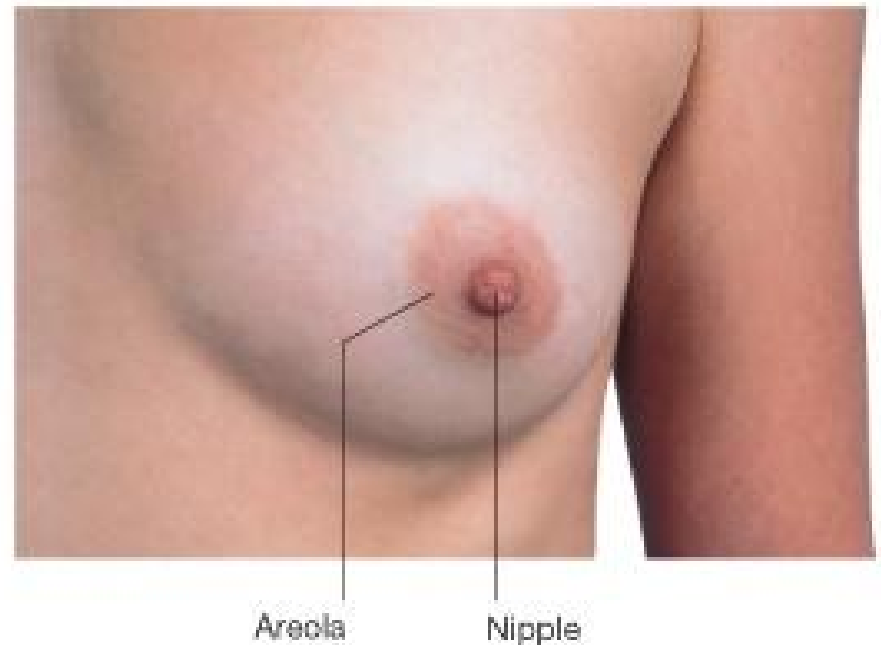


- ◆ These compound alveolar glands pass milk into the **lactiferous ducts**, which open to the outside at the **nipple**.
- ◆ Just deep to **the areola**, each lactiferous duct has a dilated region called a **lactiferous sinus**.
- ◆ Milk accumulates in these **sinuses** during **lactation**.



The areolae

- ✓ contain numerous **sebaceous glands**, which enlarge during pregnancy and secrete an oily substance - **lubricant** for the areola and nipple
- ✓ Are variable in size, and are pink in white nulliparous women but changed to brown during first pregnancy
- ✓ There is **no fat** beneath the areola



The Nipples

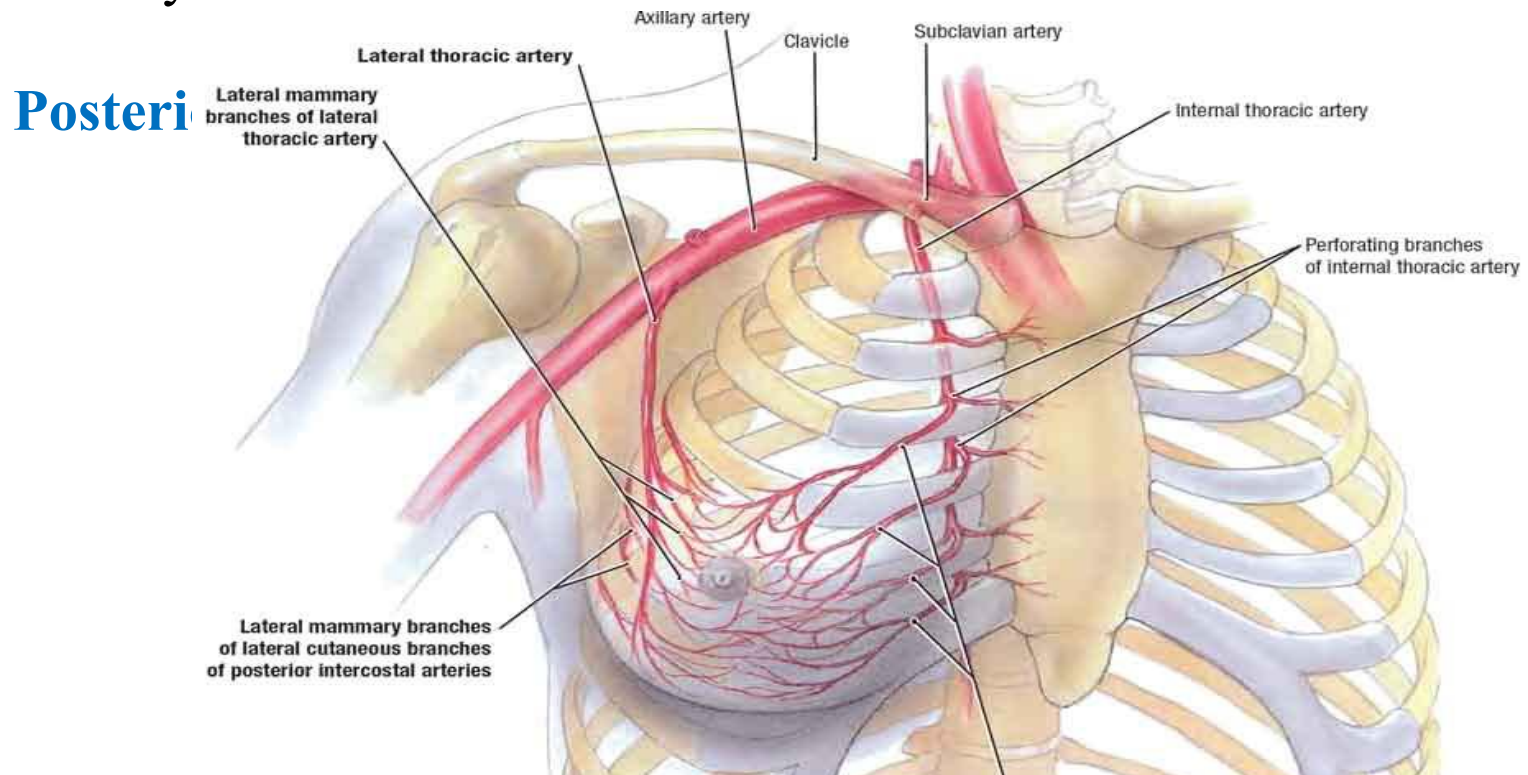
- ✓ are conical or cylindrical prominences in the centers of the areolae.
- ✓ have no fat, hair, or sweat glands.
- ✓ The tips of the nipples are fissured with the lactiferous ducts opening
- ✓ are composed mostly of circularly arranged smooth muscle fibers that compress the lactiferous ducts during lactation and erect the nipples in response to stimulation
- ✓ Usually located at 4th intercostal space, but may position vary.



The arterial supply of the breast

Perforating branches **internal thoracic** aa from the subclavian aa.

Lateral thoracic and **thoracoacromial aa** branches of the axillary aa.



Venous drainage

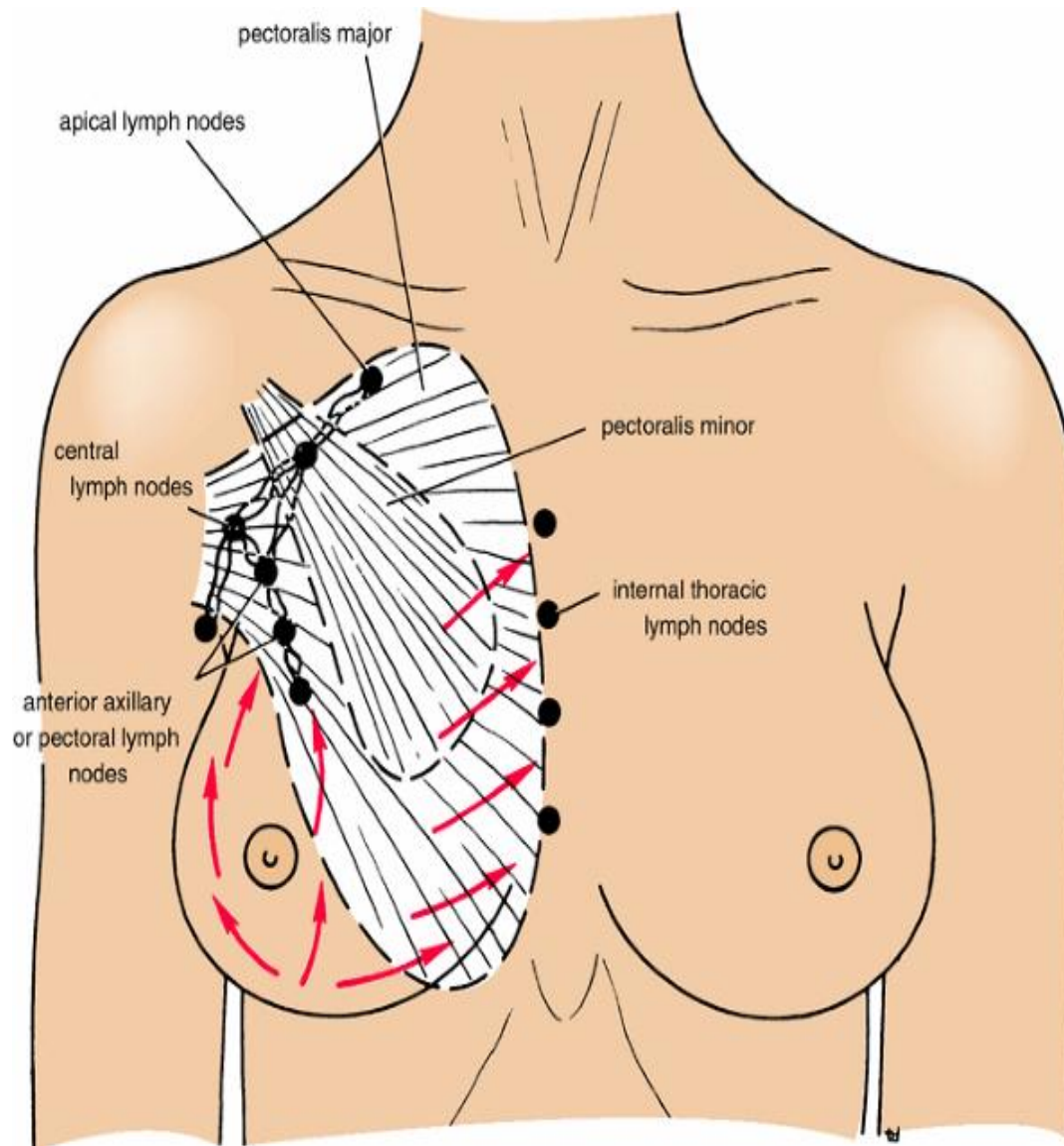
- Veins from breast drain into the axillary, internal thoracic, and intercostal veins.

Lymphatic drainage:

- > 75%), especially from the lateral breast quadrants, drains to the axillary lymph nodes
- from medial and inferior part drains to parasternal and abdominal lymph nodes

Innervation

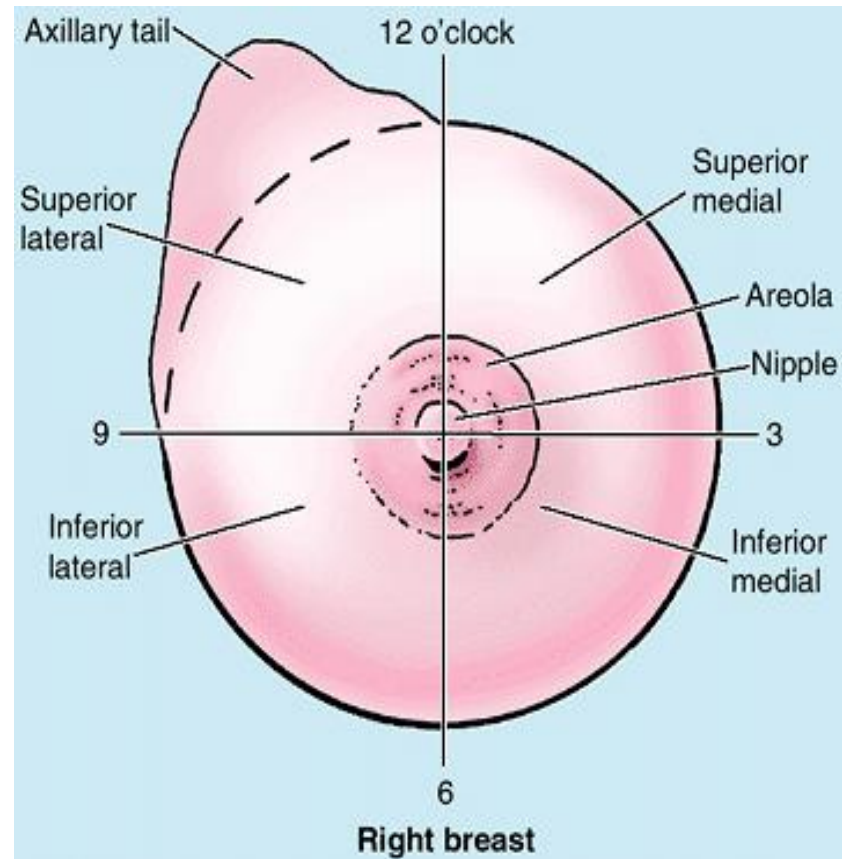
- ✓ anterior and lateral cutaneous branches of the 2nd to 6th intercostal nerves.
- ✓ The nipple is innervated by the fourth intercostal nerve



Lymph drainage of the breast

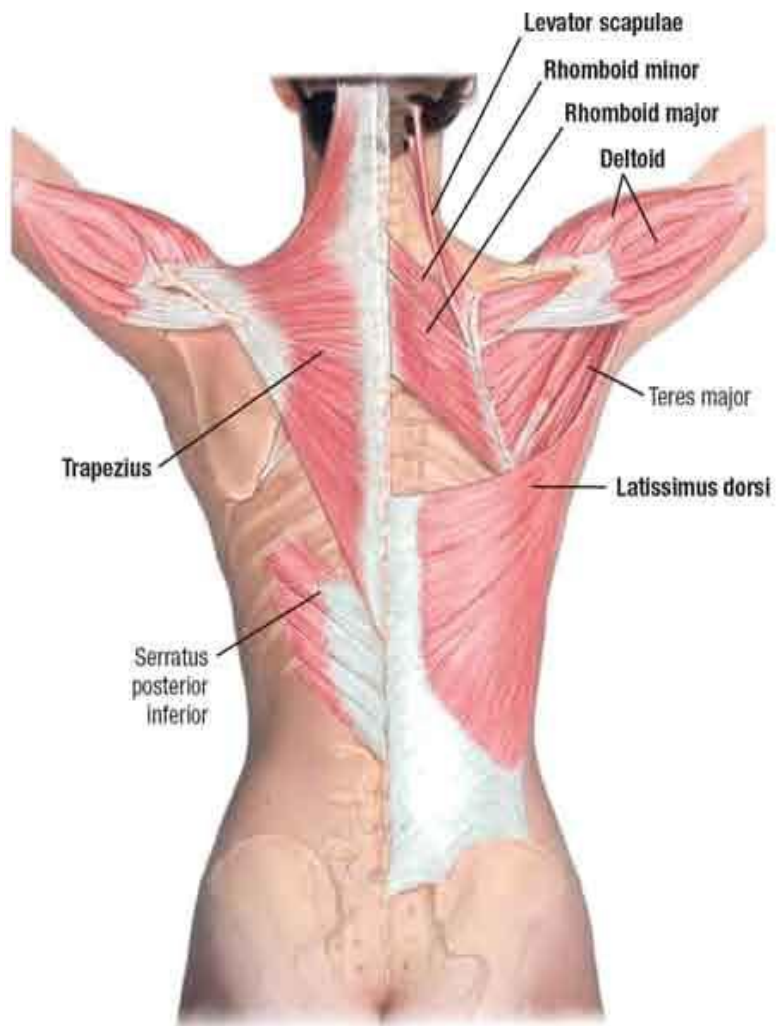
Breast Quadrants

60% of carcinomas of the breast occur in the upper lateral quadrant

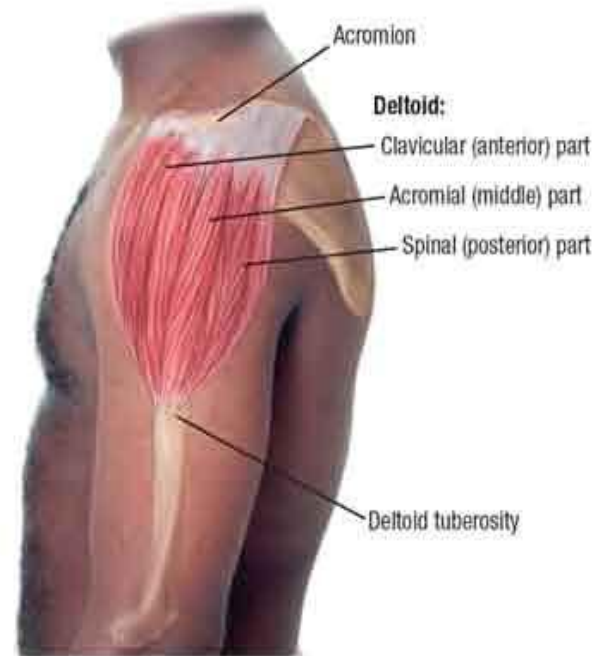


Posterior axioappendicular muscles

- ◆ Attach the superior appendicular skeleton to the axial skeleton.
- ◆ Divided into three groups.
 - **Superficial** (extrinsic shoulder) muscles : trapezius and latissimus dorsi
 - **Deep** (extrinsic shoulder) muscles : levator scapulae and rhomboids
 - **Scapulohumeral** (intrinsic shoulder) muscles: deltoid, teres major, and the four rotator cuff muscles (supraspinatus, infraspinatus, teres minor, and subscapularis)



Posterior View



Lateral View

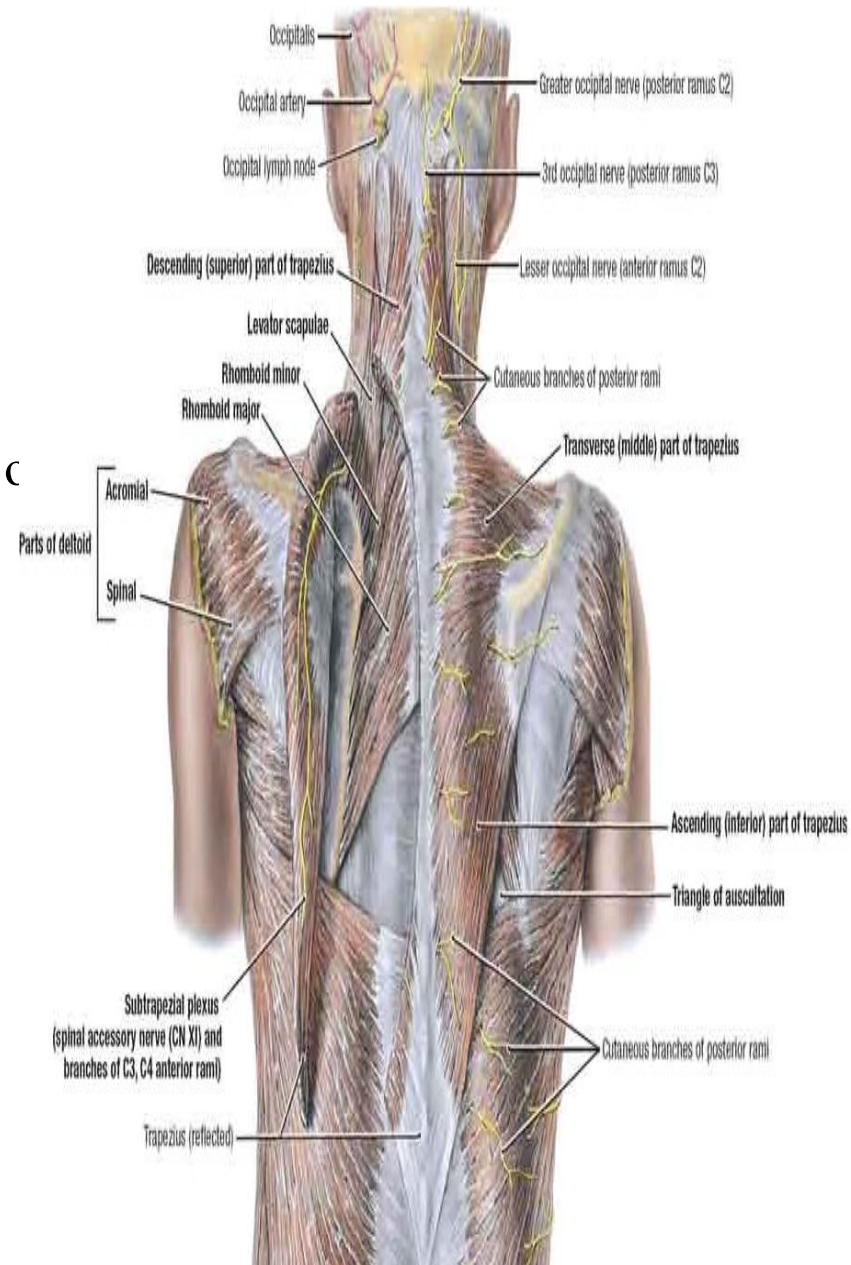


Anterior View

Superficial muscles

Trapezius

- Provides direct attachment of pectoral girdle to trunk.
- Covers posterior part of neck and superior half of trunk
- Attach pectoral girdle to cranium and vertebral column
- Fibers are divided into 3 parts
 - Superior part
 - Middle part
 - Inferior part



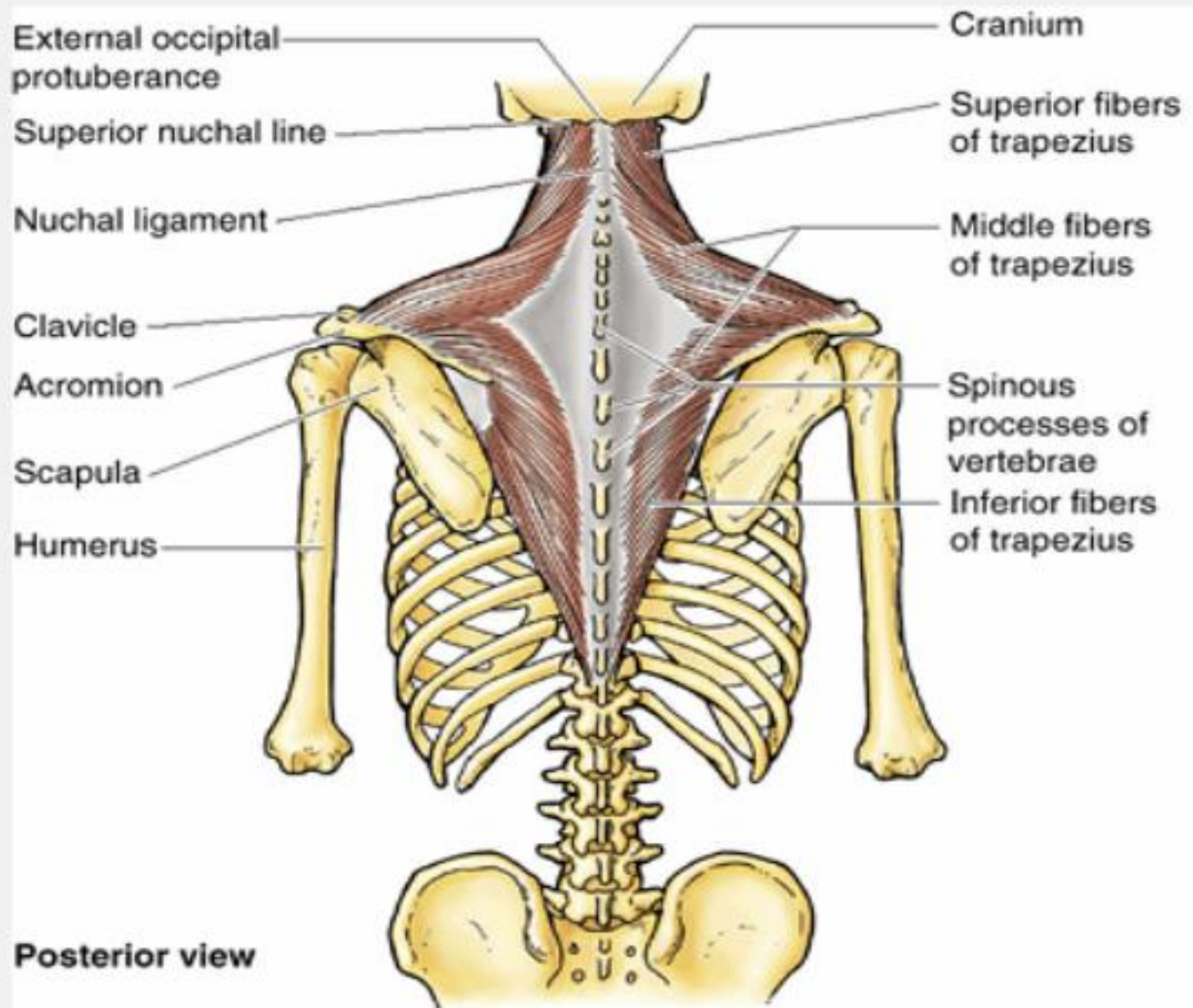
Origin: Medial third of superior nuchal line; external occipital protuberance; nuchal ligament; spinous processes of C7 to T12 vertebrae

Insertion: Lateral third of clavicle; acromion and spine of scapula

Innervation: Accessory nerve (CN XI) (motor fibers) and C3, C4 spinal nerves (pain and proprioceptive fibers)

Action:

- descending part elevates scapula
- ascending part depresses scapula
- middle part (or all parts together) retracts scapula
- descending and ascending parts act together to rotate glenoid cavity superiorly.



Latissimus dorsi

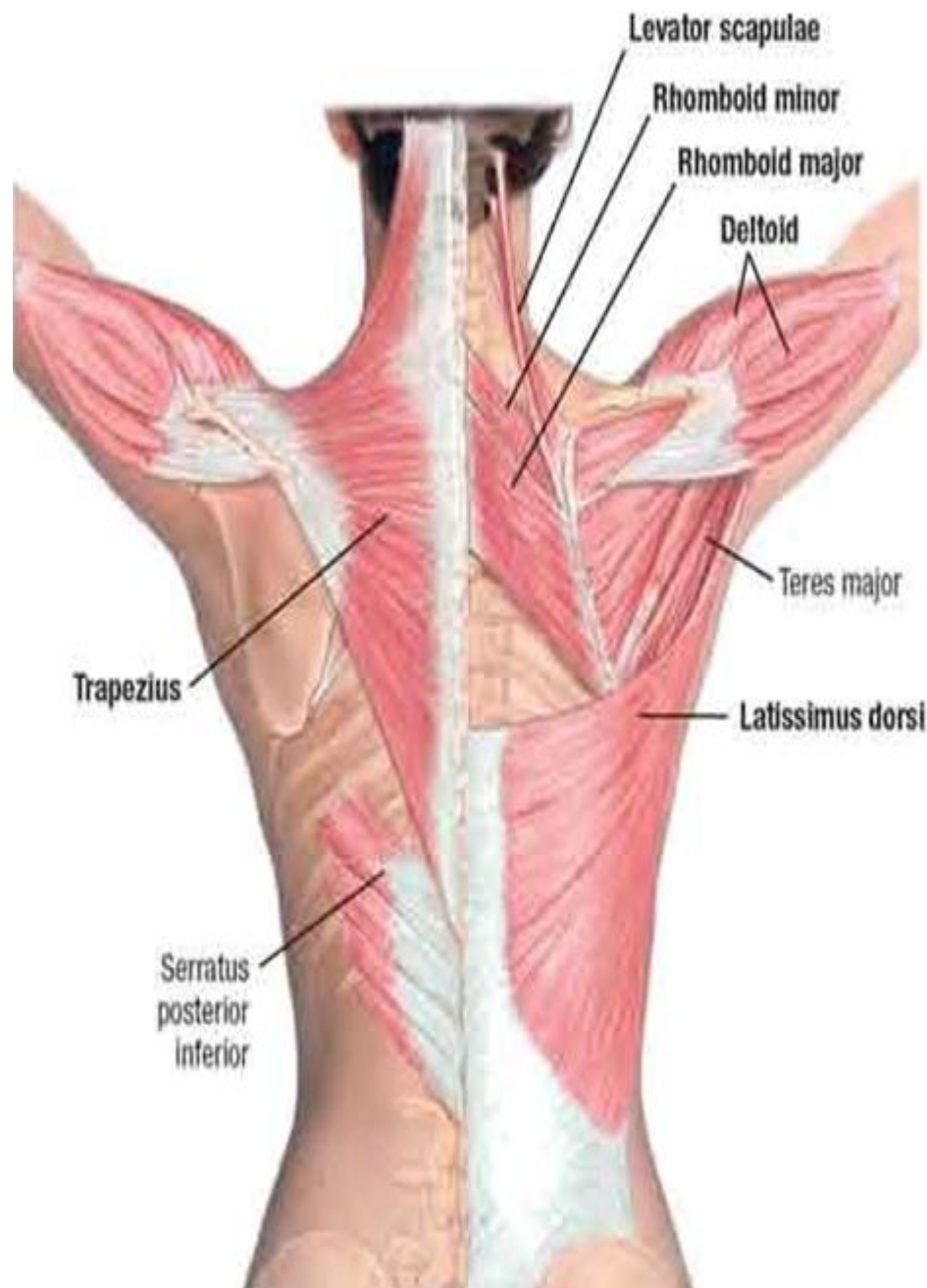
- Fan-shaped, covers wide area of back
- Passes from trunk to humerus
- Raises trunk to the arm

Origin: Spinous processes of inferior 6 thoracic vertebrae, thoracolumbar fascia, iliac crest, and inferior 3 or 4 ribs

Insertion: intertubercular groove of humerus

Innervation: Thoracodorsal nerve

Action: Extends, adducts, and medially rotates humerus; raises body toward arms during climbing



Deep muscles

Levator scapulae

Lies deep to SCM and trapezius

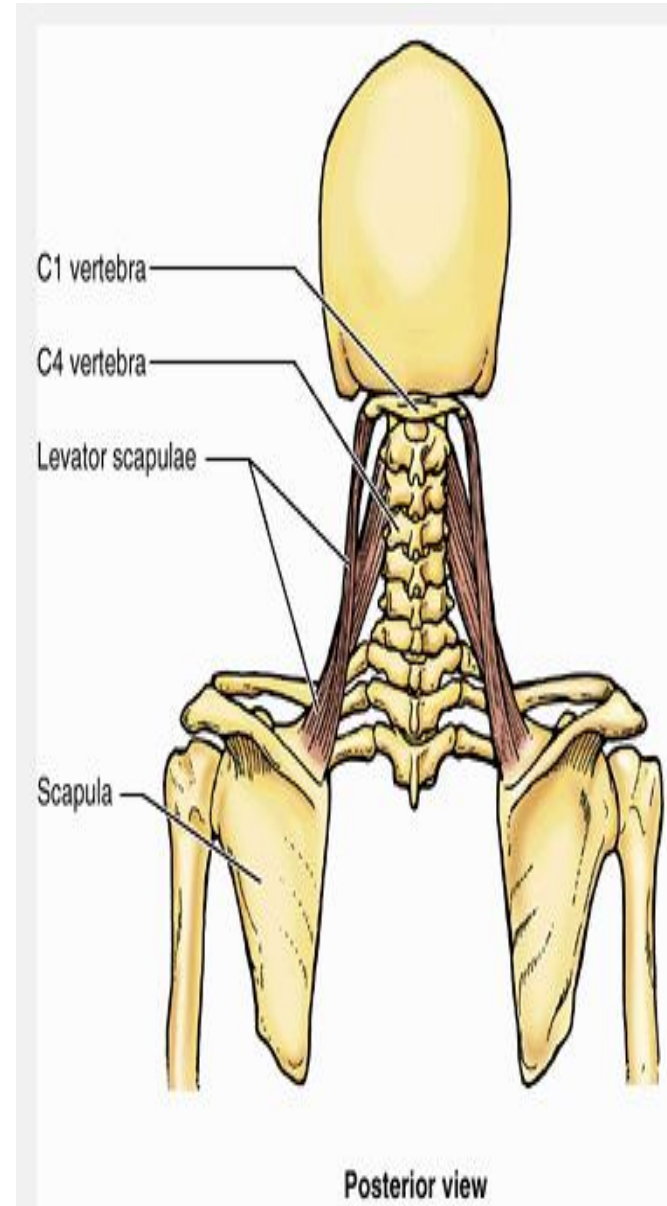
Origin: Transverse processes of C1 to C4 vertebrae

Insertion: Medial border of scapula superior to root of spine

Innervation: Dorsal scapular (C5) and cervical (C3, C4) nerves

Action:

- Elevates scapula and tilts its glenoid cavity inferiorly by rotating scapula with the rhomboid and pectoralis minor
- Acting bilaterally, extend the neck
- Acting unilaterally, contribute to lateral flexion of the neck



Rhomboids (major and minor)

Lie deep to trapezius

Form parallel bands that pass inferolaterally from vertebrae to medial border of the scapula

Origin:

- Minor: nuchal ligament; spinous processes of C7 and T1 vertebrae
- Major: spinous processes of T2 to T5 vertebrae

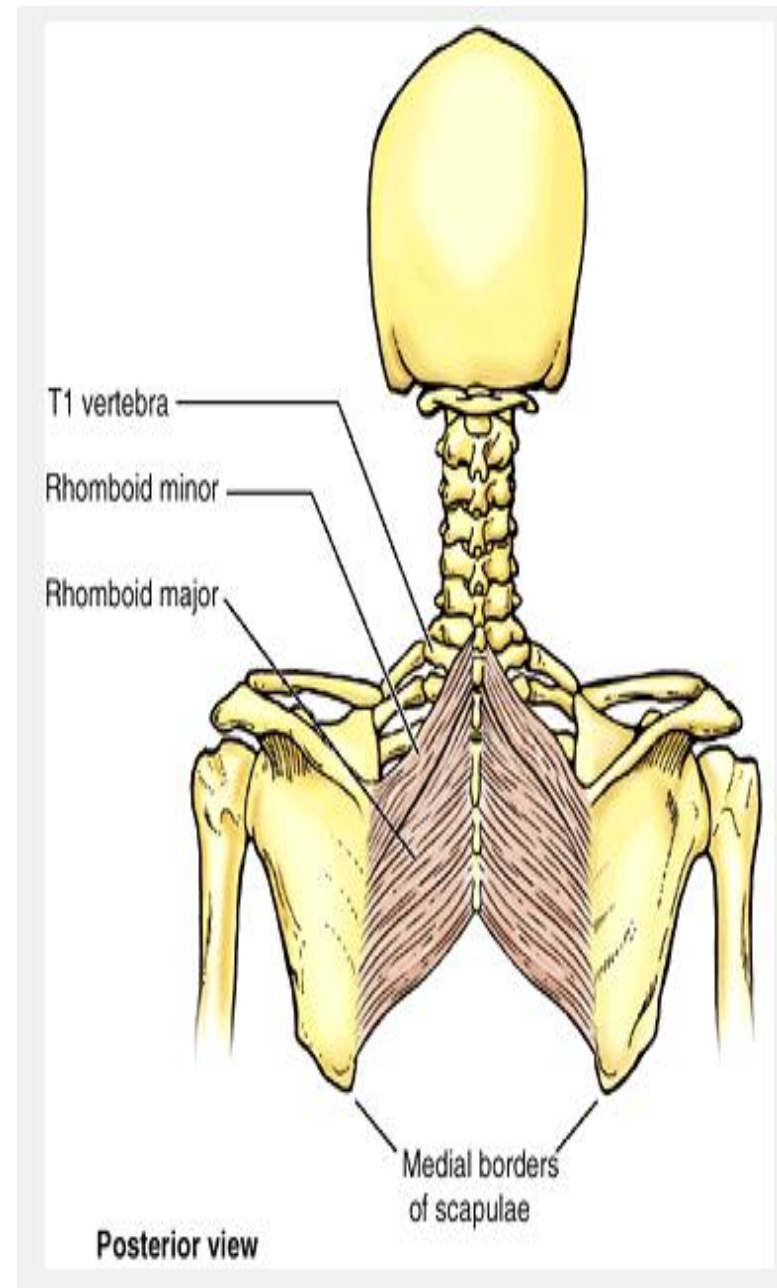
Insertion:

- Minor: medial end of scapular spine
- Major: medial border of scapula from level of spine to inferior angle

Innervation: Dorsal scapular nerve

Action:

- Retract scapula and rotate it to depress glenoid cavity
- fix scapula to thoracic wall during movements of upper limb



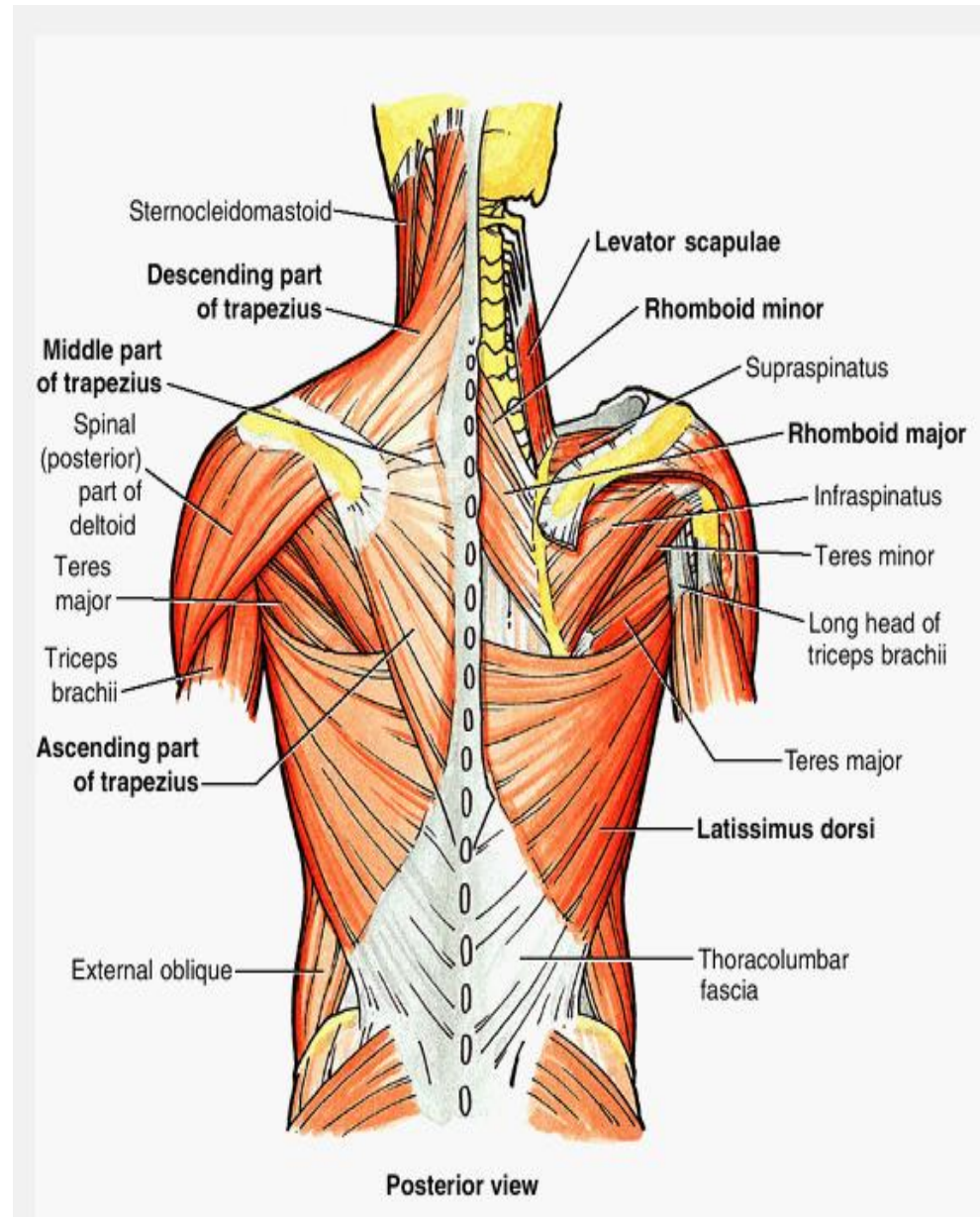
SCAPULOHUMERAL MUSCLES

Six muscles

- Deltoid
- Teres major
- Supraspinatus
- Infraspinatus
- Subscapularis
- Teres minor

Short muscles that pass from scapula to humerus

Act on glenohumeral joint



Deltoid muscle

Thick powerful muscle forming the rounded contour of the shoulder

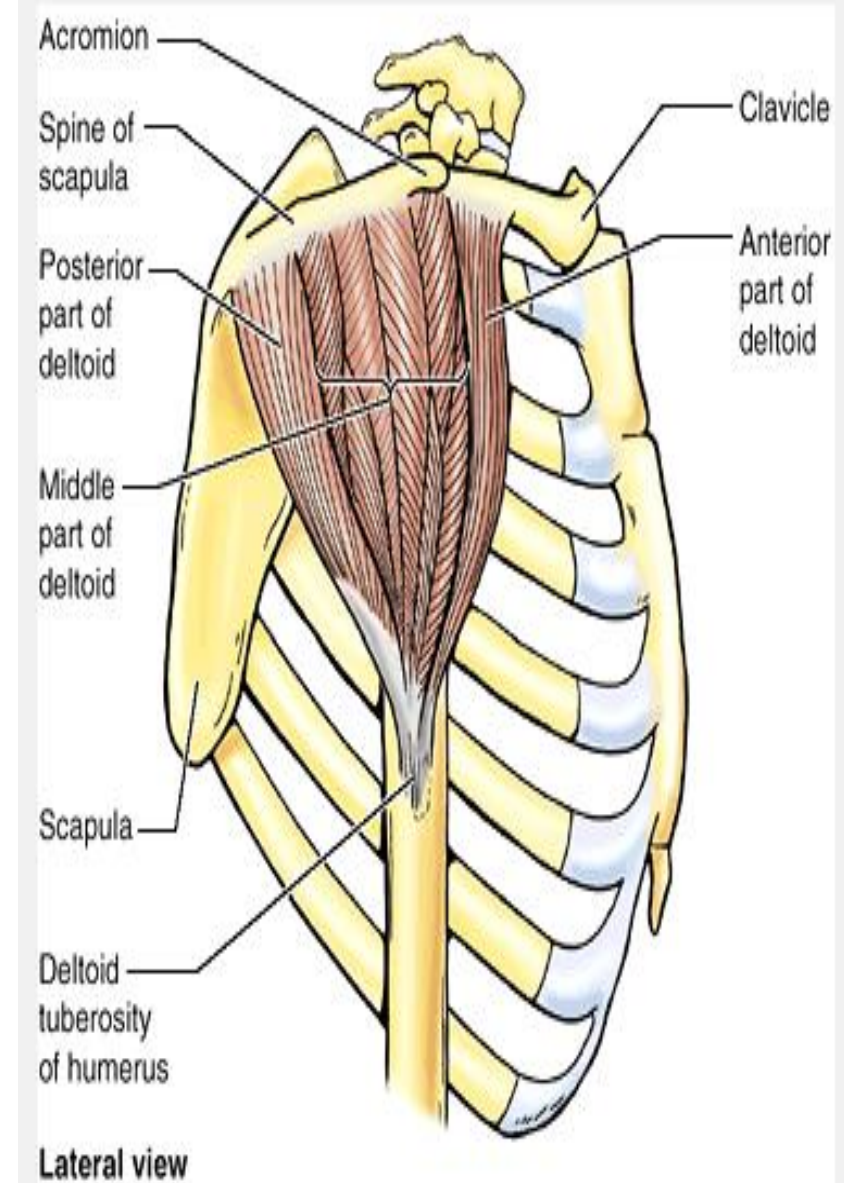
Divided into

- **clavicular (anterior),**
- **acromial (middle), and**
- **spinal (posterior) parts** that can act separately or as a whole

When **all three parts** contract simultaneously, the arm is **abducted**

The clavicular and spinal parts steady the arm as it is abducted

Act as a **shunt muscle**, resisting inferior displacement of the head of the humerus from the glenoid cavity



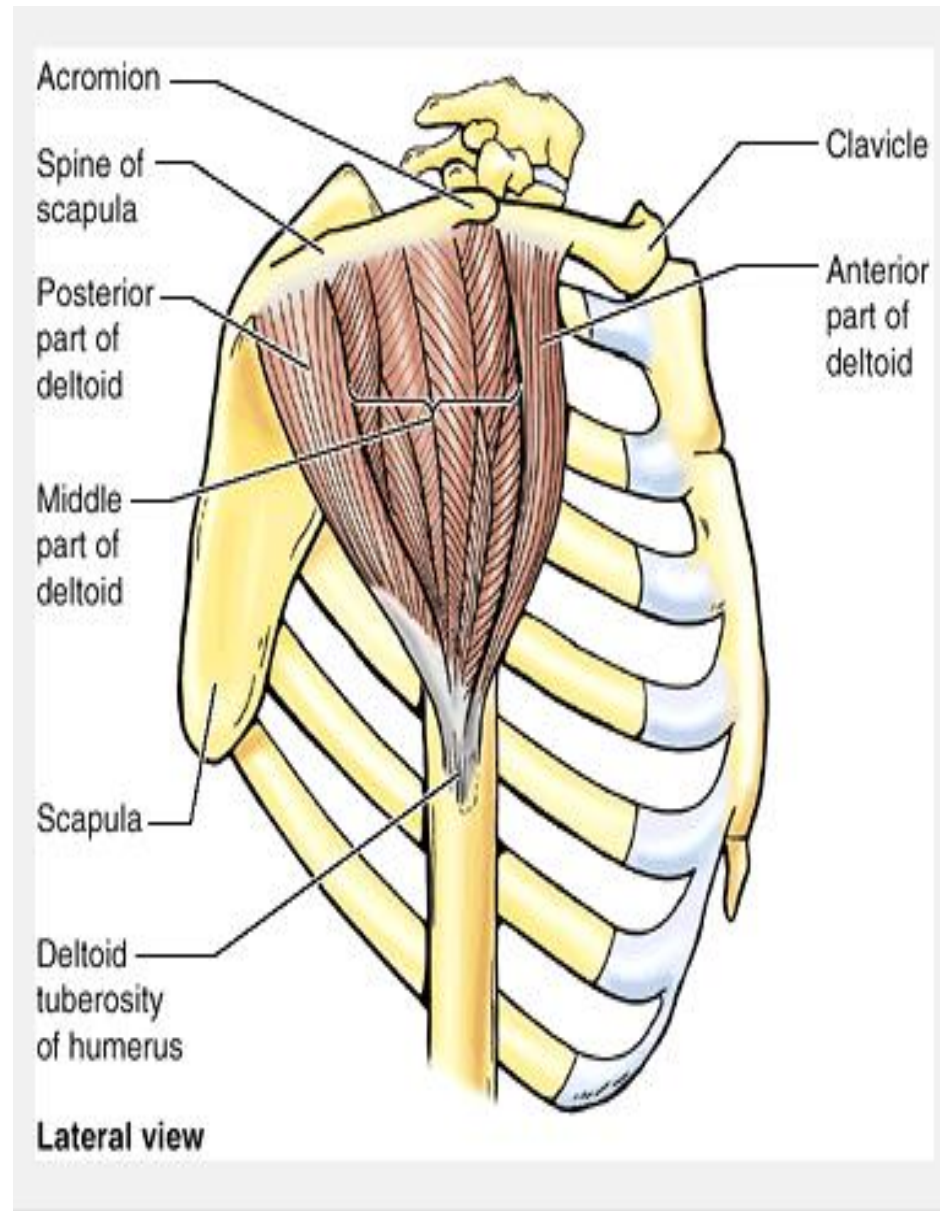
Origin: Lateral third of clavicle;
acromion and spine of scapula

Insertion: Deltoid tuberosity of
humerus

Innervation: Axillary nerve

Action:

- **Anterior part:** flexes and medially rotates arm
- **Middle part:** abducts arm
- **Posterior part:** extends and laterally rotates arm



Teres major

Thick rounded muscle that lies on the inferolateral third of the scapula

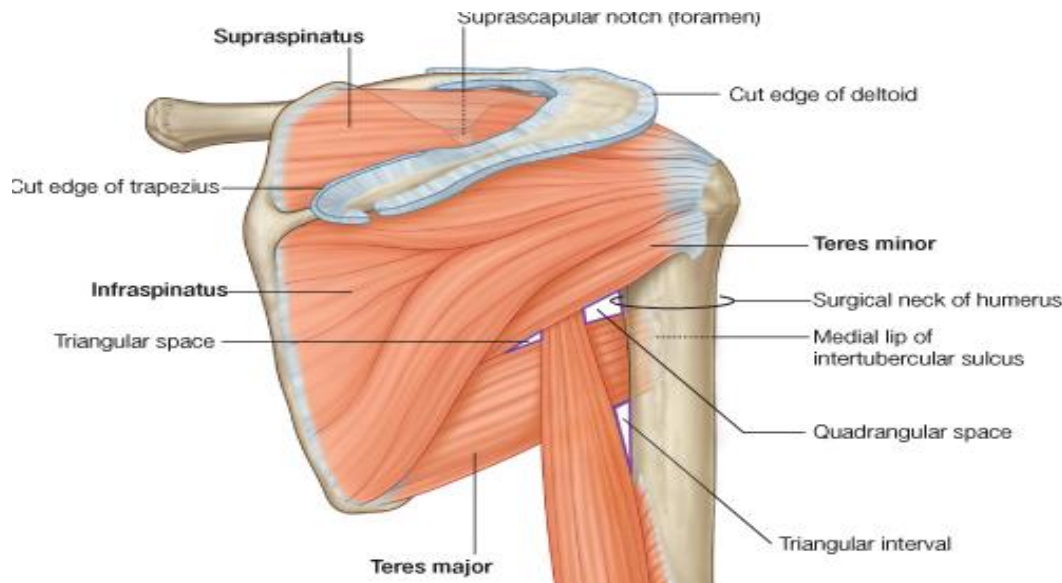
Along with the deltoid and rotator cuff muscles it is an **important stabilizer of the humeral head in the glenoid cavity** during movement

Origin: Posterior surface of inferior angle of scapula

Insertion: Medial lip of **intertubercular** groove of humerus

Innervation: Lower subscapular nerve

Action: Adducts and medially rotates arm



Rotator cuff muscles – SITS mm

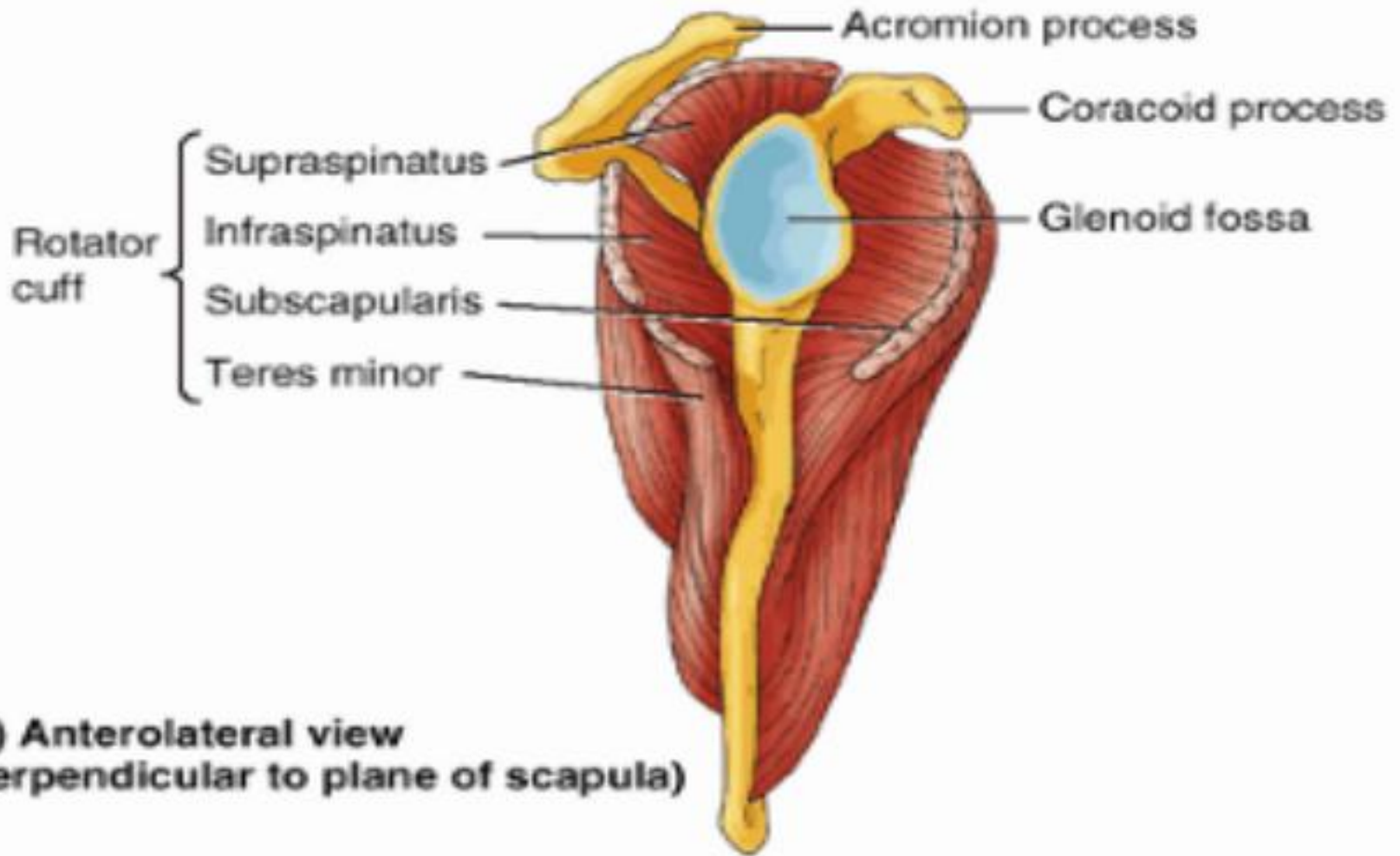
- Four of the scapulohumeral muscles: **supraspinatus, infraspinatus, teres minor, and subscapularis.**
- Rotator cuff because they form a **musculotendinous cuff** around glenohumeral joint.
- All except the **supraspinatus** are rotators of the humerus

The **supraspinatus** initiates and assists the deltoid in the abduction of the arm

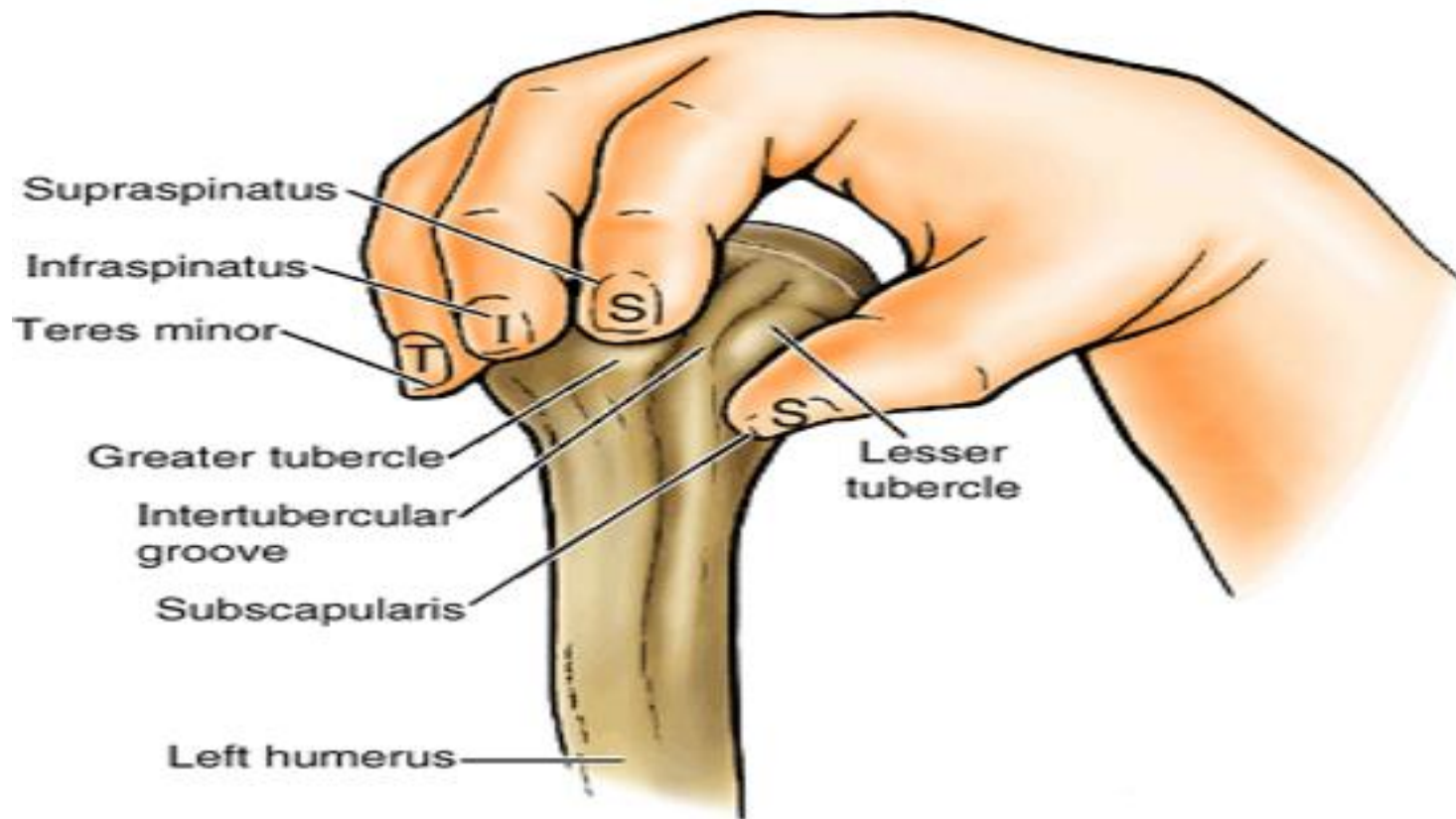
The tendons of the rotator cuff muscles blend with the joint capsule of the glenohumeral joint, which **protects the joint** and gives it **stability**

Strengthen the shoulder joint anywhere, **except inferiorly**

Tonic contraction of these muscles holds the relatively large head of the humerus **firmly against the small and shallow glenoid cavity** during arm movements



Rotator cuff muscles that provides active support for the glenohumeral joint



Supraspinatus

Origin: Supraspinous fossa of scapula

Insertion: **greater tubercle** of humerus

Innervation: Suprascapular nerve

Action: Initiates and assists deltoid in abduction of arm and acts with rotator cuff muscles

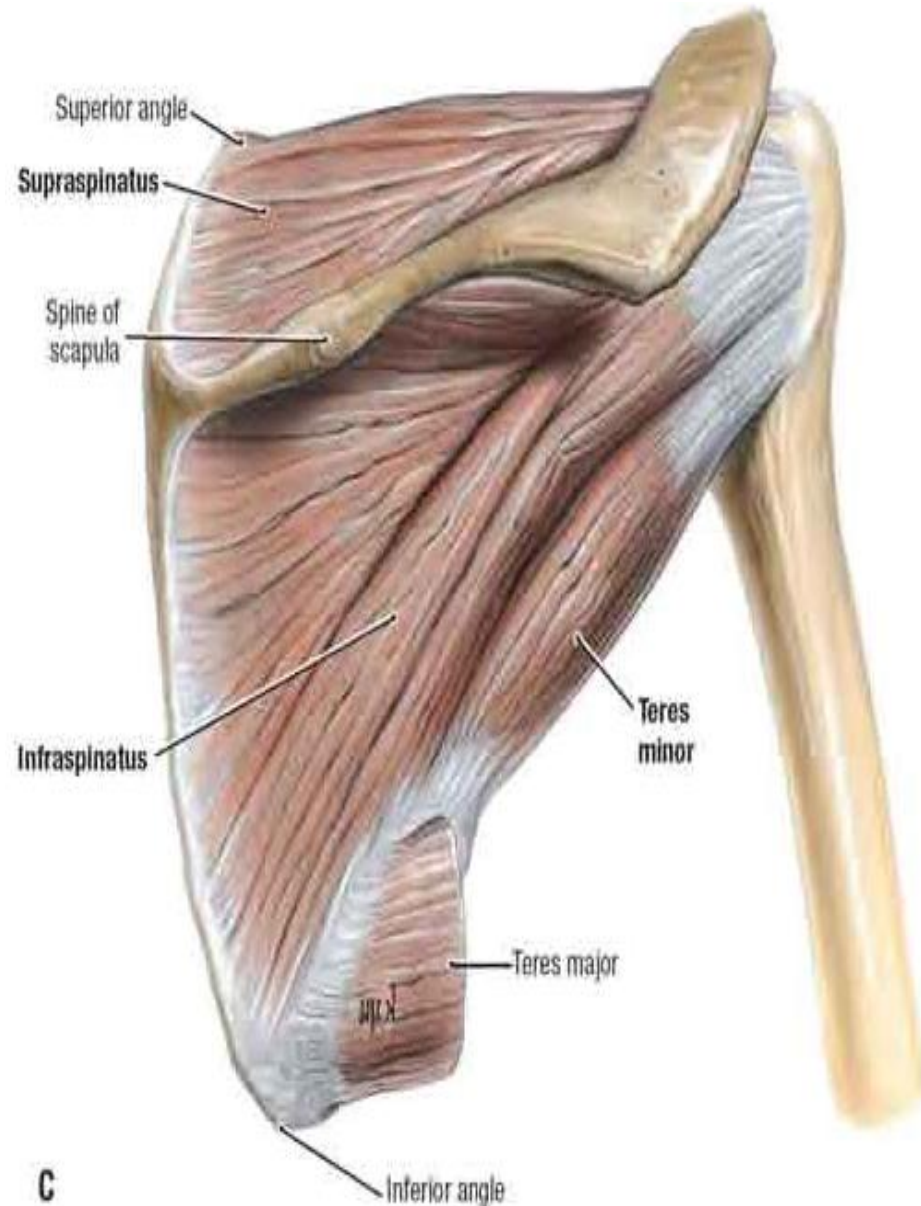
Infraspinatus

Origin: Infraspinous fossa of scapula

Insertion: **greater tubercle** of humerus

Innervation: Suprascapular nerve

Action: Laterally rotate arm; help hold humeral head in glenoid cavity of scapula



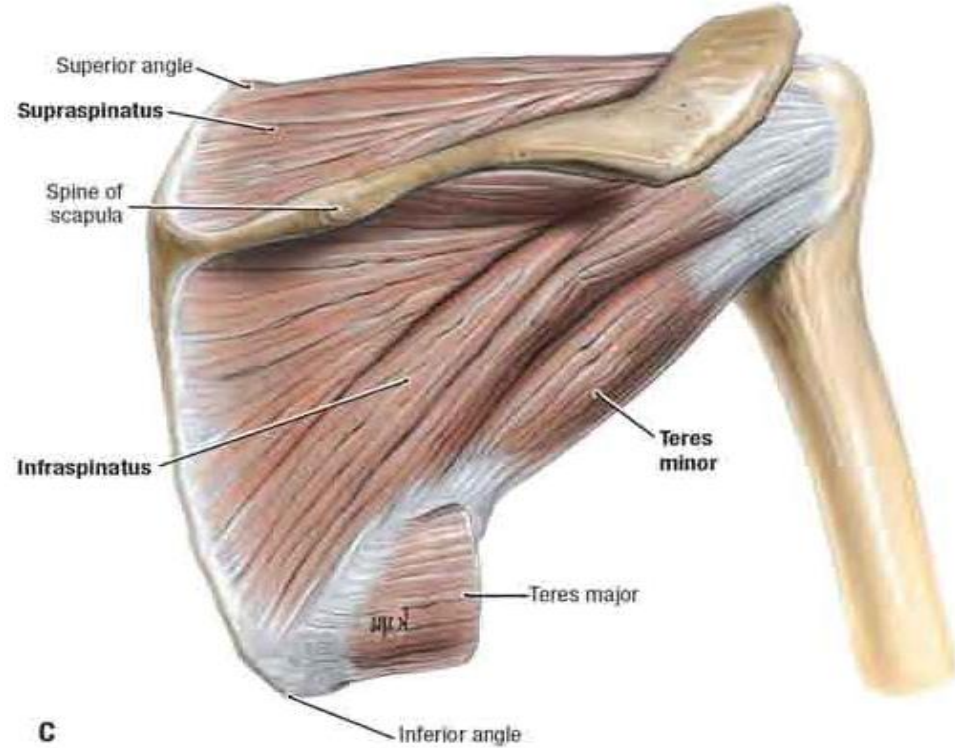
Teres minor

Origin: Middle part of lateral border of scapula

Insertion: greater tubercle of humerus

Innervation: Axillary nerve

Action: Laterally rotate arm; help hold humeral head in glenoid cavity of scapula



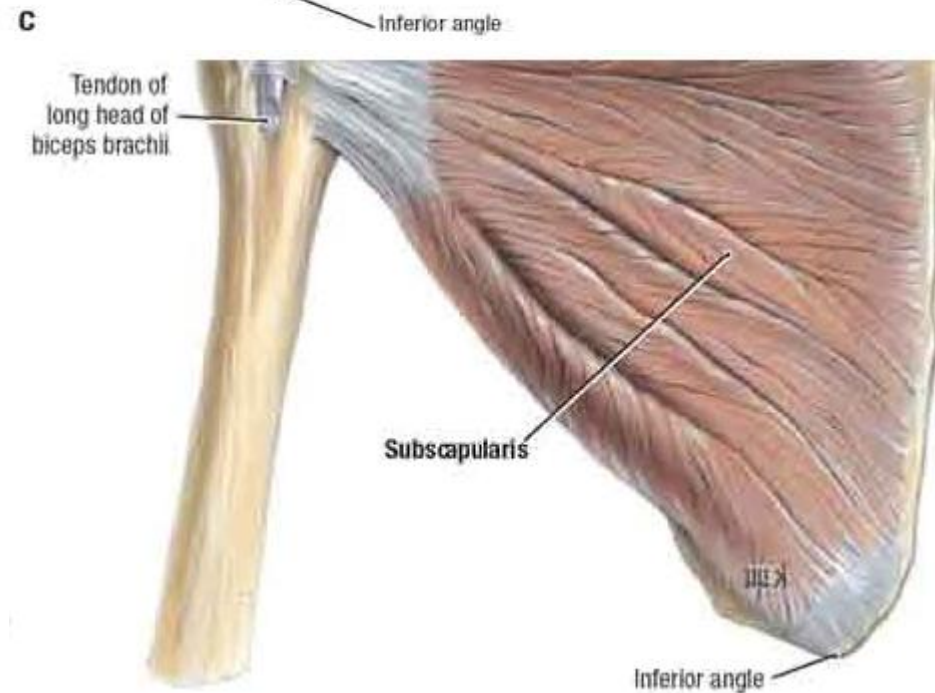
Subscapularis

Origin: Subscapular fossa

Insertion: Lesser tubercle of humerus

Innervation: Upper and lower subscapular nerves

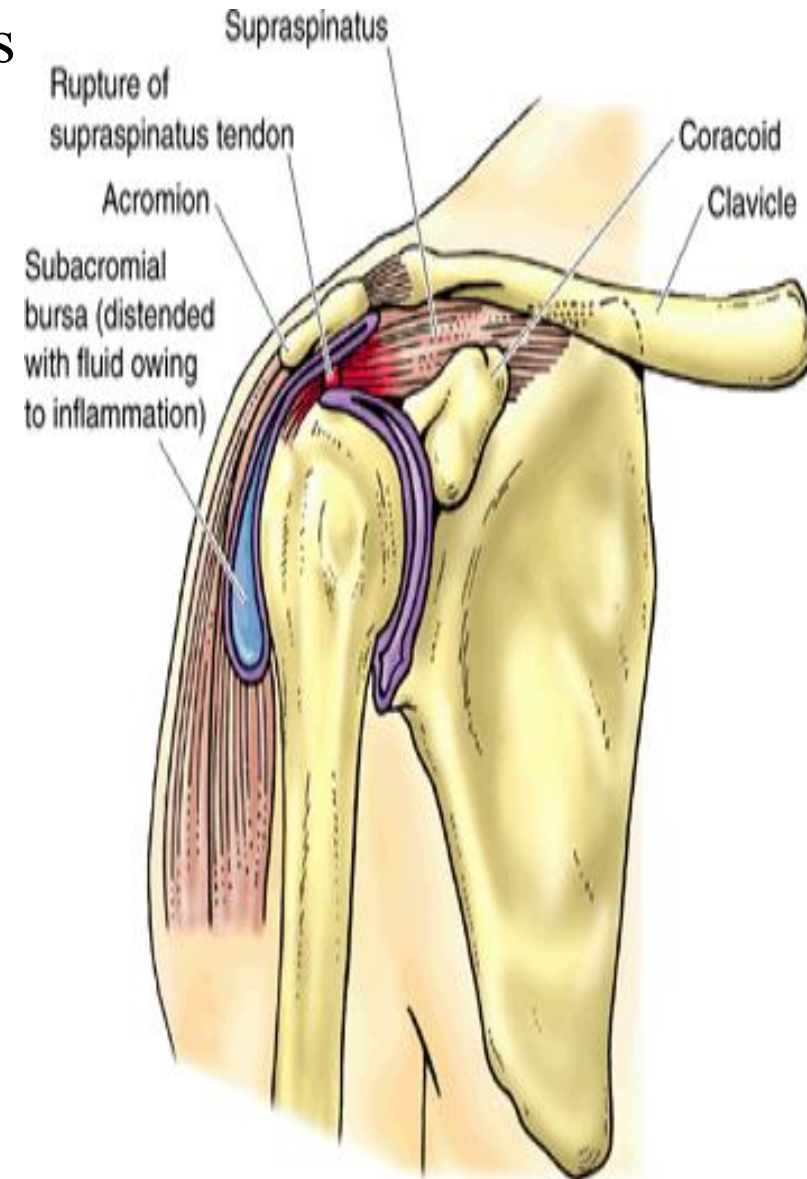
Action: Medially rotates and adduct arm; helps hold humeral head in glenoid cavity



Injury to axillary nerve

- ◆ Occur when surgical neck of humerus is fractured.
- ◆ Results in atrophy of deltoid.
- ◆ Rounded contour of shoulder disappears
- ◆ To test deltoid (function of axillary nerve) the arm is abducted against resistance

Rotator cuff injuries: instability of glenohumeral joint. Rupture of **supraspinatus** tendon is the most common injury



Anatomical space on the back of the shoulder

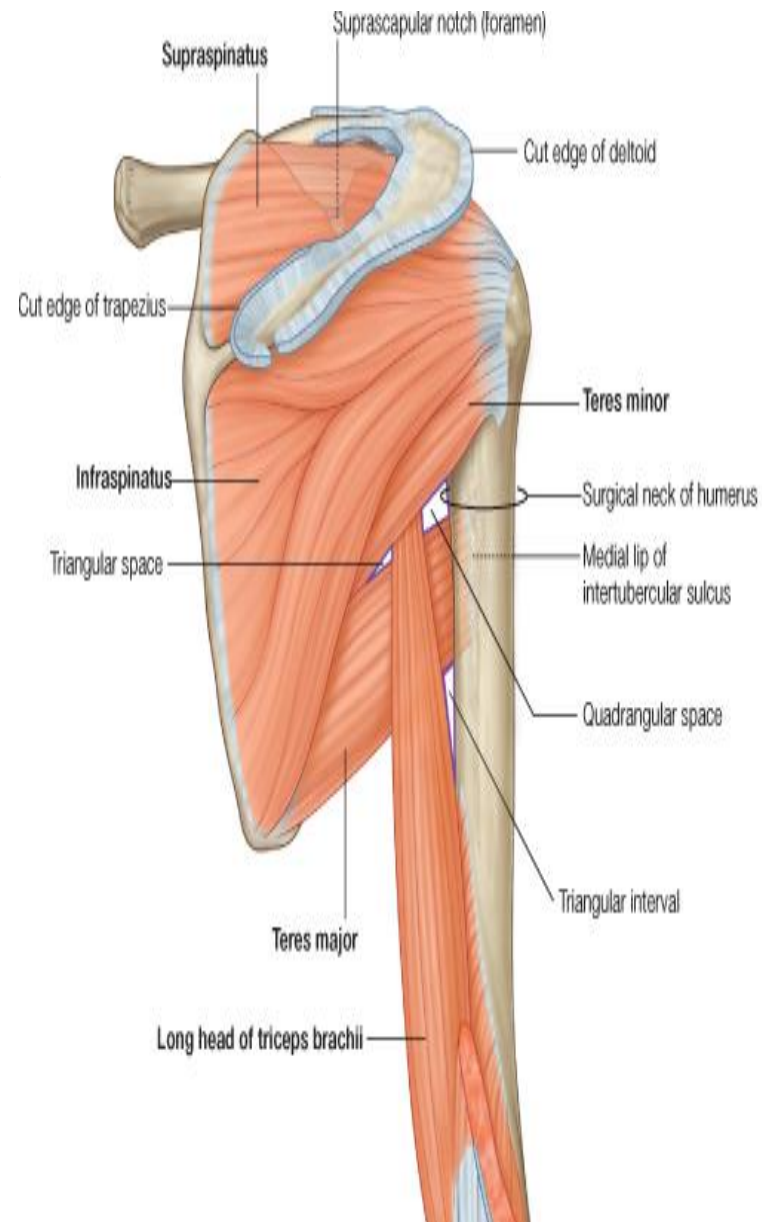
1. **Quadrangular space:** a potential intermuscular space

Boundaries

- Superiorly- Teres minor
- Laterally- Surgical neck of the humerus
- Inferiorly - Teres major
- Medially- Long head of triceps brachii

Contents

- ✓ **Axillary nerve**
- ✓ **posterior circumflex humeral aa pass**



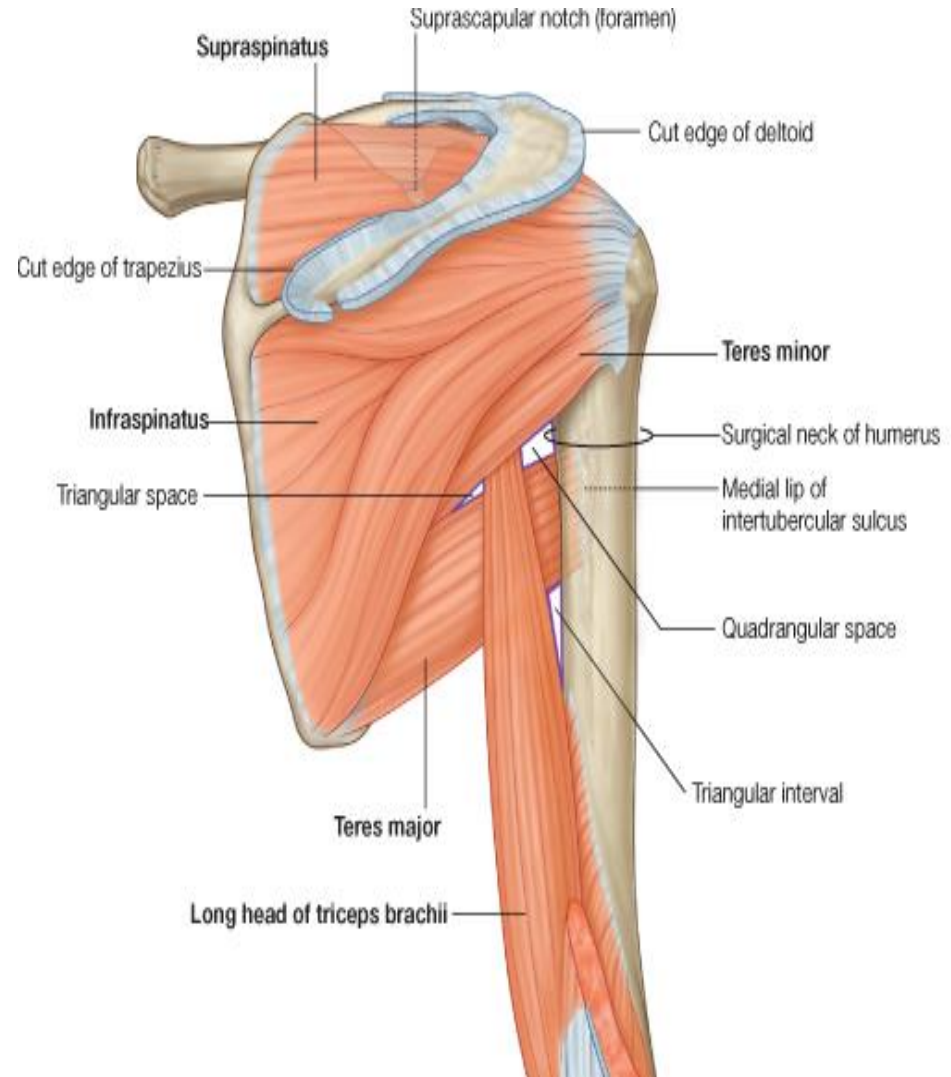
2. Triangular space

Boundaries

- ✓ superiorly- **Teres minor**
- ✓ inferiorly – **Teres major**
- ✓ laterally - **long head of triceps**

Contents

- ✓ **circumflex scapular aa**
pass through it



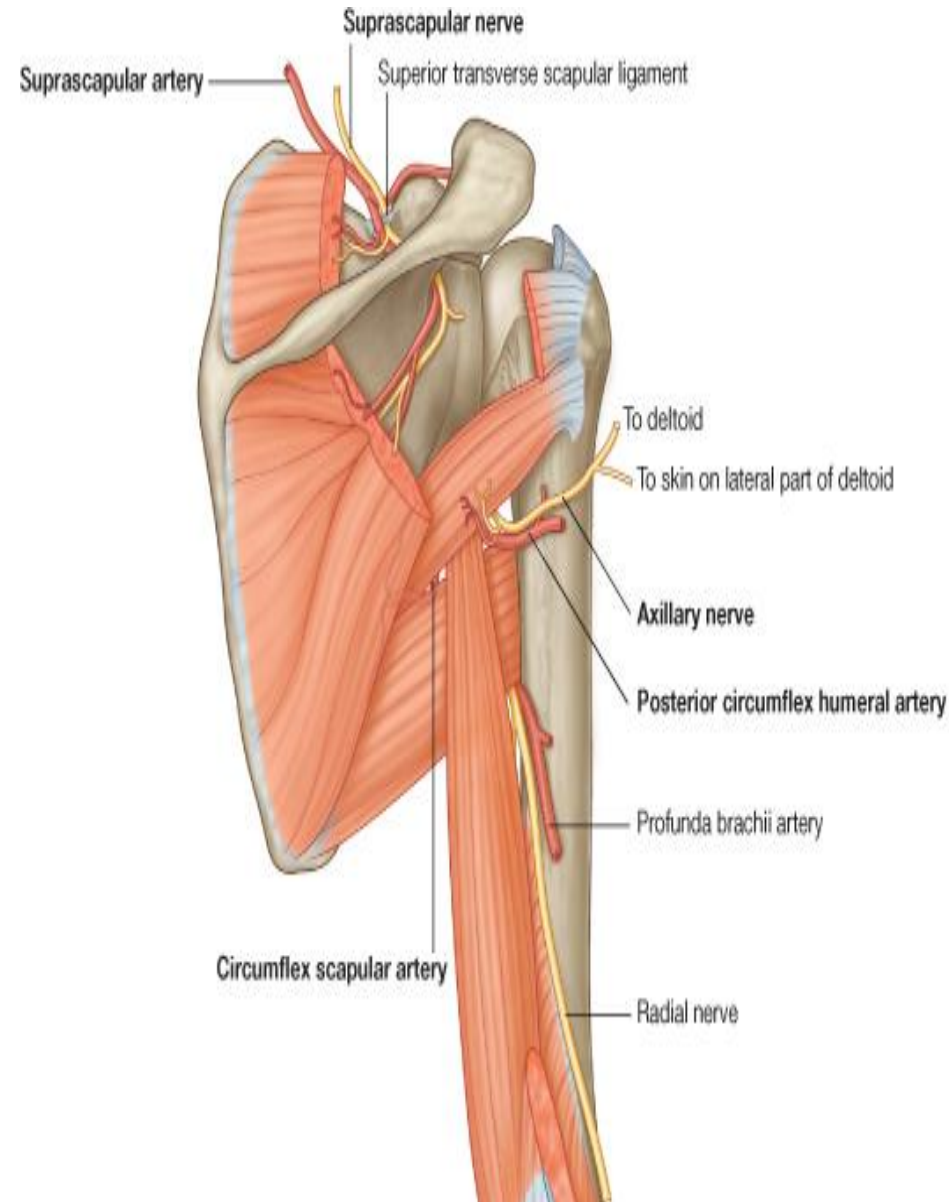
3. triangular interval

•Boundaries

- ✓ **superior** – Teres major
- ✓ **medial**– long head of triceps
- ✓ **lateral**- lateral head of triceps / humerus

•Contents

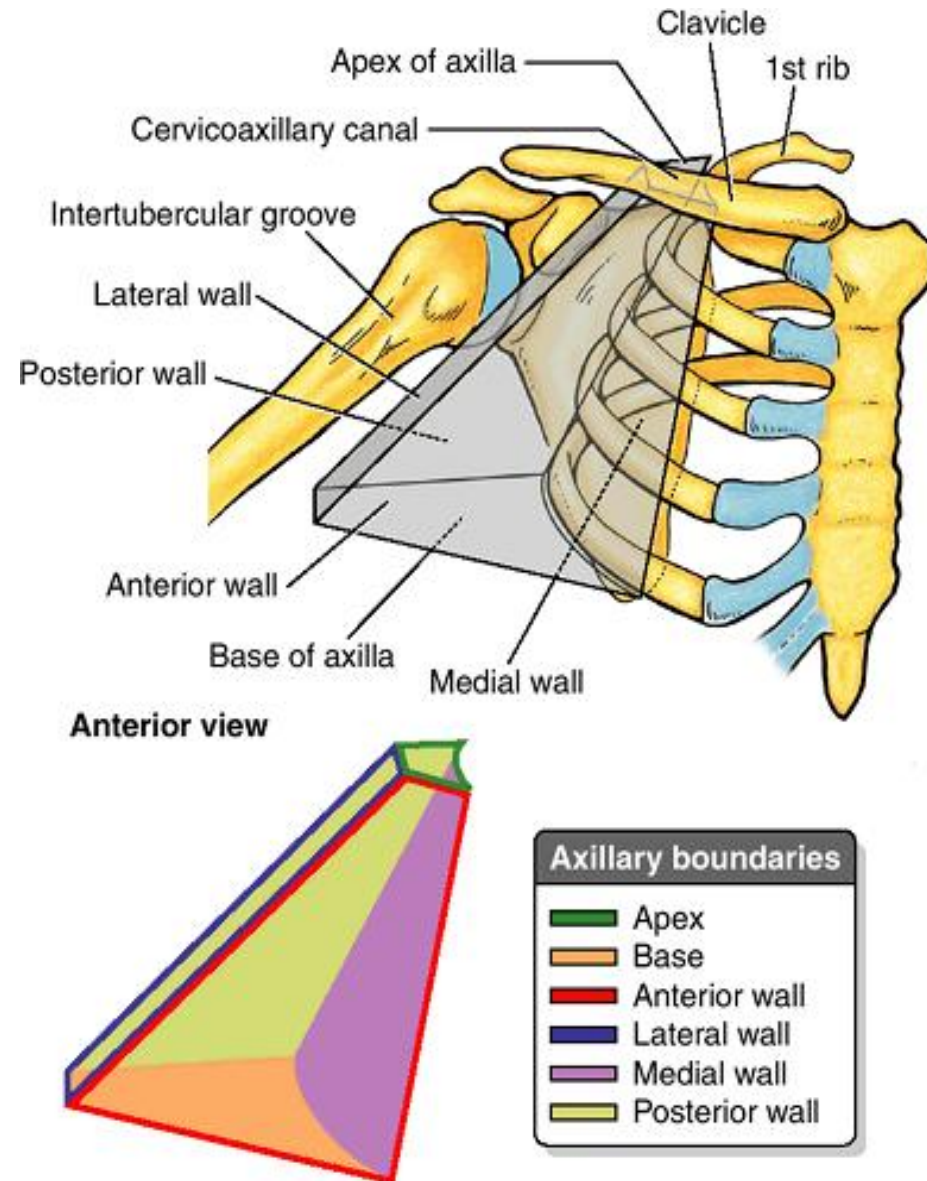
- ✓ **Radial** n and **profunda** brachii aa pass in it





Axilla & Brachial Plexus

- Pyramidal space b/n glenohumeral joint and skin and axillary fascia
- provides a passageway for vessels and nerves going to and from the upper limb
- Has an apex, base, and four walls
- **Apex**
 - Is cervicoaxillary canal
 - Bounded by **1st rib**, **clavicle** and **superior edge of scapula**
- **Base** (Axillary fossa)
 - Formed by concave **skin**, **subcutaneous tissue**, and **axillary fascia**



Axilla

Anterior wall

- Formed by pectoralis major and minor with their fascia

Posterior wall

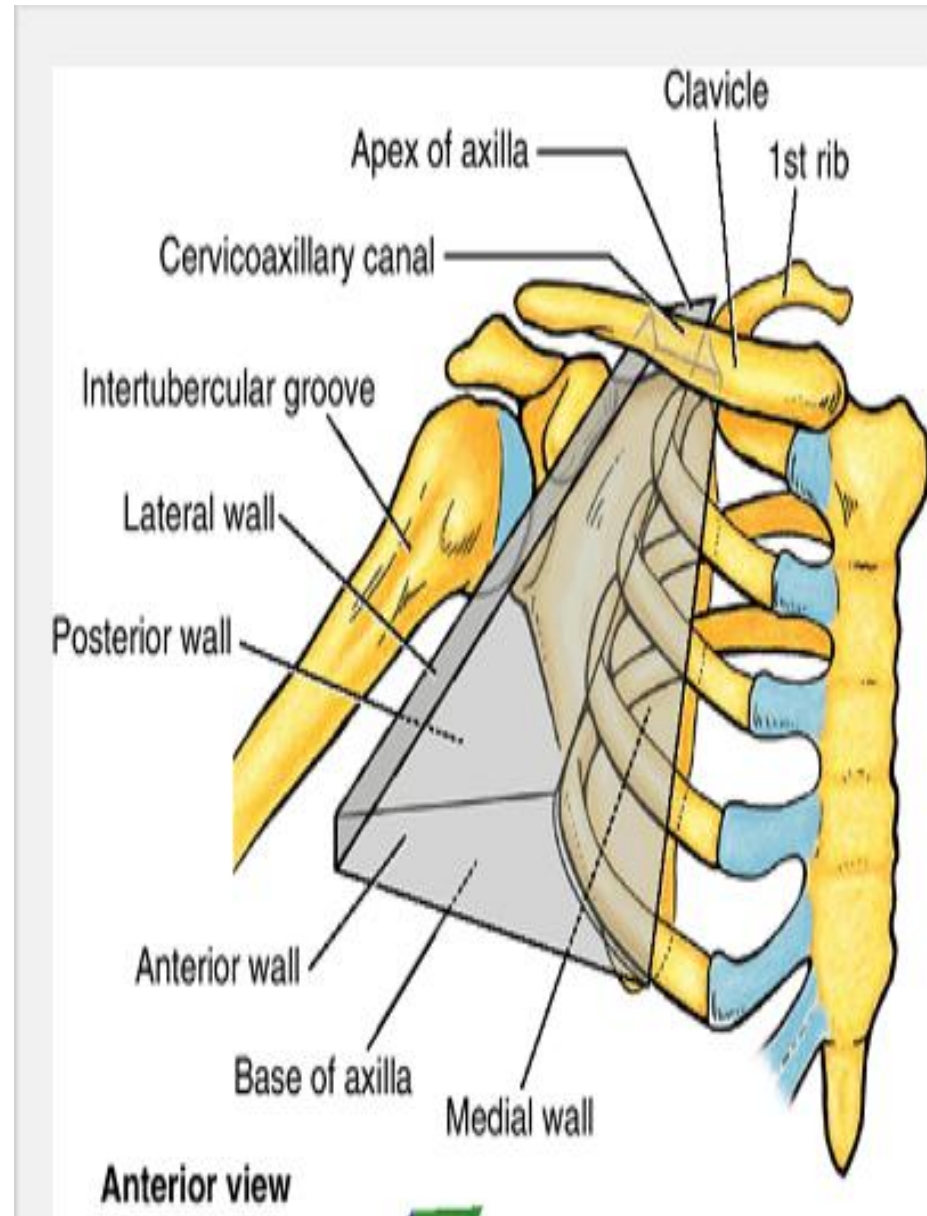
- Formed by scapula and subscapularis and inferiorly by the teres major and latissimus dorsi

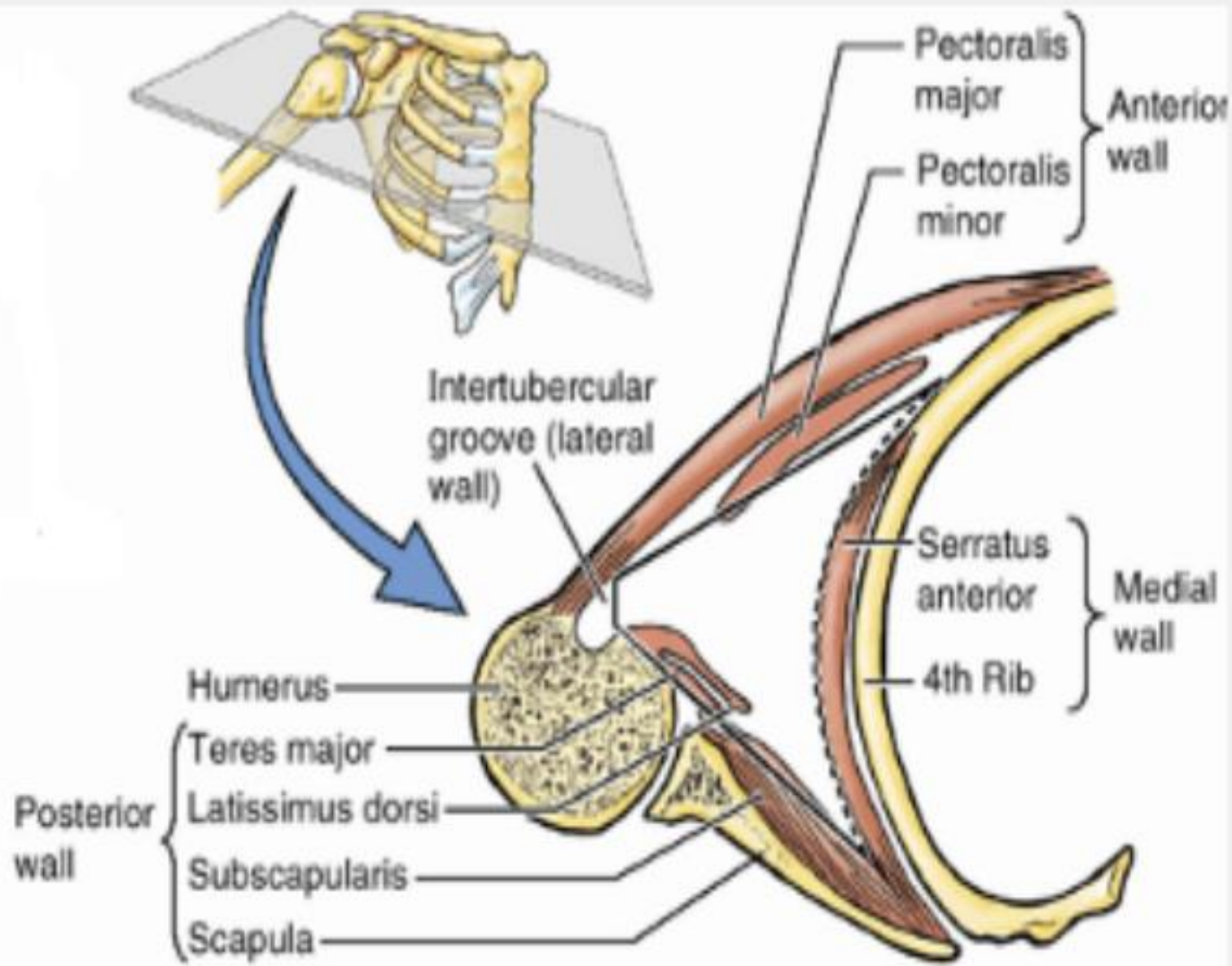
Medial wall

- Formed by the thoracic wall (1st-4th ribs and intercostal muscles) and serratus anterior

Lateral wall

- Formed by humerus



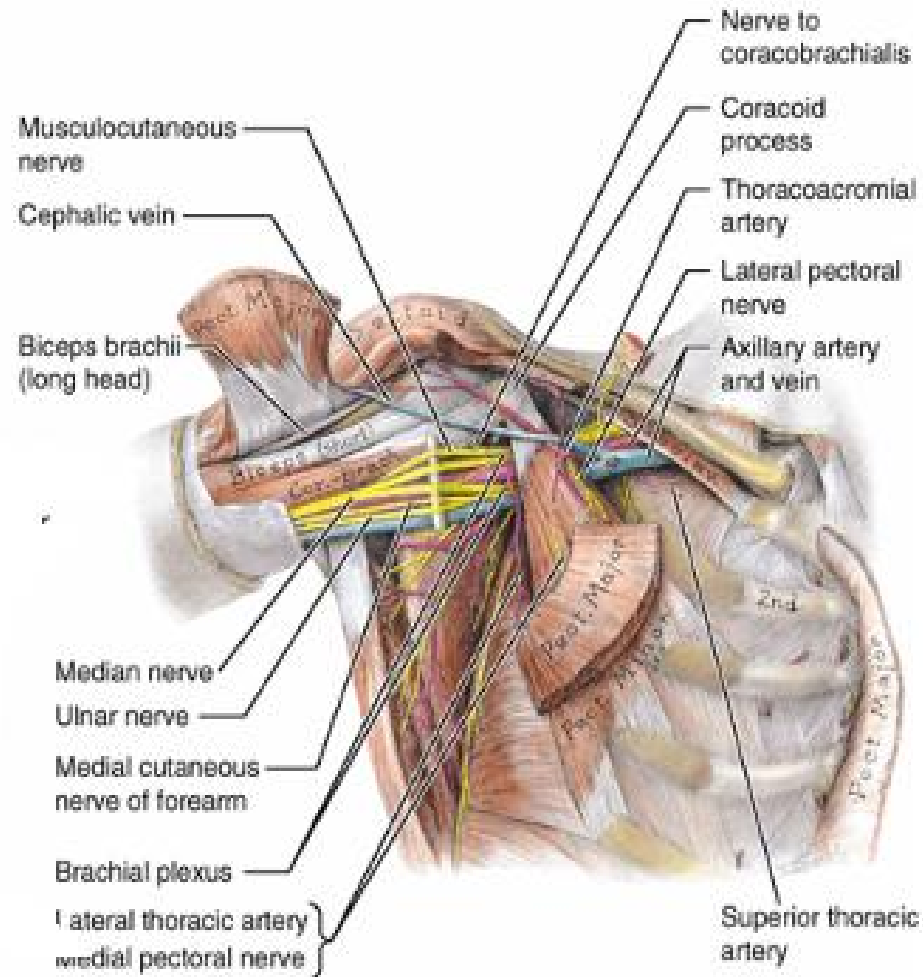


(B) Inferior view of transverse section

Contents of the axilla

The axilla contains

- ✓ The **axillary artery** and its **branches**
- ✓ **The axillary vein** and its tributaries
- ✓ The nerves of the cords and branches of the **brachial plexus**,
- ✓ lymphatic vessels, and several groups of **axillary lymph node**



Contents cont...

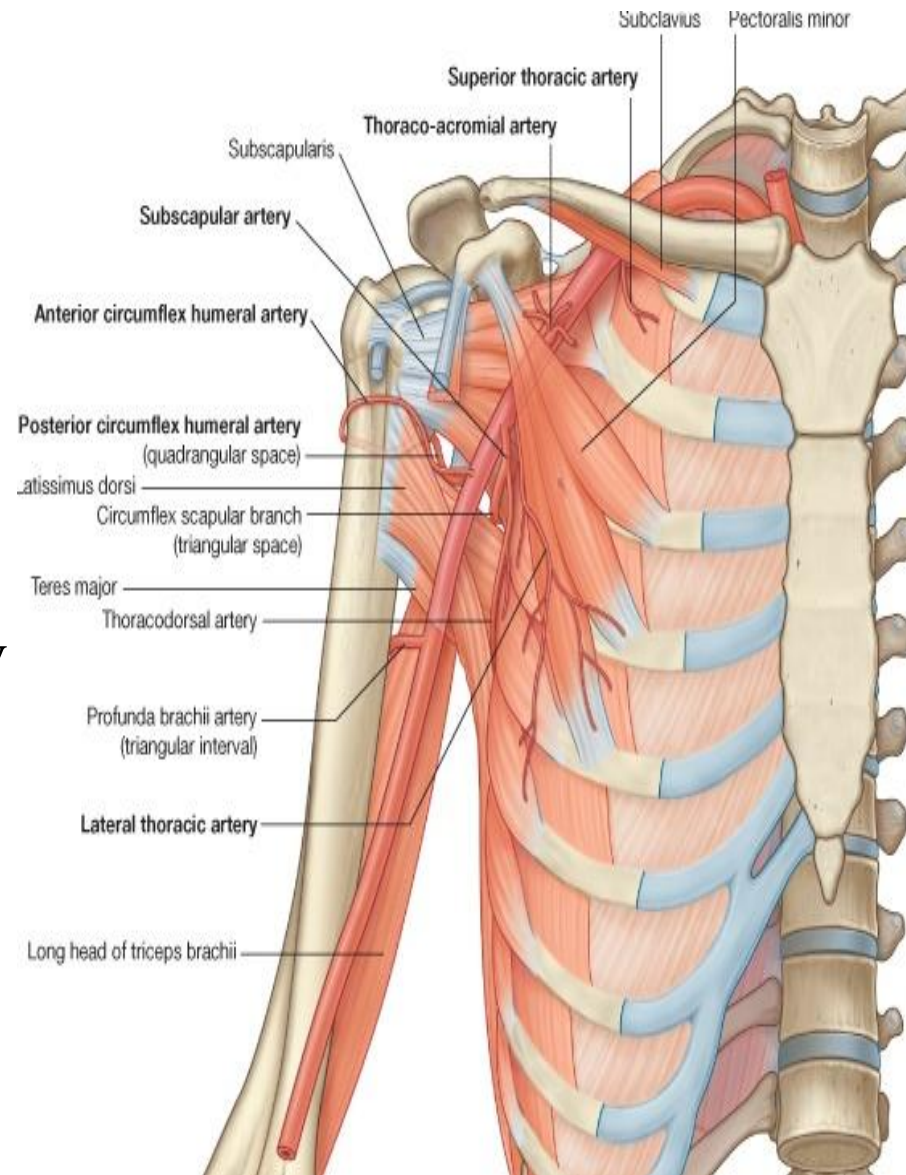
Axillary artery

- Extend from the first rib to **inferior border of teres major**
- divided into **three parts** relative to the pectoralis minor

1st part -superior thoracic artery

2nd part - thoraco acromial artery and lateral thoracic artery

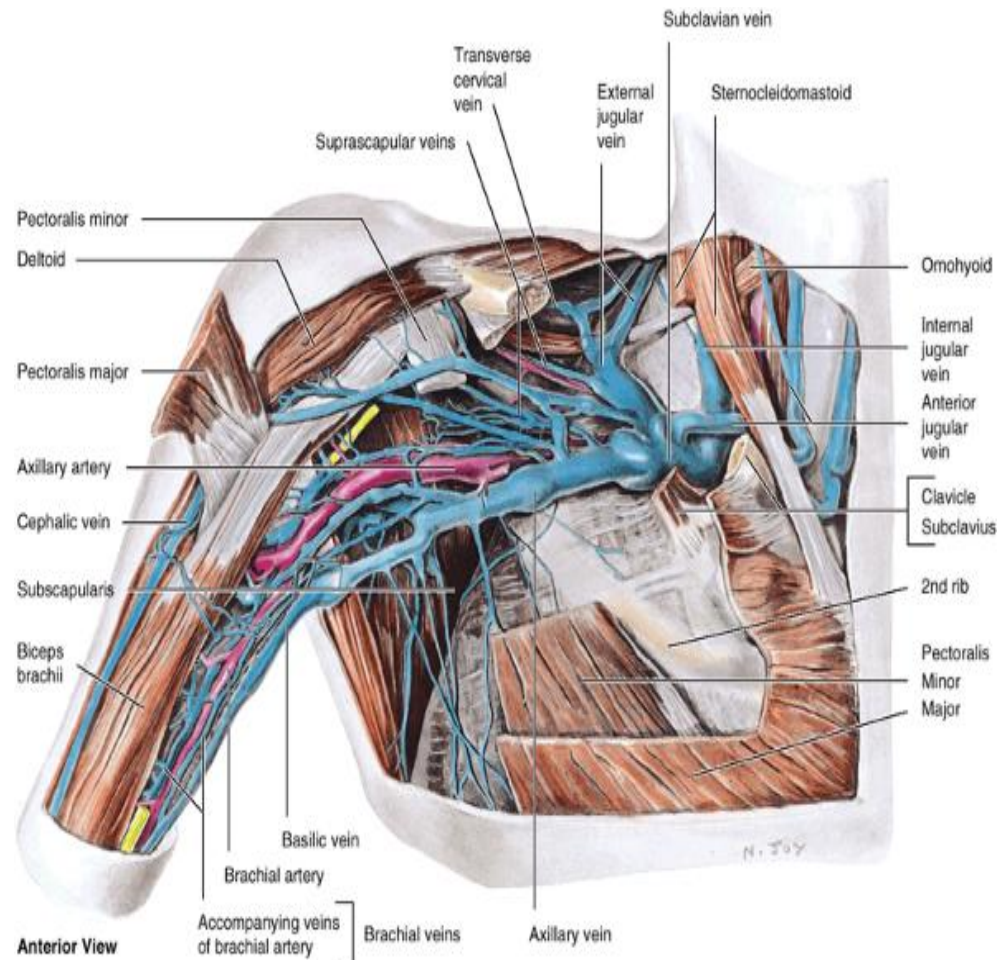
3rd part - subscapular artery, anterior/posterior circumflex humoral artery



Contents cont...

Axillary vein

- Anteromedial to axillary artery
- Formed by the union of accompanying brachial vein and basilic vein
- Ends at the lateral border of the 1st rib, where it becomes **the subclavian vein**

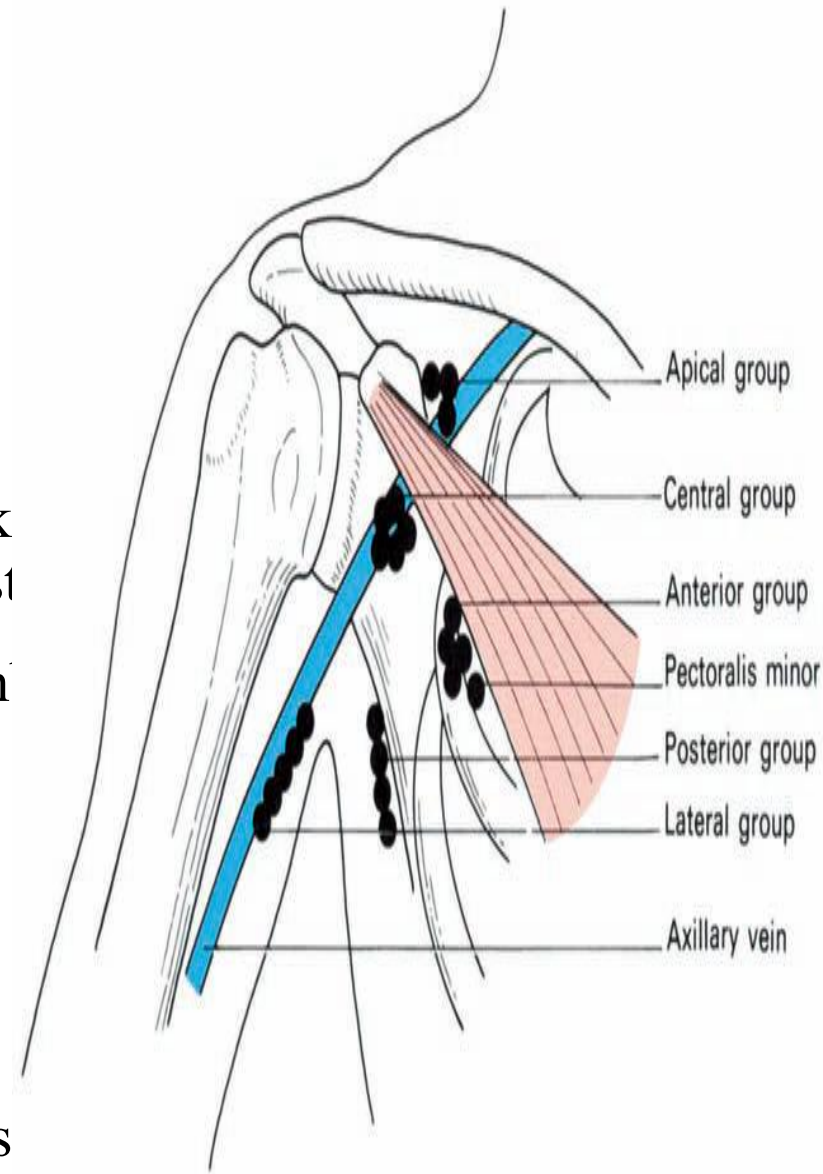


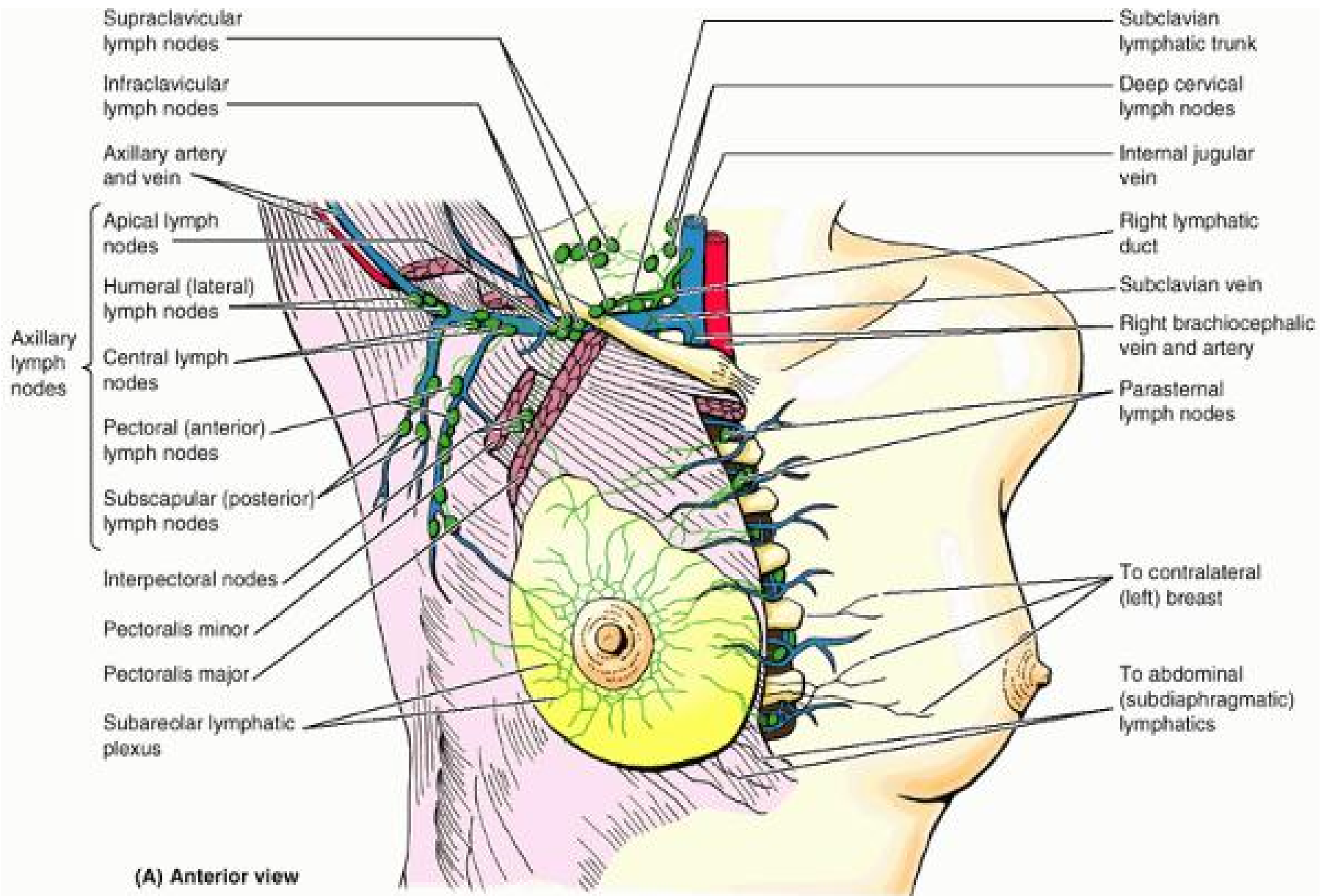
Contents contn...

AXILLARY LYMPH NODES

➤ groups of axillary lymph nodes

1. **pectoral (anterior)** - from the lateral quadrants of the breast and superficial anterolateral abdominal wall above the umbilicus.
2. **Subscapular (posterior)** from the back down as far as the level of the iliac crest
3. **Humeral (lateral)** - from the upper limb
4. **The central nodes** - deep to the pectoralis minor near the base of the axilla, from the above three groups
5. **The apical nodes** - at the apex of the axilla , receive the efferent lymph vessels from all the other axillary nodes





Contents contn...

BRACHIAL PLEXUS

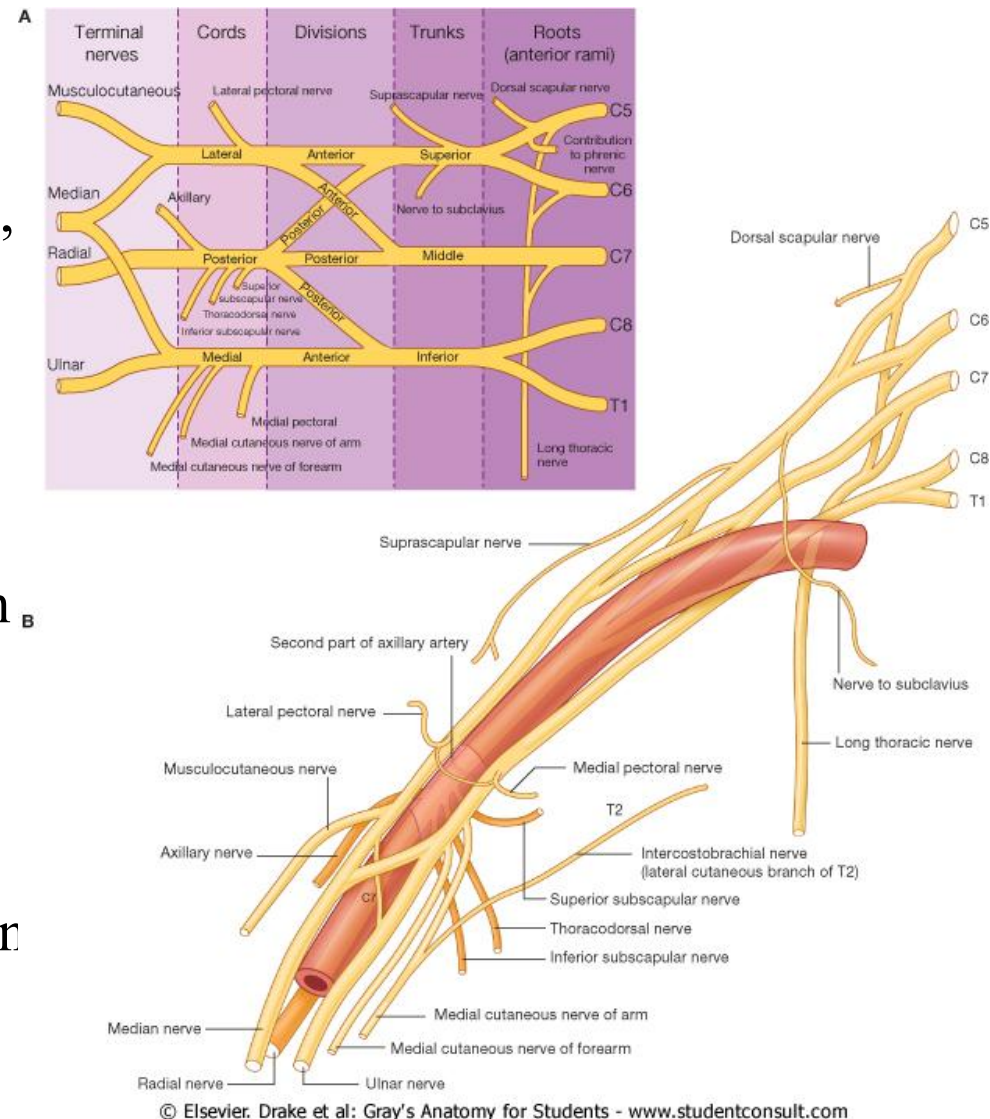
is formed by the union of the anterior rami of the C5-T1 nerves, which constitute the **roots of brachial plexus**

the roots unite to form three **trunks**

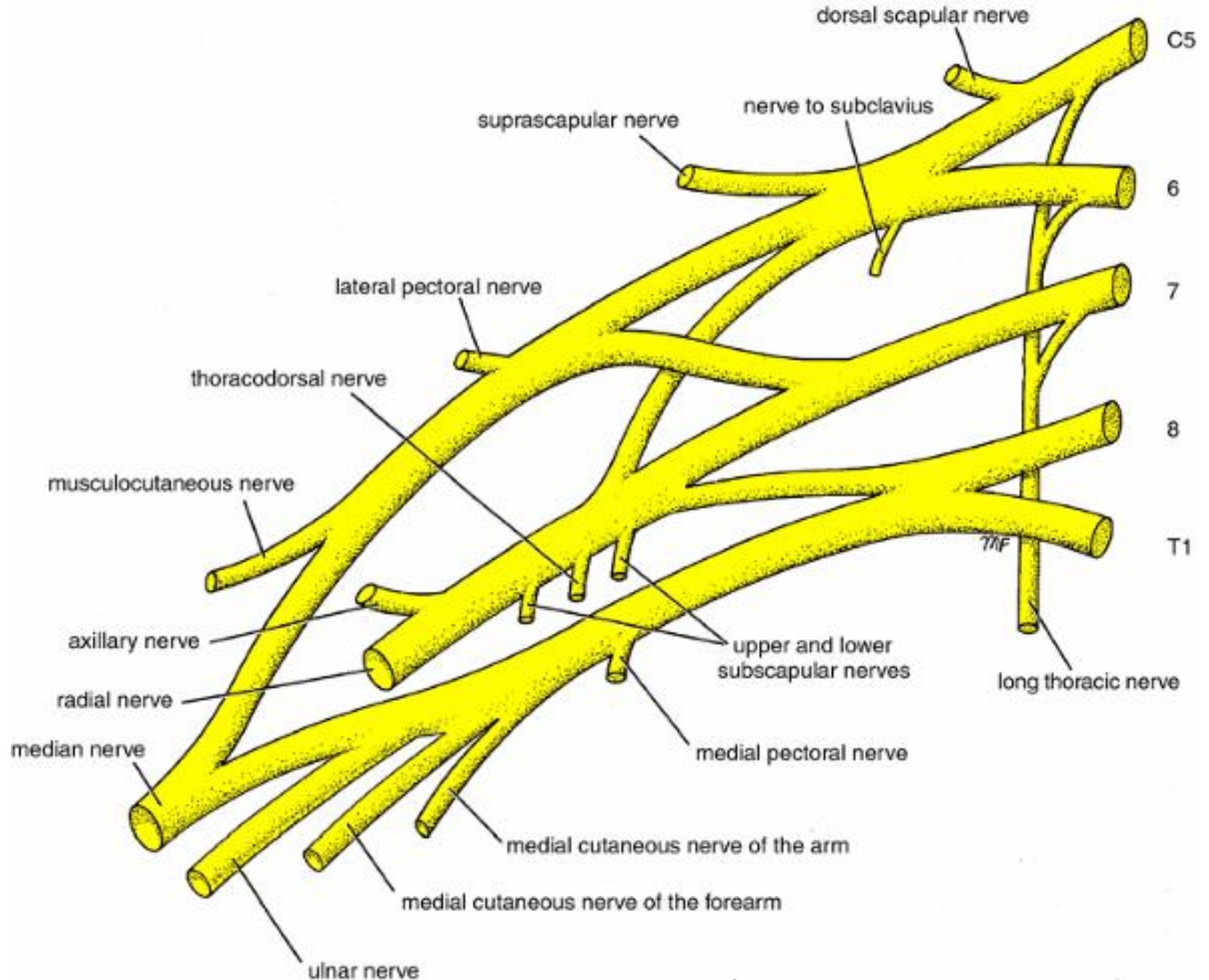
A superior trunk, from the union of the C5 and C6 roots.

A middle trunk, which is a continuation of the C7 root.

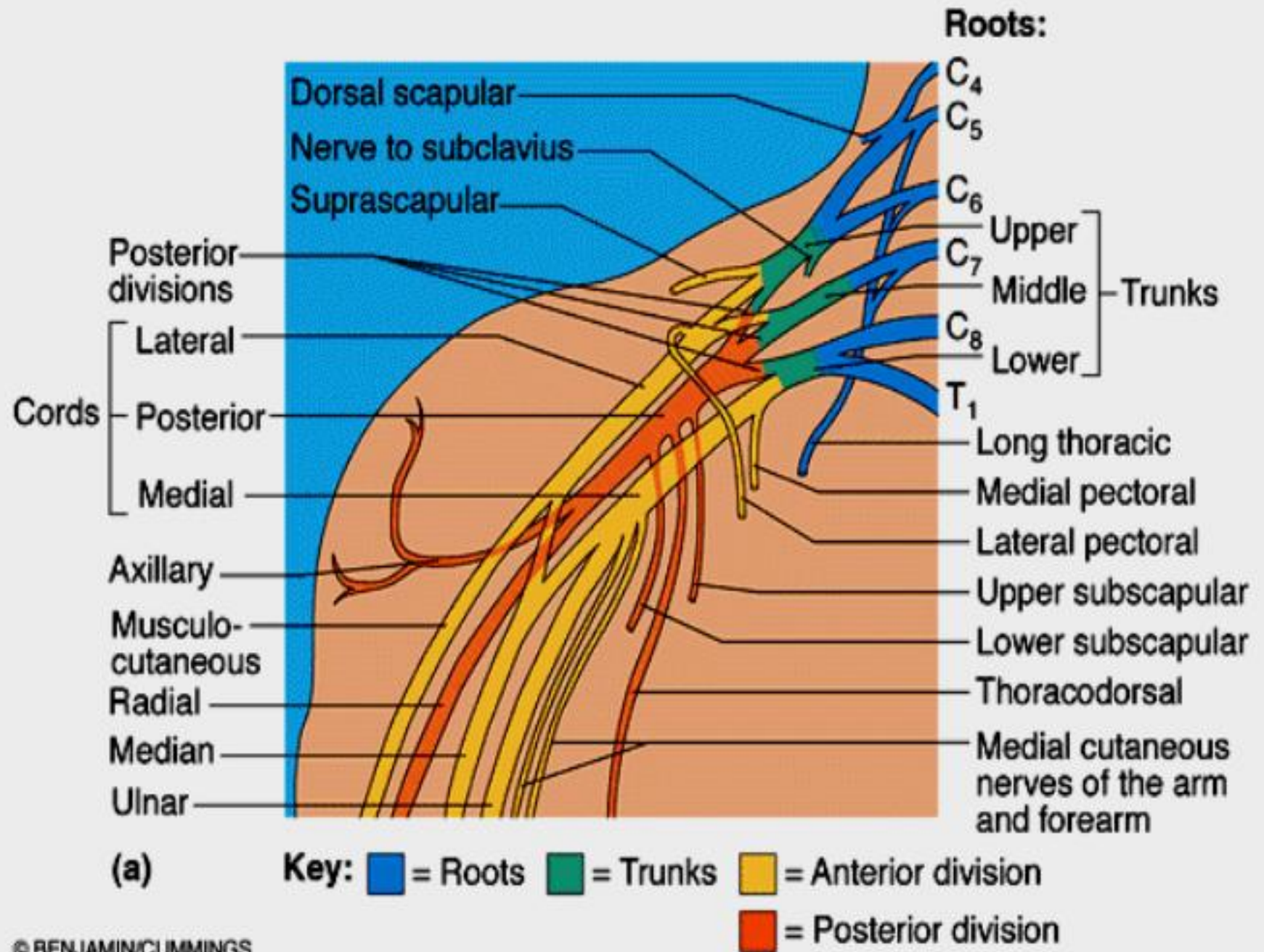
An inferior trunk, from the union of the C8 and T1 roots



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Roots, trunks, divisions, cords, and terminal branches of the brachial plexus





Thank you!!